

PATHWAYS TO PROSPERITY

Safeguarding biodiversity
in development

Marie A Brown

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Summary of recommendations

This report demonstrates that better outcomes for the natural environment can be achieved, and achieved in an efficient and effective way in a development context, through the adoption of a range of tools and improvements. The report's recommendations are summarised according to priority in Table 1. The four key areas for innovation and improvement are then grouped and discussed in further detail.

Priority category	Recommendations
1 Urgent priority	National policy on indigenous biodiversity Impact management policy Adaptive management guidance
2 Sustained, long-term improvements	Improving biodiversity information Regional biodiversity plans Guidelines for biodiversity impact assessment of vulnerable taxa and ecosystems Third party delivery of mitigation, offset and compensation requirements Conservation tenders for mitigation trust funding
3 Discretionary improvements	Tools for strategic planning Improving knowledge of existing legal mechanisms to protect biodiversity Outcomes-based conditions Enhanced use of RMA bonds Ecosystem service assessments in decision-making Safe Harbour agreements

Table 1: Recommendations set out in order of priority

1. Strengthen national leadership on the protection of indigenous biodiversity in development

A dominant theme that arose during our assessment of biodiversity management in development was the lack of national leadership on biodiversity. The effects of this lack of direction are evident in the relative absence of policy innovation in lower levels of government, the variation in the interpretation of biodiversity-related matters, and the resulting poor outcomes for biodiversity. As well as addressing these matters, consistent national instruments could help lessen agency capture at regional and district levels, which constrains the implementation of the Resource Management Act 1991 (RMA). Therefore we recommend that immediate attention be directed towards addressing these policy gaps, including promulgation of the following:

1. National policy on indigenous biodiversity
2. National policy on impact management
3. Guidelines for adaptive management

Policy processes are often slow-moving, but their ability to influence outcomes is potentially significant. Therefore we recommend that the Ministry for the Environment leads a national biodiversity policy programme to provide well-overdue guidance on the protection and management of indigenous biodiversity. This key task would require the engagement of the Department of Conservation – which we suggest should lead the development of impact assessment guidelines for vulnerable taxa and ecosystems – and the participation of civil society and resource users in the development process.

2. Formulate plans and policies that recognise and protect indigenous biodiversity now and in the long term

We currently lack a clear pathway for how biodiversity will be managed in a development context, as well as tools to advance along that pathway. These problems are exacerbated by a deficit of biodiversity information. Three key innovations are outlined:

1. Improving biodiversity information (including its accessibility)
2. Improving strategic planning through wider use of decision-support tools
3. Developing regional biodiversity plans to guide action towards achieving regional biodiversity objectives

The development of cooperative plans could help further engage communities in achieving better outcomes for nature. They could also assist with ensuring that conservation outcomes are maximised and that the activities of groups are coordinated towards clear strategic outcomes where possible.

3. Improve the use of existing tools

Supporting reliable and effective outcomes from resource management processes, and improving compliance, are urgent and crucial challenges. Refined use of existing tools will enable more efficient and secure outcomes for nature from development projects. Potential improvements include:

1. Utilising the full range of existing legal tools to protect biodiversity through increased use where appropriate
2. Providing guidelines for assessing development impacts on vulnerable taxa and ecosystems in assessments of environmental effects
3. Stronger requirements to achieve ecological outcomes in resource consent conditions
4. Using correctly costed and robustly administered RMA bonds to provide more reliable security for outcomes

4. Develop innovative and adaptive policy instruments to improve biodiversity outcomes and efficiencies

Policy innovation enables the development of new and flexible instruments which can safeguard nature in development in more effective and efficient ways. New tools canvassed include:

1. Third party delivery of offsets and compensation
2. Ecosystem services assessments for biodiversity in development
3. Conservation tendering by mitigation trusts
4. Safe Harbour agreements



Overview

In EDS's recent publication *Vanishing Nature: Facing New Zealand's Biodiversity Crisis* (2015), the authors argued that indigenous biodiversity and ecosystem services were declining due to private interests in extraction and degradation usually prevailing over the public interest in safeguarding nature. In addition, many people are disconnected from nature and therefore do not invest their time and resources in seeking its protection, resulting in biodiversity often being excluded from decision-making. A suite of solutions were proposed, the most transformative of which were novel economic institutions designed to align divergent stakeholders towards undertaking actions that build natural capital rather than destroy it. *Vanishing Nature* demonstrated that the status quo is failing to generate good outcomes for nature and is reducing New Zealand's prosperity in the long term. Big changes are necessary.

However, such transformative solutions take time and biodiversity decline continues daily. There is a need to also focus on immediate efforts that can help curtail loss. Development impacts are one of many pressures on biodiversity, along with the effects of invasive species and impacts of climate change, but they are significant. Biodiversity that is vulnerable to development pressures is being lost at very rapid rates and is generally poorly represented in protected areas. In the absence of large-scale change, such losses are likely to continue. But their extent can be reduced through innovation and change in policy and practice, and that is the focus of this report.

The report identifies improvements and extensions to the array of tools currently available in New Zealand to manage the impacts of development on biodiversity. They can all be implemented with no or limited legislative change. The changes we recommend will not only yield improved outcomes

for nature, but will do so in an effective and efficient manner. There is much opportunity for agencies, communities and the private sector to better manage the impacts of development in order to achieve biodiversity protection goals without significantly constraining economic development. For resource users and developers, more adaptive policy will enable greater flexibility in achieving economic outcomes.

Purpose and scope

The purpose of this report is to identify planning and policy tools that could help better achieve the effective protection, maintenance and enhancement of biodiversity and ecosystem services in New Zealand while still enabling economic development to occur. The report aims to answer several key questions:

1. What are the tools and methods in New Zealand that are used to address the impact of development on biodiversity and how can they be improved?
2. What gaps exist in the current availability of tools, and are there options available that can fill these based on experience in other jurisdictions?
3. How could those tools be operationalised, taking account of the New Zealand situation?

The scope of the investigation has been limited to tools that are likely to deliver improved outcomes within the current context, without requiring substantial legislative or organisational change. They include several different types of policy instruments. The focus has been on identifying practical opportunities for agencies and resources users to innovate, and consequently the solutions focus on the activities of these two stakeholders. The report does not

address the need to provide incentives for conservation on private land beyond interaction with development activities, nor does it address conservation issues more widely.

Research approach

The research approach included a mix of primary and secondary research. The primary research consisted of interviewing some 80 people including agency frontline staff, industry representatives, people in the conservation community, scientists and policy makers throughout New Zealand and Australia. There was broad recognition that existing law and policy is simply not generating the outcomes for nature required, nor the outcomes for society that are desired. All interviewees provided crucial insight and experience which was drawn on to formulate the ideas and conclusions in this report. The secondary research involved reviewing the many existing reports and articles and web-based sources on biodiversity and development. For example, an extensive domestic review of possible policy instruments was conducted by Greenhalgh et al. (2012), which also analysed the extent to which they could be harnessed to protect ecosystem services.¹

Based on interviewee observations and experience and our secondary research, a number of potential tools were identified. Some of these are currently operational on a small scale in New Zealand but have potential merit at a national level. Others had demonstrated their capacity to enhance outcomes in other jurisdictions and are potentially applicable to the New Zealand context.

Throughout our analysis it also became clear that, in addition to applying new tools, there is considerable potential to improve the implementation of existing tools. We identified a wide range of instruments that currently exist but which are rarely used and do not seem to be well known by many practitioners. As a result, approaches to improve the application of existing tools (and the knowledge of practitioners about what they are and how they can be applied) were also included in the investigation.

The potential tools identified were then assessed for their likely applicability to the biodiversity-development nexus in New Zealand. A large number were discarded either because their immediate utility and influence on outcomes was unclear or because, although seemingly promising at first glance, further analysis indicated that they were not the most effective means of generating good outcomes. Tools that would likely require significant legislative change (e.g. dedicated threatened species legislation) were also set aside. A list of tools investigated in detail but ultimately discarded, and our reasons for not including them further in this report, is included in Appendix 2.

In some cases, exclusion was a judgement call, based on the researchers' experience and knowledge in the field. This process resulted in a short list of prospective tools which are described in this report. They have only been included where the research, particularly the analysis of known outcomes, was compelling enough to suggest that their introduction could generate improved outcomes in practice.



Marokopa Beach, Waitomo District. Bryce McQuilian

The biodiversity-development nexus

This report focuses on the nexus between biodiversity and development, so we start by defining biodiversity, describing its current state in New Zealand, and exploring the challenge of managing development impacts upon it.

New Zealand's biodiversity and how it is faring

Biodiversity underpins ecosystem function, which in turn underpins our prosperity. Biodiversity is defined as *the variability among living organisms from all sources including, inter alia, terrestrial, marine and other aquatic ecosystems and the ecological complexes of which they are part; this includes the diversity within species, between species and of ecosystems.*² Managing this public good is fraught with difficulty due to its complexity. Biodiversity is non-fungible (one species or habitat type cannot be replaced by another), meaning it is difficult to achieve fair and mutually beneficial biodiversity exchanges, presenting an additional (and some would argue intractable)³ policy challenge compared to other forms of natural capital (e.g. carbon).⁴

New Zealand's unique biota is globally significant both for its uniqueness and the level of extinction threat it faces. New Zealand ranks second for endemism to family level.⁵ Examples of high endemism in different taxa are 100 per cent of lizards, bats, frogs and conifers; 96 per cent of freshwater fish and spiders; 81 per cent of flowering plants and 56 per cent of birds.⁶ The biota is so unique, that New Zealand was identified as one of 35 international biodiversity hotspots by Mittermeier et al. (2004).⁷ High rates of endemism and a very high proportion of threatened species make for a fragile and vulnerable biota. Examples of the species and ecosystems that make New Zealand unique can be found throughout this report (called 'Nature Snippets').

New Zealand has suffered significant historical loss of biodiversity. Loss of species through habitat removal and modification, harvest and pollution is compounded by modern development pressures and the impact of biological invasions including mammalian predators, weeds, pests and diseases. Of the remaining biota, nearly 40 per cent of our plants, 40 per cent of our birds, 74 per cent of our freshwater fish and more than 85 per cent of our lizards are threatened with extinction.⁸ A greater proportion of New Zealand's native species is at risk of or threatened with extinction than those of any other country.⁹

In October 2015, the government released the first mandatory national report on the state of the environment: *Environment Aotearoa*. This outlined biodiversity loss and climate change as key areas of concern. The report demonstrated the shortcomings in our knowledge of biodiversity, both in terms of our fundamental understanding of species, ecosystems and how they interact, and information on their current states and trends. Many species are so poorly understood as to be categorised as 'Data Deficient' – meaning that we don't know enough about them to identify whether they are threatened or not. Since 2005, seven per cent of our species have become more threatened.¹⁰

Biodiversity is affected by the acute and chronic impacts of human activities through habitat loss, degradation and fragmentation; direct harvesting of nature; pollution from sediment, heavy metals and nutrients; and ongoing disturbances. The loss of large tracts of habitat, and ongoing fragmentation of what remains, has reduced the presence of nature on the landscape.¹¹ Research demonstrates that areas of the country (e.g. lowlands and river plains) which have suffered the greatest historical losses continue to be the most threatened by development today.¹² Simply retaining

Box 1 The importance of ecological restoration

Ecological restoration has been defined in numerous ways, including as *the process of assisting the recovery of an ecosystem that has been degraded, damaged, or destroyed*.¹³ Ecological restoration is core to conservation in New Zealand because of the rapid and large-scale environmental change that followed the waves of human settlement. If a representative array of our biodiversity is to be protected, a great deal more ecological restoration will be necessary. The degraded state of many of our remaining ecosystems and species demonstrates the ecological overshoot which has occurred and from which we must recover. The biodiversity-related activities of community groups, research organisations, regional and local councils, landowners and many corporates revolve around ecological restoration projects in urban and rural areas.

Ecological restoration includes the reinstatement of hydrological systems, planting of indigenous species, control or eradication of mammalian pests, creation of other habitat (e.g. artificial reefs in marine areas), removal of invasive plant matter and active interventions to save specific species. The background decline of biodiversity in the absence of active management is a common consideration in a development context. Ecological restoration techniques are commonly called upon by developers when carrying out mitigation, offset or compensation projects. The knowledge amassed by the field of ecological restoration helps to inform environmental impact assessment and biodiversity monitoring. It also assists in determining the feasibility of proposed projects.

biodiversity at current levels is insufficient: ecological restoration is critical to meeting our international, national, regional and local biodiversity goals. Therefore the need for ecological restoration is an important consideration in a development context (Box 1).

Managing the impacts of development: The challenge

Economic development is necessary for human society and the benefits from resource extraction and development are shared to some extent. In many cases mining, farming, quarrying, residential development and the construction of infrastructure such as roads and wastewater systems result in biodiversity loss – both temporary and permanent. Development activity occurs across all of New Zealand's land, freshwater and marine systems, even in protected areas. For the purpose of this report, 'development' includes all human activities that have an acute impact on nature through disturbance. This includes the full gamut of activities from low-impact tourism through to large-scale habitat clearing for activities such as agriculture, forestry and urban development.

Trade-offs between the loss of biodiversity and the benefits of these activities are common. A growing population, declining natural capital and many remaining economic development opportunities being located in biodiverse areas has led to the conflicts between economic development and nature protection becoming more acute.¹⁴ The extent of biodiversity loss in New Zealand makes it clear that biodiversity protection and economic development must be far more carefully managed in the future. In theory this is a straightforward endeavour. In practice it is complex and likely requires more than traditional governance approaches. The challenge is to find a way to foster development which

is environmentally sensitive while still enabling it to be financially viable.

In some cases, the profit-making concerns of developers and the interests of the broader public in biodiversity conservation are moving into greater alignment. This is largely due to evolving consumer markets and an increasing focus on corporate social responsibility and the so-called 'social licence to operate'. For example, Case Study 1 demonstrates that extraction companies will, in some cases, explore opportunities to reduce environmental impacts, particularly where financial gains are also prospective. Other examples include the purchase of hybrid or fully electric fleet vehicles, the installation of energy-efficient fixtures in commercial premises, and the rise of eco-labelling.

In most cases, however, government policy is required to safeguard biodiversity, particularly in a development context. Although such policy can be effective in slowing biodiversity decline, it can also impose significant costs. The overall cost of policies are comprised of the direct costs of implementation (e.g. increased staffing requirements to undertake new responsibilities), the indirect costs that can arise from implementation (e.g. predation by recovering native species on orchards) and opportunity costs (e.g. lost economic possibilities through policies that restrict access to natural capital). Opportunity costs are generally greater than the other categories.¹⁵ If new policy approaches are likely to significantly increase direct or opportunity costs for developers, they generally have to overcome sizeable opposition. That aside, new policy approaches will be most acceptable for the business community when they are fit for purpose, logical and do not entail unreasonable costs.

The development community has voiced significant concern in recent years over the narrow array of policy tools implemented in New Zealand. Developers are also vocal about the cost associated with gaining approval across multiple

Case Study 1 Waikawa Fisheries and Precision Seafood Harvesting

New Zealand operates a scampi fishery within the Quota Management System. Scampi is fished via bottom trawling, a method of fishing that can damage life on the seafloor. In addition to the benthic impacts, the scampi fishery nets five times as much bycatch as target fish. Around the world, scampi-like species are generally fished with potting methods. Those methods are cost-efficient compared with bottom trawling, and yield live catch which is more valuable. They also reduce bycatch and benthic impacts.

The high level of bycatch has been of concern for some years, and one quota owner has investigated a lower-impact approach. Waikawa Fisheries, a small family-owned company in Nelson, has engaged the Cawthron Institute to develop an alternative method of fishing that substantially reduces bycatch, using suspended pots. The research is being supported by funding from the Ministry of Business, Innovation and Employment.¹⁶ Unfortunately, none of the types of pots used overseas work with New Zealand scampi. They are bait-shy and 26,000 hours of experimental fishing on the Chatham Rise netted just two scampi.¹⁷ Work is continuing on the design of a custom pot for New Zealand scampi.

Another project developing new fishing technology designed to increase financial returns (while helping to reduce environmental impacts) is Precision Seafood Harvesting, which is being supported by the Ministry for Primary Industries' Primary Growth Partnership programme. The purpose of the project is to enhance the quality of landed fish caught by trawl nets while also reducing bycatch. The fish are landed on board alive, potentially enabling higher market prices to be obtained for them, and the survival chances of bycatch returned to the sea are increased.¹⁸



Reducing bycatch through innovations in seafood harvesting will help protect marine biodiversity. *Precision Seafood Harvesting*

regimes for the same project. The policy tools on offer are mainly regulatory and are frequently highlighted as being too restrictive, thereby limiting the ability of resource managers to respond to a generally dynamic socioeconomic context.¹⁹ As a result, the tools fail to generate the desired environmental outcome because they are often not fit for purpose.²⁰ The dissatisfaction of resource users has been a key driver for the continuing reforms to the RMA, demonstrating the influence of this sector. The state of biodiversity in New Zealand sends a clear signal that existing policy settings are not addressing biodiversity loss sufficiently, which makes them difficult to defend and sustain.

A framework for action: *Vanishing Nature*

This report follows on from the 2015 EDS publication *Vanishing Nature: Facing New Zealand's Biodiversity Crisis* and builds on its analysis. *Vanishing Nature* set out the history of biodiversity management in New Zealand from the time of first human settlement. It catalogued the decline of our biodiversity and the response to that by communities, agencies, businesses and politicians. Biodiversity loss is a chaotic issue – with multiple drivers and impacts – so that many struggle to grapple with the complexity of it. The

second chapter of *Vanishing Nature* cut out much of the noise to articulate what the key problems really are. The profound rates of loss were traced back to three fundamental drivers:

- Biodiversity is generally excluded from markets
- The power of those wishing to damage nature is greater than that of the public who have an interest in safeguarding it
- People do not understand the extent of their dependency upon nature

All other conventionally understood drivers of harm are 'proximal' drivers or immediately obvious impacts, of which there are many. All such proximal drivers have their roots in the fundamental drivers, and those fundamental drivers are the real conservation challenge.

Vanishing Nature explained why it is so difficult to manage human impacts on biodiversity and ecosystem services.²¹ They are common pool resources which are freely and easily obtained and so it is difficult to guard against their extraction or depletion. Hardin's classic 'tragedy of the commons' descriptor is relevant here. Those that benefit from the degradation of nature can do so handsomely, while bearing only minor depletion costs, if any. They tend to be highly motivated to protect their access to nature and have resources to deploy to this end.²² The public interest bears the cost of depletion, and none or few of the benefits, but the cost that falls on any individual person is small. Without the incentive of potential economic gain, there is less to motivate the wider public to engage in safeguarding their interest in nature conservation. This power imbalance is called a 'collective action problem' and explains why private interests tend to have far more overall influence on regulatory design and the workings of public agencies than the more diffuse public interest.²³ This influence constrains the effectiveness of conventional management approaches as agencies are often 'captured' by private interests.²⁴

Vanishing Nature goes on to set out what this power imbalance means on public land, private land, in freshwater and marine environments, and for community conservation. On public land the profound funding shortfall imperils the biodiversity perceived to be most secure – with underinvestment in conservation science, statutory advocacy and protected area management all having negative consequences for nature. On private land, active destruction and benign neglect result in constant attrition of natural capital, with limited requirements to address most externalities. Weak implementation of law and policy and an absence of polluter pays approaches, have contributed to serious cumulative impacts on the environment. A disjointed legislative framework around the management of freshwater biodiversity, and a more limited (but evolving) one in the marine environment, is equally concerning. Habitat loss, fragmentation, modification, pollution and the ever present impact of biological invaders make for a perilous context for our natural heritage. Community conservation

and the efforts of private landowners are significant and are definitely impressive. But they are generally scattered and uncoordinated and often struggle to maintain momentum. They are thus unlikely to address opposing drivers of harm.

But solutions are possible. Those set out in *Vanishing Nature* occupied three tiers: strategic, tactical and practical (Figure 1).



Figure 1: Diagram of three solution types outlined in *Vanishing Nature*

- 1. Strategic solutions** are the game-changers that have immense impact, such as the proposed environmental consumption tax and rebate system. Strategic solutions align the very divergent interests of different stakeholders towards common goals, giving them all a good reason to do the right thing, even if for different reasons. They increase the capacity for conservation of nature. They also reduce the need for, and increase the effectiveness of, tactical and practical solutions. Strategic solutions are few in number and difficult to implement, but they generate far better and more sustainable outcomes than lower-level solutions.
- 2. Tactical solutions** are those which seek to better balance private and public interests – legislation is a typical example. Tactical solutions are often not fit for purpose and can contain ambiguities and deficiencies that agencies are slow to address. In the absence of strategic tools to align stakeholder interests, tactical solutions are usually only modestly successful at safeguarding nature. The fundamental drivers discussed earlier typically constrain the effectiveness of these interventions.
- 3. Practical solutions** enable us to do more with the resources we already have to safeguard biodiversity. They include covenants, captive breeding methods, pest control techniques and surveying approaches. They are politically popular, easy to implement, and they increase efficiencies.

The solutions focused on in this report are either tactical or practical. They are designed to sit within the wider landscape of change recommended by *Vanishing Nature*, and to contribute specifically to the situation where indigenous biodiversity – particularly that which is vulnerable to loss – is threatened by development activities.

Current policy for safeguarding biodiversity in development

This section identifies the types of policies that can be deployed to manage the impacts of development on biodiversity and then reviews the current application of these policy types in the New Zealand setting. Throughout this review opportunities for improvement are identified.

An overview of policy types

A policy is a course or principle of action adopted or proposed by an organisation or individual to achieve an objective or desired outcome. Given that biodiversity is a common pool resource, policy is generally implemented by public

agencies. However, policy tools can be both implemented and supported by all actors.

There are five key types of policy approaches to protecting any public good, including biodiversity (and included within that, ecosystem services) and these are set out in Table 2. While regulatory approaches predominate globally, experience tells us that they often fail, or sometimes where good outcomes are achieved it is at disproportionate social and economic cost. In recent years, there has been growing recognition of the importance of having a diverse array of

Type	Description	Examples
Regulatory instruments	A penalty will be incurred if harm or previously determined acceptable levels of harm to the environment are exceeded.	Command and control, permits and technological prescriptions
Economic instruments	Price and market-based initiatives that render environmentally harmful activities expensive or incentivise good practice through economic impacts on the operation. They may act to replace or more usually, support regulatory requirements.	Taxes, tariffs, subsidies, tradable permits and ecosystem services accounting
Procedural instruments	Processes that assess compliance and demonstrate alternative ways of doing things.	Audits and environmental impact assessments
Cooperative instruments	Collaborative processes that draw together a range of different actors towards a common goal.	Commitments, agreements, strategies, action plans, research programmes and partnerships
Persuasive instruments	Primarily information-based processes that do not bind actions and entail minimal intervention.	Information, education and advocacy

Table 2: Types of policy approaches which can be applied to biodiversity management

policies and policy approaches that are flexible and adaptable to changes over time.

In recent decades there has been international recognition of the need to deploy a diverse array of biodiversity policy approaches (adaptive policy) but this has been slow to catch on in New Zealand. Adaptive policies are more likely to be robust and respond well to changing circumstances than rigid approaches that leave little room for innovation or quick response to changing needs. A range of policy tools that are combined, so as to be complementary, is considered optimal although a single effective policy may also be sufficient.²⁵ Importantly, policy tools must be clearly and logically matched with the issue they are attempting to address in order to be fit for purpose.²⁶

Evaluating policy outcomes

The acid test of any biodiversity policy of course lies in the difference that it makes to behaviour and by extension to the environment. Monitoring is required to understand the

impact and effectiveness of policy, and without investing in it the inertia of the status quo tends to persist with little compulsion to re-evaluate or change tack. Insufficient effectiveness monitoring may also mean that approaches that are 'working' are difficult to defend and are thus removed and not replaced (e.g. Auckland tree protection).²⁷

In New Zealand, effectiveness monitoring has been somewhat patchy despite statutory requirements for agencies to monitor policy outcomes. Under section 35 of the RMA, decision-makers are required to evaluate the outcomes of policy, specifically *the efficiency and effectiveness of policies, rules, or other methods in its policy statement or its plan*.²⁸ Similar evaluation processes are in place under other legislation such as the Local Government Act 2002. Under the RMA, a report is required to be produced at least every five years.²⁹ The depth and quality of these reports varies significantly and there has been little oversight from central government in this regard. In *Vanishing Nature*, the need for increased agency reporting and accountability was identified and this need remains.

Black mudfish (*Neochanna diversus*)

The native black mudfish are found only in the northern part of the North Island. They are small fish, generally growing to about 90 millimetres in length, that prefer gently flowing (swampy streams) or still water (wetlands). Within their range, their habitat has been severely reduced due to drainage, drain clearance, herbicide use, reduced water quality and the impacts of invasive pest fish. Mudfish prefer shallow water with low turbidity and plenty of vegetation cover to give them shelter during the day from potential predators (such as the kingfisher). They rely on large wetland complexes with intact hydrology.³⁰

Despite the conservation concerns surrounding mudfish they are well adapted to their environments. They are able to survive long periods on land after their habitat dries out (an adaptation that may have enabled them to survive in the presence of potential aquatic predators). When their habitat is permanently destroyed by development, they are commonly translocated to other areas. For example, the construction project for the Waikato Expressway through Rangiriri included a three-year translocation programme which saw the mudfish held in tanks for a number of years until they were released into a newly restored habitat.³¹ Release sites are also under pressure, with the number of potential sites reducing over time, meaning that the creation or restoration of destination wetlands is sometimes necessary.



Black mudfish (*Neochanna diversus*). Jennifer Price

Biodiversity policy in New Zealand: The actors

There are three key types of actors in the biodiversity-development setting in New Zealand: proponents of development (referred to as 'resource users' for the purpose of this report); agencies such as councils, the Department of Conservation and other permitting entities such as New Zealand Petroleum and Minerals; and 'non-vested interests'. Non-vested interests comprise the wider public and its interest in nature protection and the preservation of cultural and intrinsic values. 'Non-vested' implies that the public has no stake in the matter, which of course is not true – everyone benefits from the protection of nature. The meaning of 'non-vested' is restricted to those having no financial stake after Walker et al. (2008).³² This category generally has little power, while those with a vested interest in securing access to a resource (i.e. resource users) have more influence. Agencies – the 'middle men' – frequently side with the most powerful of those players, meaning that they often enable development activities at the cost of the public interest.³³ The different interests and preoccupations of these groups have significant implications for how policy is formulated and implemented.

A number of government agencies are involved in managing development impacts on biodiversity (see Appendix 1). But the prime responsibility falls on regional and district councils (as well as unitary authorities which combine regional and district council functions) and the Department of Conservation. The councils largely manage the impacts of development on privately owned land, in freshwater and in the marine space under the RMA. They do this through establishing a regulatory planning framework and determining resource consent applications (in the first instance) within this framework.

The Department of Conservation is mainly concerned with activities undertaken on the conservation estate and some which impact on protected species elsewhere. It administers three consenting regimes: concessions and authorisations for activities conducted on the conservation estate under the Conservation Act 1987 (access arrangements are issued under the Crown Minerals Act 1991); and the permitting regimes under the Wildlife Act 1953 and the Marine Mammals Protection Act 1978. In considering permits to allow activities that will or could harm biodiversity, the Department exercises the property right of the Crown on behalf of the people of New Zealand. Decisions are made primarily by staff under delegation, and otherwise by the Minister of Conservation, sometimes in association with other Ministers. Departmental decision-makers follow an internal standard called the Consent Application Processing Standard Operating Procedure which lays out requirements for processing and timeframes. Decision-making considers the degree and magnitude of effects; whether the proposed activity is consistent with the 'purpose for which the area

is held'; and whether it is consistent with the relevant conservation management strategy or plan.

Other agencies also play a role in managing the impacts of development (see Appendix 1) such as the Environmental Protection Authority which was established in 2011. In addition, the Courts have an important role in decision-making, especially under the RMA, as many cases are directly referred or appealed to them. The specialist Environment Court resolves 75 per cent of its cases via mediation and is actively involved in improving professional practice, for example by participating in regular professional development programmes.

Regulatory instruments

New Zealand has a comprehensive suite of environmental regulation that is implemented by the range of statutory agencies described above. The application of the various regulatory tools to biodiversity management is explored below. Our analysis reveals that law and policy goals are often unrealised because they lack clarity and a clear pathway for implementation; inappropriate tools are relied upon to reach the goals; and follow-up is typically lax and lacks funding.

National policy statements

The RMA makes provision for the promulgation of national policy statements *to state objectives and policies for matters of national significance that are relevant to achieving the purpose of this Act*.³⁴ There is no dedicated national policy statement on biodiversity although a range of non-statutory supporting documents are in place. A lack of leadership and guidance is an enduring challenge in biodiversity management, and is perhaps best demonstrated when one examines the multiple attempts to promulgate a national policy statement on indigenous biodiversity (see Case Study 2).

The New Zealand Coastal Policy Statement is the only compulsory national instrument under the RMA and has been in place since 1994. It has contained consistently strong provisions for the protection of biodiversity. The current Policy 11 (in force since 2010) directs that impacts on indigenous taxa that are listed as threatened or at risk in the New Zealand Threat Classification System lists are 'to be avoided'.³⁵ The findings of the Supreme Court in the 2014 *Environmental Defence Society v King Salmon Company* case mean that this requirement is now more stringently interpreted than it was in the past.³⁶

Subnational instruments

In the absence of overriding national policy, subnational documents dominate in the form of regional policy statements, regional plans and district plans under the RMA (administered by councils) and conservation management strategies and plans under the Conservation Act 1987 (administered by the Department of Conservation). These two Acts are the primary vehicles for any strategic and

Case Study 2 The biodiversity national policy statement challenge

A priority action arising from the New Zealand Biodiversity Strategy (discussed later) was the introduction of a National Policy Statement on Indigenous Biodiversity (one of the identified core 'building blocks' necessary for its implementation).³⁷ In 2000, the Ministerial Advisory Committee on Biodiversity noted its importance, but considered that other actions were more pressing.³⁸ An attempt to promulgate such a national policy statement took place a few years later. This process faltered, and in 2006 the Ministry for the Environment and the Department of Conservation jointly released non-statutory guidance on protecting biodiversity on private land (the *Protecting Our Places* document). A further attempt in 2011 similarly failed, despite relatively broad support across most submitter groups.

The bulk of those submitting against the provisions of the document were private landowners, businesses and other entities that derive economic gains from nature at minimal costs. While the costs of the policy were frequently highlighted, the private benefit derived from the use and damage of nature did not receive similar attention. The power of private interests prevailed over the public interest in nature protection, and the proposal was sidelined. In order to bring the document to fruition, government and other stakeholders will need to demonstrate leadership and political bravery.

The proposed National Policy Statement on Indigenous Biodiversity 2011 would have enabled the following: the implementation of the goal of no net loss at a national scale (currently reflected already in several second-generation regional policy statements); a common definition of ecological significance and common minimum criteria to assess this; clearer direction to local and regional government on implementing their existing mandates; and guidance on emerging tools such as biodiversity offsets. In *Vanishing Nature*, we outlined potential improvements to the 2011 version, noting that it remains deficient in a number of respects but that overall it would help to increase the policy visibility of biodiversity.³⁹ Current signals from the Ministry for the Environment indicate that the Department is supportive of a national policy statement on biodiversity and work is underway to reinvigorate the process. At the 2015 EDS Conference, Minister for the Environment Hon. Dr Nick Smith announced his support for a collaborative process to finalise the policy.



Strong national direction on biodiversity will help ensure decisions safeguard our most vulnerable species such as the South Island Takahē (*Porphyrio hochstetteri*). *Manu Bird*

landscape-scale planning for biodiversity by agencies and the communities they serve. The RMA is much more dominant in practice and is presently undergoing reform, which may have an impact on the solutions proposed later in this report.

Planning documents prepared under the RMA are the primary means by which councils exert regulatory control over biodiversity-harming activities. Overall though, a lack of leadership has meant that each individual council devises different policies and plans to address biodiversity and related policy matters. Each document is distinct, typically goes through a separate planning process, and may

contain very different provisions to that of even adjacent jurisdictions. The contest between private interests and wider environmental protection mandates is generally fought out at a local or regional level, through often cumbersome and energy-intensive planning processes that can take many years. As well as inconsistency between documents, there is broad recognition that these subnational instruments are failing to address adherence to environmental bottom lines, the management of cumulative impacts, and the long-term and complex consequences of human impacts on the environment.⁴⁰

District and regional plans protect biodiversity by assigning activities different statuses, with rules attached, depending on their potential effects on the environment (including biodiversity). These controls may include restrictions on the clearing of vegetation, requirements for erosion control to reduce sedimentation, restrictions on the damming or diverting of waterways, and other land use controls to address potential sources of harm (e.g. nutrient limits). Resource consent is required to breach any rule in a plan. Most activities not controlled by the plan are 'permitted' (except for rules applying to freshwater and the coastal marine area where the reverse presumption applies) or may be exercised via existing use rights.⁴¹ The depth, scope and quality of these provisions with respect to biodiversity vary significantly throughout the country.

Opportunity Statement **The provision of national leadership and direction would help to focus biodiversity policy, improve consistency and reduce costs of policy formulation and implementation.**

Development approvals

Permits to allow development activities can be issued under four key Acts of Parliament (see Table 3) and this legislation is the focus of this report. Approval can also be sought to carry out activities that may have a negative impact on biodiversity under other legislation, although consideration of biodiversity may be more limited. For example, Land Information New Zealand routinely administers activities on parcels of Crown

land, but is not generally required to seek advice on whether the effects on biodiversity are likely to be deleterious.

Many consents and permits are issued with conditions that help manage the impact of the activity on the environment. The relevant consenting agency is generally responsible for ensuring that the requirements of the approval are met and for taking enforcement action if they are not. Conditions are crucial elements of an approval because lack of compliance with them is the basis of enforcement, and as such they are the principal active tool for enforcing the legislation. If the intent of the condition and its desired outcome is not clear it can constrain enforcement options and environmental outcomes. A lack of clear conditions is a common complaint of resource users, communities and enforcement officers. Widespread concern expressed from the bench, and throughout the planning profession, signals that more attention needs to be paid to formulating clear and coherent conditions to support both compliance and enforcement. In 2014, the Environment Court in association with the Resource Management Law Association coordinated a nationwide road show on the role and importance of consent conditions, in recognition of the barrier that unclear conditions posed to achieving positive environmental outcomes under the RMA.

Opportunity Statement **Ensuring that consent conditions focus on measurable outcomes would improve their clarity and effectiveness, and therefore likelihood of implementation.**

Legislation	Permit process	Administering agencies	Examples of requirements
Resource Management Act 1991	Resource consent Designation	Environmental Protection Authority Regional and district councils and unitary authorities	Consents are commonly required for stream modification, habitat removal and for construction in significant areas (e.g. outstanding natural landscapes).
Conservation Act 1987	Concessions Authorisations	Department of Conservation	Concessions are required for activities such as grazing of land managed by the Department of Conservation and activities ancillary to mining. ('Access arrangements' are required for mining conservation land under the Crown Minerals Act 1991. ⁴²)
Wildlife Act 1953	Permits Authorisations	Department of Conservation	Permits are required for activities that impact upon indigenous fauna listed in the schedules of the Act including salvage and translocation of wildlife affected by development.
Marine Mammals Protection Act 1978	Permits	Department of Conservation	Permits are required for any 'take' of marine mammals controlled by the Act. Take is defined as actions that harm, harass, injure and attract. An example would be whale- and dolphin-watching activities (controlled specifically under the Marine Mammals Protection Regulations 1992).

Table 3: Permitting processes for development

Taurepo (*Rhabdothamnus solandri*)

Taurepo is an endemic coastal shrub that grows to 2 metres tall, fruits throughout the year and has yellow or orange flowers.⁴³ It is the only representative of its genus and family (Gesneriaceae) in New Zealand.⁴⁴ Taurepo was the subject of an ecological investigation into the impact of the cumulative effects of a reduction in bird abundance. A study compared three offshore sanctuaries with mainland sites for rates of pollination, seed production and plant density of taurepo. The key difference between the sites was the numbers of bird pollinators such as bellbird and tūi present. The study demonstrated that the decline of birds in a given area has real and measurable consequences for crucial ecological processes like pollination.⁴⁵ At present taurepo is not threatened, but evidence of poor seed set on the mainland indicates that a reduction in abundance may well become apparent over time. Most consenting processes do not consider the flow-on effects of developments on ecosystems and the influence of critical ecological thresholds.



Taurepo (*Rhabdothamnus solandri*). Bryce McQuillan

Managing uncertainty

Biodiversity is complex and there is inherent uncertainty about the impact that any activity may have on it. The public interest in the protection of nature commonly bears the weight of ambiguity, uncertainty and a lack of strong baseline information. The information deficit for many of our species and ecosystems is a constraint on the effectiveness of conservation and the quality of decision-making where biodiversity could be at risk. This is exacerbated by poor access to much of the information that does exist.

The Terrestrial and Freshwater Biodiversity Information Service Fund was designed to improve this situation when the New Zealand Biodiversity Strategy was released in 2000. The overarching intent was to *support seamless access to essential biodiversity data, information and knowledge to achieve the goals of the New Zealand Biodiversity Strategy*. Later on, a strategy was developed that more precisely identified the need for the community and government to easily access data from all sources.

More recently, a lack of accessible biodiversity information was acknowledged at a strategic level by the National

Science Bioheritage Challenge which was launched in 2013. A key theme of the challenge is 'real time biological heritage assessment'. The goal of this theme is *that biodiversity and biosecurity decision makers can have accurate, comprehensive information at their fingertips*. With a combination of new tools and 'big data' approaches, the programme aims to overcome the currently fragmented understanding of the identity and location of priority native and introduced species. Present projects include *next generation sequencing of environmental DNA to more accurately describe biodiversity in a given area in a cost effective way*.⁴⁶

Opportunity Statement **Uncertainties in biodiversity management would be reduced if decision-makers and stakeholders could more easily access accurate and relevant biodiversity information to inform the management of development.**

Managing the inherent uncertainty of impacts on biodiversity and ecosystem services – where decline of a proposal is not considered warranted – sometimes relies on the use of ‘adaptive management’ (Box 2). Adaptive management is well recognised as an appropriate consenting paradigm for the management of inherently uncertain activities, but risks to the environment and the wider public need to be minimised through effective management and robust feedback mechanisms. The naturally iterative process of adaptive management provides space to test different management tools in practice within a controlled environment, giving opportunities to learn, innovate and garner insight.

Adaptive management requires active governance strategies whereas predominant models of consenting are largely passive, with follow-up monitoring typically lax or absent. Establishing what the ongoing management and monitoring frameworks are at the outset of a project may induce more active models of governance. Novel governance and consenting strategies (e.g. a specific consent type for projects applying an adaptive management approach) may need to be investigated and used, where appropriate, to ensure enduring regulatory oversight and transparent management of compliance and enforcement.

Much jurisprudence discusses the possible thresholds and criteria for what adaptive management is and is not. The Exclusive Economic Zone and Continental Shelf (Environmental Effects) Act 2012 (EEZ Act) includes explicit reference to adaptive management⁴⁷ on account of the inherent uncertainty of activities likely to be applied for (e.g. deep-sea mining) and the context in which they would be carried out (i.e. the poorly understood marine environment). An adaptive management approach has been applied under the RMA despite no explicit provision for it in the Act. But no

common principles have been prescribed, leading to a variety of interpretations in policy and practice that have varying capacity to manage risk to the environment.

Compensating for ecological loss

Where consent is granted permitting adverse environmental effects, it is not uncommon for some form of mitigation, offset or compensation to be required as a condition. Such conditions often aspire to meet a goal of ‘no net loss’ or even ‘net gain’ of biodiversity. The risks and benefits to biodiversity of such measures have been examined in extraordinary depth in the scientific literature, including contributions from New Zealand. Most risk exists where the biodiversity affected is vulnerable and irreplaceable, and where the risks of poor or non-implementation are insufficiently managed. The benefits relate to the ability of resource users and agencies to implement a more flexible regime for development and potentially garner gains for conservation in the process.

The advent of mitigation, compensation and biodiversity offsetting as explicit trade-offs in planning has spread rapidly throughout the world, but implementation remains largely ad hoc in New Zealand. The policy vacuum is slowly filling at a subnational level. While there is no national policy on these increasingly common trade-offs, some second-generation regional policy statements include principle-based decision-support frameworks. The proposed National Policy Statement on Indigenous Biodiversity 2011 also included guidance, but was never made operative. A whole of government guidance document was released in 2014 entitled *Guidance on Good Practice Biodiversity Offsetting in New Zealand*. This provides insight into key issues but stops short of providing policy direction.⁴⁸ Frequent discussion of the concept in case law has also been instructive, although somewhat inconsistent.

Box 2 Adaptive management

Adaptive management is an approach that attempts to address uncertainty so that risk to the environment is fairly and equitably managed. In a broad sense, adaptive management consists of the iterative management of a resource or a process. The approach has much utility in environmental management generally. A more acute setting for its application is where the impacts of a proposal on biodiversity are unclear and the project is allowed to proceed on the condition that an adaptive management regime is put in place. Consenting under an adaptive management scenario may involve one or both of the following tools being employed: staging of consents, including triggers as conditions which require review in certain situations, and long-term monitoring frameworks.

Adaptive management was defined in the Environment Court in 2013 as comprising five features: the development is staged; the existing environment is established via robust baseline monitoring; there are clear and strong monitoring, reporting and evaluation mechanisms so that steps can be taken before significant adverse effects eventuate; mechanisms are supported by enforceable resource consent conditions that require certain criteria to be met before the next stage can proceed; and the development project is at least partly reversible.⁴⁹

Much debate over what an adaptive management regime really requires to fairly manage the risk to biodiversity while not ‘unfairly’ curtailing development opportunities has arisen under the EEZ Act. For example, the failure of applications to gain approval for sand mining (Trans-Tasman Resources) and seabed mining (Chatham Rock Phosphate) led to significant controversy about whether the regulatory test for certainty was so high as to be unreasonable.

Several decisions of the Environment Court have considered the relationship between the terms used in our legislation and the definitions inherent within the internationally derived mitigation hierarchy (i.e. mitigation, offsets and compensation). The government guidance draws on these decisions to clarify the distinction between these three categories. In summary, the broad distinction is that 'mitigation' is any work that directly addresses environmental damage at the point of impact; offsets are positive actions undertaken away from the site of impact that are explicitly calculated and seek to at least achieve 'no net loss' of biodiversity values; and compensation is any positive offering from a development proponent that does not meet the other two definitions.⁵⁰ It is worth noting, however, that in practice many of these activities are combined and presented as 'packages' so may not be clearly distinguished. The policy vacuum at a national level leaves significant space for debate and for the application of differing interpretations, and this is likely to persist until national policy clarifies the hierarchy and its locally relevant meanings in full.

The ambiguous and inconsistent context for managing an already risky policy tool could affect biodiversity in a number of ways, including:

- Weak implementation of the mitigation hierarchy (described more fully later), including a failure to emphasise the avoidance step, leading to the loss of vulnerable and irreplaceable biodiversity
- The 'exchange' of biodiversity values which are significantly different in type or amount

- A failure to observe a requirement for additionality such that compensation delivers gains that would have occurred anyway and do not present a net benefit, such as where offsets monies are subsumed into business-as-usual funding for conservation agencies

All these problems have occurred in the context of consenting under the RMA. They are also relevant when conservation land is swapped under the provisions of the Conservation Act 1987 (see Box 3). Land swaps have some similar characteristics to offsets and compensation as they involve the 'exchange' of two sets of values. This means that some of the assessment criteria and methods commonly used in best practice offsetting are likely applicable to land swaps. A land swap pertaining to the Ruahine Ranges was recently approved by the Department of Conservation to enable the Ruataniwha Dam project to proceed and is the subject of judicial review (proceedings lodged mid-October 2015). The Court's determination should provide guidance on how land exchanges should be undertaken in the future.

Opportunity Statement **Provision of national policy on impact management would help improve consistency, strengthen consideration of exchanges, increase the number of projects that fully address their impacts, and improve environmental outcomes.**

Box 3 Land swaps

A land swap may consist of a simple boundary adjustment or a large area of land changing hands. Requirements differ depending on whether the land is gazetted as a specific type of reserve or as stewardship land. Land swaps can occur for reserves managed by the Department of Conservation or a local authority and must only occur where the reserves being swapped are of the same classification. In the case of reserves managed by the Department, there is no requirement to publicly advertise the proposed exchange. For reserves owned by local authorities, the Department carries out the exchange on behalf of the Crown, and all transactions must first be notified. If the land to be exchanged is held for a specially protected purpose, such as a conservation park or ecological area, then it must first have that status revoked by a separate process, which involves public consultation.

The requirements applying to the exchange of stewardship land are not as rigorous. Stewardship land comprises the one-third of conservation land held by the Department that is not formally classified as a specific category of reserve. Such land may well have high biodiversity values, as it includes areas purchased by the Nature Heritage Fund for conservation purposes. Some iconic areas are all or partly stewardship land such as Lake Waikare in the Waikato Region.

Under the Conservation Act 1987, stewardship land may be exchanged for any other type of conservation land; and broad discretion is provided for under section 16A of the Act. There are few requirements guiding such exchanges. One important test is: *The Minister shall not authorise any such exchange unless the Minister is satisfied, after consultation with the local Conservation Board, that the exchange will enhance the conservation values of land managed by the Department and promote the purposes of this Act.*

Similar concerns about biodiversity offset equivalency abound with land swaps. Once land has been 'swapped out' of the public conservation estate, it is no longer subject to the Conservation Act 1987. The Parliamentary Commissioner for the Environment assessed land exchanges in 2013 and determined that the weak protection status of the land, and the opaque exchange process, entailed risks to biodiversity. She recommended (among other things) the *development of clear principles and processes for assessing net conservation benefit*.⁵¹

Securing regulatory outcomes

A key concern, where development consent is conditional on compensatory actions being undertaken, is that those requirements are unclear, are not enforced, or do not succeed. New Zealand has a poor history, as has the rest of the world, in ensuring that mitigation, offset and compensation requirements are implemented correctly and that the effect on biodiversity is appropriately addressed. The most detailed analysis of compliance with such conditions was undertaken during the summer of 2010/2011. This found that just two-thirds of 245 requirements (across 81 consents) nationwide were undertaken and that less than half of the requirements with material ecological benefits eventuated.⁵² While most experience with such requirements is under the RMA, the Conservation Act 1987 also enables the Department of Conservation to require mitigation or compensation as a condition of concession agreements (although offsets are not specifically mentioned).⁵³ Analysis to be published shortly revealed that, from a study of 20 projects under that Act, similar levels of compliance were observed.⁵⁴

It is clear from the outcomes being observed that greater support for compliance and monitoring functions is necessary, particularly where requirements are long term and require sustained regulatory oversight. High turnover, poor institutional memory and limited resourcing for compliance and monitoring functions in local government (despite their potential for cost-recovery) can exacerbate the risks to biodiversity of long-term projects. The increasing use of biodiversity offsets and compensation means that long-term requirements are often large in scale and highly technical. Even self-monitoring requirements placed on developers need to be vetted by the consenting agency. Several developers interviewed for this project noted that those (often costly) reports frequently disappeared into a 'black hole' of an underfunded agency with insufficient time or expertise to consider them. Sustained regulatory oversight is an important component of improving compliance rates.⁵⁵ Support for this may include the development of additional administration capacity, introduction of computer-based monitoring systems, and engagement of sufficient compliance staff to adequately monitor all relevant consents. While poor compliance is the result of multiple factors, poor administration of compliance mechanisms is one of them.

A range of tools can help improve the likelihood of compliance. These include covenants and bonds under section 108 of the RMA. Bonds are used as a form of insurance to secure consenting outcomes, usually medium- to long-term requirements. The relevant legal provision under the RMA is:

A bond required under section 108(2)(b) may be given for the performance of any 1 or more conditions the consent authority considers appropriate and may continue after the expiry of the resource consent to secure the ongoing performance of conditions relating to long-term effects, including—

- (a) a condition relating to the alteration or removal of structures:*
- (b) a condition relating to remedial, restoration, or maintenance work:*
- (c) a condition providing for ongoing monitoring of long-term effects.⁵⁶*

Conditions are mainly bonded where it will take some time to demonstrate compliance (e.g. maintenance of new planting over a set period or until a goal is reached such as canopy closure) or where there is significant risk to the community. Bonds may be relatively small (e.g. several thousand dollars to secure a small-scale planting programme) or comprise many millions of dollars where activities have a high likelihood of generating adverse effects from non-compliance. As an example, the Waikato Regional Council holds a bond of \$40 million to secure compliance with certain conditions applying to the Newmont Waihi gold mine and a bond of \$1.2 million is held in respect of the Whangamatā Marina conditions.

Opportunity Statement Strengthening the implementation of RMA bonds by ensuring they are used when appropriate and that they are properly costed and administered would secure better resource management outcomes for biodiversity.

In New Zealand, permit holders are generally responsible for giving effect to their mitigation, offset and compensation requirements. Sometimes other entities are established that are independent and charged with distributing agreed sums of money over a long period of time (usually called 'mitigation trusts'). In all other instances, the resource users must undertake the works themselves, which has been material to the reasons for poor compliance: they simply may not have the desire or the expertise to do so.⁵⁷

Mitigation trusts are independent entities that carry out conservation work or distribute contestable funding to address adverse effects. They are usually established in response to a large-scale project with diffuse effects over a wide area (e.g. the Waikato Catchment Ecological Enhancement Trust) or by an agency to manage effects of a number of projects (e.g. the Taranaki Tree Trust). When the consent or other permission is granted, agreement is normally reached on the high-level objectives of the trust (which are not necessarily linked to conservation priorities) and its annual disbursements. Assessments of compliance of the trust are often based on ensuring that the correct quantum has been distributed, rather than on analysing any resulting improvements in biodiversity. The costs of setting up and running a mitigation trust are often significant, including legal fees, reimbursement of trustees, distribution of monies, liaison with stakeholders and monitoring

outcomes. As a result of such factors, mitigation trusts are not as effective in achieving biodiversity gains as they might be because money is spent on administration instead of conservation. Given the declining state of biodiversity, enhanced implementation of mitigation trusts is an opportunity for innovation.

Opportunity Statement Improving the ecological outcomes achieved by mitigation trusts by carefully directing funding and monitoring the difference made would maximise the positive impacts of their funding.

Unlike other jurisdictions, there are limited mechanisms available in New Zealand to transfer liability to third parties to undertake mitigation, offset and compensation delivery on a consent holder's behalf. New Zealand has little experience of third party arrangements, although research in the past has demonstrated their potential utility.⁵⁸ In practice, cash transfers to conservation organisations in lieu of conservation work being undertaken by resource users have happened for some time. For example, the Department of Conservation has recently begun accepting cash in lieu of conservation

work ('conservation credits') by some concession holders, and the money is used at the discretion of the relevant unit of the Department. More informally, community groups and other organisations are often supported by resource users as a condition of their consents. Third party delivery of offsets and compensation generally functions as a market or quasi-market internationally, where gains and losses are traded to meet regulatory requirements. Third party structures enable the aggregation of small project contributions, reducing transactions costs and the costs of delivery and potentially enhancing ecological outcomes.

Opportunity Statement Providing a flexible process for resource users to contribute to third party delivery of offsets and compensation in lieu of delivering their own requirements would enable the gains to be delivered by actors with relevant skills, and would increase the likelihood of compliance while not necessarily alienating developers from the outcomes of their contributions.

Werewere-kōkako/sky blue mushroom (*Entoloma hochstetteri*)

This distinctively coloured mushroom is the reproductive stage of a decomposer fungus, the feeding stage of which lives on dead plant matter in the soil. This mushroom is also easily recognised because it is depicted on the \$50 banknote, the only mushroom to feature on New Zealand's currency. The banknote design is based on the Tūhoe story linking the mushroom and the North Island kōkako: the bird is said to rub its head against this mushroom to get its blue wattle. It is only seen in mature native forest and may take many decades to appear in a developing forest. Loss of older forest in development projects can include the loss of many species only found or specially adapted to mature ecosystems.



Werewere-kōkako/sky blue mushroom (*Entoloma hochstetteri*). Peter Buchanan

Unregulated impacts

Regulation in New Zealand does not address all biodiversity: much of it is excluded for various reasons and certain activities are themselves exempt from regulatory requirements. The cumulative impacts of excluding even minor effects from consideration can be significant. For example, the protection of wetland ecosystems is considered a matter of national importance under the RMA. However, recent analysis by Myers et al. (2013) demonstrated that in 60 per cent of regional plans, effects were only controlled on wetlands deemed to be 'significant', with little protection provided for small or ephemeral wetlands. Given the acutely important biodiversity and ecosystem services values of all wetlands, and the fact that so many of them have been lost over time to land drainage, the analysis went on to conclude that the protections in regional and district planning instruments were generally inadequate for protecting these critical environments.⁵⁹

Many activities are also excluded from regulatory requirements because they are considered to have only minor impacts that can be controlled via standards (permitted activities) or because they have historic privileges (existing use rights). Permitted activities are those that do not require consent, but they may still have restrictions applying to them such as operating standards.

The right to continue carrying out activities that previously did not require permission - but now do - is widely held for purposes such as water abstraction and mining. The RMA also explicitly provides for certain existing uses that were lawfully established to continue without the need for additional permission.⁶⁰ No consent is required for these existing use rights, although confirmation from the council is sometimes sought by proponents for peace of mind. Analysis has demonstrated that councils have been reluctant to challenge purported existing use rights even where legal grounds to do so are available.⁶¹ Permitted activities and existing uses contribute to cumulative effects and are generally poorly accounted for.

Managing cumulative impacts

Biodiversity is at serious risk from the cumulative impacts of development - or 'death by a thousand cuts'. Small incremental changes to the environment often compound and can have enormous aggregated effect. Cumulative effects are referred to by several statutes, but none contain a clear stand-alone definition. The inclusion of 'cumulative effect' within the more general definition of 'effect' in the RMA is the most comprehensive: *Any cumulative effect which arises over time or in combination with other effects—regardless of the scale, intensity, duration, or frequency of the effect.*⁶² Cumulative impacts on biodiversity are best addressed at a strategic level rather than individual project level. Without adequate strategic-level control and

oversight, cumulative effects can result in the degradation of significant natural values.

Cumulative effects, disturbance and other less visible impacts are only weakly addressed in law and practice, compared with many acute impacts, and there is much potential to improve in this area. Some level of disturbance is natural, but human-induced disturbance is often at a scale, or over a length of time, that may be unsustainable for some species and populations. There are some examples of policy mechanisms that seek to directly address disturbance on a precautionary basis, such as seismic surveying (see Case Study 3).

The RMA explicitly provides for local and regional authorities to manage cumulative effects, and consent can be refused if a proposed activity is likely to be the 'straw that will break the camel's back'.⁶³ Recent analysis demonstrates that cumulative effects are often poorly managed owing to a range of barriers such as a lack of adequate strategic planning at a national or regional level and the unwillingness of local government to absorb the costs of initiatives required to better manage such effects.⁶⁴ The result of these factors is that agencies attempt to address cumulative effects on a consent-by-consent basis which, given their sometimes non-linear and unpredictable nature, is rarely effective.

Opportunity Statement Tools to support strategic planning for biodiversity would contribute to improved outcomes from policy and better recognition of cumulative effects.

Economic instruments

Economic instruments are those which use monetary approaches to address sources of harm or remediation thereof. Economic instruments are popular for their ability to yield good environmental outcomes at less cost than traditional regulation and their use has been described as 'booming' internationally.⁶⁵ Market-based instruments are a discrete subset of economic instruments. The inordinate complexity of the elements and interactions that make up biodiversity (with different values not being interchangeable) means that there is a high risk of losses being obscured in market-based systems. As such, any economic instrument that addresses biodiversity must be very carefully designed.

However, economic instruments are virtually absent in New Zealand⁶⁶; no true markets exist that are fully focused upon biodiversity conservation or the protection of biodiversity from development impacts. No government guidance on the use of economic mechanisms has been provided to agencies managing biodiversity in development.

Biodiversity co-benefits do arise out of some economic instruments deployed for other purposes in New Zealand, such as the fisheries Quota Management System, the

Case Study 3 Seismic testing and whales

Disturbance can modify the behaviour of animals in ways that can be imperceptible without detailed study by experts. An example of this is the effects of seismic surveying on marine mammals. Whales and dolphins commonly use sonar to detect prey and other members of their own species. Seismic surveying typically entails an underwater air gun being shot at regular intervals from a slow-moving ship in order to detect oil and gas deposits. Potential impacts on individual animals include hearing loss, habitat abandonment, disruption of lifecycles and death. Population and ecosystem level impacts may also occur.⁶⁷

In New Zealand, prospecting and exploration permits issued by New Zealand Petroleum and Minerals under the Crown Minerals Act 1991 often include a requirement to undertake seismic surveying. Such activities also come under the jurisdiction of the EEZ Act, where seismic surveying is a permitted activity (i.e. does not require marine consent), but those who carry it out must comply with the Department of Conservation's 2013 Code of Conduct for Minimising Acoustic Disturbance to Marine Mammals from Seismic Survey Operations. The Code includes requirements to assess the potential effects of the activity and to develop a mitigation plan in advance of the survey activities.

While the majority of industry players were supportive of the introduction of the Code in 2013, some reacted negatively on the grounds that they considered there to be insufficient proof that sonar guns were harmful to whales.⁶⁸ However, if biodiversity is to be adequately safeguarded from poorly understood impacts, introduction of controls on a precautionary basis is precisely what needs to happen. This is particularly important where the taxa that are potentially affected have low resilience, such as the critically endangered Maui's dolphin. The introduction of this Code on a precautionary basis illustrates how pre-emptive policy can be used to safeguard biodiversity when the impacts of an activity on them are uncertain.



Humpback Whale (*Megaptera novaeangliae*). Manu Bird

Emissions Trading Scheme and the use of a nutrient cap and trade system in Lake Taupo to control diffuse pollution from farming. Some subdivision policies could also be classified as economic instruments, where in exchange for the protection of biodiversity, additional subdivision rights can be granted (e.g. former Rodney District Council). However, in this case, analysis demonstrates that poor management has resulted in the loss of many of the values the land was set aside to protect.⁶⁹

The limited uptake of economic instruments to protect biodiversity is in spite of specific direction under the RMA for the Minister for the Environment to consider the use of economic instruments to achieve the purpose of the Act.⁷⁰ The lack of policy innovation has been in part blamed on the devolution of environmental responsibilities to local government, where a lack of resources and central government leadership slows progress.⁷¹ However, the adoption of economic instruments in a development context

is possible – rates relief and postponement is widely adopted (see Case Study 4) and demonstrates that communities do respond to even minor financial incentives.

Economic instruments are attractive for business as they allow greater flexibility in meeting requirements than rigid regulatory prescription, and they may help encourage the more efficient use of resources. A large number of potential applications for economic instruments have been canvassed by a range of authors over the years,⁷² but few have engaged specifically with biodiversity. Improvements in the quantification of biodiversity in recent years should make the application of economic instruments to its protection more achievable. The value of biodiversity is increasingly expressed in terms of the ecosystem services that nature provides (see Box 4 overleaf).

Ecosystem services can provide a useful framing for decision-making that affects the environment. Some of

those services are easier to understand, and can be more easily and transparently costed, than others. Internationally, the merits of ecosystem services approaches to drive business decision-making and inform impact assessment are increasing, but application in New Zealand is currently limited. The use of economic instruments is likely to increase in freshwater management, as the requirement to put in place nutrient limits on rivers kicks in via the National Policy Statement for Freshwater Management 2014 (which includes a National Objectives Framework for freshwater quality). While these instruments do not currently address biodiversity in any substantive way, future iterations are likely to do so, and economic instruments may become important tools to support freshwater biodiversity. At present, a lack of support and guidance for agencies to implement economic instruments where they could be effective is likely constraining uptake.

Case Study 4 Far North District Council rates postponement policy

The Far North District Council administers two different policies for rates relief – a rates remission and a rates postponement approach. Rates remission is available for landowners who enter into agreements to protect land in perpetuity. For each year that the site is protected by covenant, no rates are payable. In contrast, rates postponement applies to limited-life covenants that exceed 10 years in duration. For the first 10 years, rates are postponed, and they are remitted year by year after a decade. If the covenant is amended significantly, or uplifted, the postponed rates must be repaid.

The Council administered 63 agreements with landowners as of December 2014, of which seven are in perpetuity and 56 are under temporary protection. The total area protected is approximately 1,838 hectares, with 24 hectares in perpetuity and 1,814 hectares in temporary protection. The average size per covenant is 29 hectares, with the largest being 611 hectares and the smallest 400 square metres. Approximately five new covenants are processed each year. Monitoring is carried out on the fifth and ninth year of each 10-year cycle and the Council advises that there have been no compliance issues. The costs of visits to the site by a professional ecologist for advice and monitoring purposes are absorbed by the Council, but landowners are liable for some costs such as surveying and conveyancing.



Endemic plants such as the coastal tree daisy (*Olearia solandri*) are commonly protected through covenants on private land.
Angela Simpson

Box 4

What are ecosystem services?

Ecosystem services fall into four broad categories, but are highly interdependent. They are described in many different ways; the description from the Millennium Ecosystem Assessment in 2005 is adopted here due to broad international uptake.⁷³

Provisioning	Food production Fibre production
Supporting	Soil formation Photosynthesis
Regulating	Flood mitigation Water purification
Cultural	Recreational values Spiritual benefits

Table 4: Examples of ecosystem services⁷⁴

Ecosystem services, as a way of framing environmental management, have become increasingly important in public policy.⁷⁵ Landcare Research recently released the first national assessment of ecosystem services in New Zealand and work continues on how such market-based tools can be marshalled to best effect here.⁷⁶ Launched in 2014, the review highlighted where ecosystem services are being degraded and where others are recovering. The authors state:

On the positive side, we are seeing improvements in pest management in cities and native forest, growth in indigenous forest and shrubland, and health and environmental benefits from less use of open fires and better home insulation. On the negative side, some examples are the decline in pollination services. We are losing biodiversity, particularly rare species in freshwater and wetland environments, and also water is being over-allocated in some catchments.⁷⁷

The approach yields useful information, such as demonstrating that the land-based ecosystem services in New Zealand contributed \$57 billion to human well-being in 2012.⁷⁸ There has been an upsurge in recent years in the use of ecosystem services assessments to inform decision-making.⁷⁹

Opportunity Statement Provision of guidance and support for the use of economic instruments and ecosystem services accounting in resource management and business planning could improve the management of effects on biodiversity, particularly cumulative effects.

Procedural instruments

A number of procedural instruments are highly relevant to the protection of biodiversity in a development context in New Zealand. While many are not statutory processes themselves, they explicitly link to regulatory controls, and are a common source of contention in litigation proceedings. Examples discussed below are assessments of environmental effects (AEEs) (New Zealand's version of environmental impact assessment) and observing the mitigation hierarchy (see below).

Assessments of environmental effects

An AEE is required in some form for most regulated activities, although there are exceptions (e.g. fishing). The AEE is a pivotal document for consenting activities as it sets out the nature and magnitude of effects on the environment, thereby determining factors in decision-making. A failure to undertake an appropriately detailed assessment of environmental effects may mean that impacts on biodiversity proceed without appropriate adherence to the mitigation hierarchy, and as a result the risks of development activities impacting on the environment are inadequately managed.

Globally, the genesis of environmental impact assessment occurred only in the 1960s. It was brought about by the recognition of natural limits, and growing social expectations that resource users should exercise due care. In New Zealand, environmental impact assessment was first required by the Environmental Protection and Enhancement Procedures introduced via Cabinet minute in 1974. These procedures were overseen by a Commission for the Environment until the government reforms of 1987. The procedures were frequently revised but never mandated, nor fully integrated into the planning process.⁸⁰

There is currently no detailed guidance for environmental impact assessment in New Zealand, and most interviewees voiced concern at the variation in depth, quality and veracity of AEEs with respect to biodiversity. Broad guidance on what an AEE should contain is present in relevant legislation, including: under the RMA (schedule 4(6)); the Conservation Act 1987 (section 17S(1) and RMA schedule 4(6) if the Minister requests); and the EEZ Act (section 39 and the information principles in section 61).

A somewhat outdated guidance document is available from the Ministry for the Environment,⁸¹ but it is not sufficiently detailed to be very helpful in assessing biodiversity impacts. A more recent guidance document from the Environmental Protection Authority is also available under the EEZ Act, but does not address any specific matters relating to vulnerable taxa and ecosystems.⁸² Standards for AEEs are provided by some councils and in 2015 the Environmental Institute of Australia and New Zealand released voluntary guidance on ecological impact assessment for New Zealand. Methods to monitor biodiversity in ways that can contribute to

Katipō spider (*Latrodectus katipo*)

The endemic Katipō spider is one of the most distinctive invertebrates in New Zealand. Katipō are widow spiders and have a pronounced red streak across their back. The females commonly establish lairs beneath driftwood, or at the base of plants on or near sandy beaches, and may use webs to trap their main prey (beetles). The decline of the katipō has been particularly rapid over the past 30 years.⁸³ The spider prefers to live in coastal dunes, sensitive ecosystems which have been dramatically modified by human pressures such as agricultural conversion, burning, coastal subdivision and general disturbance.⁸⁴ As a result of these impacts, and most particularly habitat loss and fragmentation, katipō are classified as being in 'serious decline'.



impact assessments are also provided by the Department of Conservation's Biodiversity Monitoring Toolbox, which includes some best practice manuals.⁸⁵

However, many of these documents do not provide sufficient scope, depth and detail for a resource management context where vulnerable biodiversity is at risk. Their use is also voluntary and there are gaps in guidance for some species and ecosystems. Where detail is included, the tools and methods may be too costly and time-intensive for the scale of many activities.

Opportunity Statement Providing comprehensive guidance on environmental impact assessment for activities that may harm vulnerable taxa and ecosystems would help to protect biodiversity by improving practice and lowering consenting risk for developers.

Mitigation hierarchy

A major international innovation in recent decades is the 'mitigation hierarchy', one of the most important procedural instruments in protecting biodiversity in development. The mitigation hierarchy is an international framework for considering the interplay of development and the environment. If effects are

not considered to be avoided, remedied or directly mitigated sufficiently, positive conservation actions may be mandated through conditions placed on permits or voluntarily adopted in jurisdictions where environmental legal process is sparse.

The mitigation hierarchy is specific and critical. The procedure, when strictly observed, requires that in the first instance harm should be avoided. This can be achieved by not carrying out activities that will damage natural values or by relocating them to less sensitive areas. The remediation of harm followed by directly mitigating the impacts of development are the second and third preferences, with offsetting and compensation of harm to be contemplated only if the prior three steps have been applied. This procedural factor is critical to ensuring that development activities entail the least impact upon nature.

While the mitigation hierarchy is not explicitly enshrined in New Zealand legislation in the form set out in Figure 2, practice tends to reflect an approximation of it.⁸⁶ The RMA and the Conservation Act 1987 include the requirement to 'avoid, remedy or mitigate' adverse effects. Second-generation regional policy statements also generally prescribe the hierarchy, set it out in full, and mandate that the final stage of 'offset' or 'compensation' is a last resort. It would seem that in practice, and increasingly in policy, the staircase nature of the mitigation hierarchy applies in New Zealand as elsewhere.

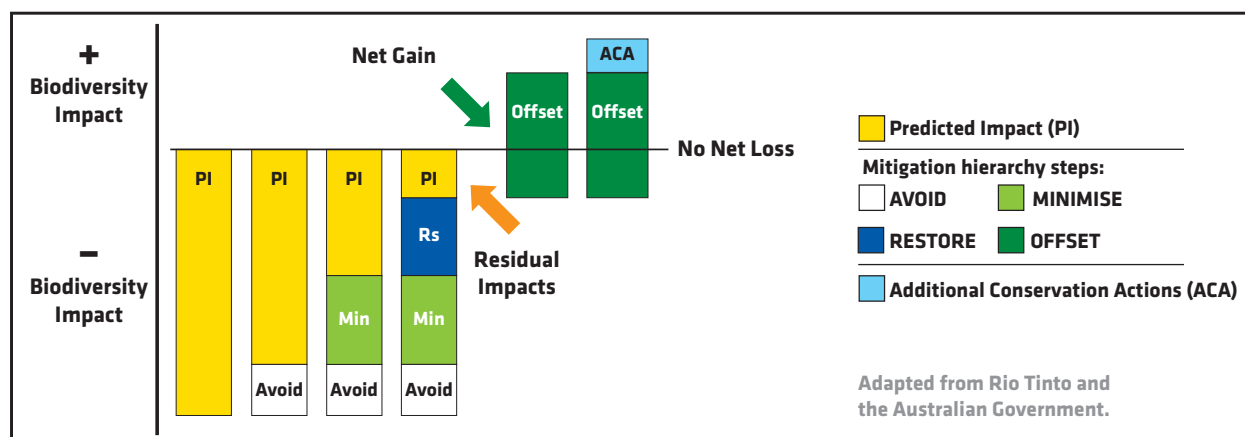


Figure 2: Mitigation hierarchy to guide impact management

Cooperative instruments

Cooperative instruments are commonly used in New Zealand, and their usage is increasing internationally,⁸⁷ along with collaborative mechanisms more generally. Cooperative instruments draw together different stakeholders and use a range of tools in an endeavour to fix policy problems without the hierarchical approach of traditional regulation.⁸⁸ Voluntary cooperative instruments are often relied upon to drive changes in industry and/or delay regulatory approaches. Eschewing regulatory approaches in favour of softer instruments which are based on voluntarism has had variable results throughout the world, depending on the context.

Cooperative instruments as global linking documents

Cooperative instruments, in the form of international treaties, are highly relevant to the protection of biodiversity in development. New Zealand is a signatory to a number of international agreements including the Convention on Wetlands of International Importance (also known as the Ramsar Convention), the Convention on Migratory Species and the Convention on Biological Diversity. The aforementioned New Zealand Biodiversity Strategy was released in 2000 in response to our ratification of the Convention on Biological Diversity. The Strategy is a cooperative instrument that contains broad national direction on biodiversity. It does not contain policy guidance per se, but includes four key goals. Goal 3 is:

Halt the decline in New Zealand's indigenous biodiversity

*Maintain and restore a full range of remaining natural habitats and ecosystems to a healthy functioning state, enhance critically scarce habitats, and sustain the more modified ecosystems in production and urban environments; and do what else is necessary to; Maintain and restore viable populations of all indigenous species and subspecies across their natural range and maintain their genetic diversity.*⁸⁹

The other three goals relate to community and individual action, the Treaty of Waitangi and genetic resources respectively. The Strategy identified a number of changes that it sought to achieve. Those relevant to the biodiversity and development nexus included shifting from:

- *the current decline in indigenous biodiversity to a level of stabilisation;*
- *ad hoc arrangements to clearly aligned and coordinated actions;*
- *little market recognition of biodiversity to market-driven rewards and sanctions.*⁹⁰

The Strategy is non-statutory and there are no legal requirements for agencies to implement it – or consequences for failing to do so. Initially, significant resources were made available to support the implementation of the Strategy, but funding directly linked to it has now ceased.

Leatherjacket (*Parika scaber*)

The indigenous leatherjacket, also known as the triggerfish and the creamfish, is a masterful ecosystem engineer of reef ecosystems. The leatherjacket nibbles the reef, causing disturbance and creating habitat for encrusting organisms. It is omnivorous and plays a key ecological role through preying on kina. In the absence of sufficient predation, kina strip the kelp off rocky reef areas creating monospecific 'kina barrens'. Leatherjackets have few young because few species prey on them – the rough skin of both juveniles and adults, and the fearsome spines on the latter, don't make for a tasty meal.⁹¹ The leatherjacket is harvested usually as bycatch in bottom trawling off the South Island, and is managed within the Quota Management System. Such management has failed to address the ecological role of the leatherjacket in rocky reef systems.



Leatherjacket (*Parika scaber*). Wade Dook

An assessment after five years of implementation made a comprehensive suite of recommendations to improve the outcomes for biodiversity and the efficacy of the Strategy itself.⁹² The review noted that while the Strategy provided an *adequate platform for tackling the decline of indigenous biodiversity*, additions were required to address the fact that it was *strong on vision, but weak on identifying quantifiable targets*. There has been a recent attempt at ‘refreshing’ the Strategy but the current status of this initiative is unclear. Publicly stated goals that have emerged from the Department of Conservation since (the ‘2025 stretch goals’) bear limited resemblance to those in the Strategy.⁹³

With only four years to run before the four key goals are meant to be achieved, it is clear that the Strategy has not been able to generate the level of change required. This is likely due to a lack of leadership, requisite funding and other matters to which Green and Clarkson (2006) refer. What has resulted is a poor strategic context for biodiversity protection, which in turn has likely weakened the policy settings for safeguarding nature, and particularly the management of cumulative effects.

In the absence of sufficient national leadership to catalyse effective cooperative instruments, many have been promulgated at a subnational level (see Case Study 5). Cooperative instruments have also been developed by industry sectors. An example of a long-running voluntary agreement is the Dairying and Clean Streams Accord and localised initiatives such as the Waiokura Best Practice Dairying Study. Cooperative instruments are particularly popular in addressing biodiversity issues, given that biodiversity values generally transcend jurisdictional boundaries and may involve many players. Recent research from the United Kingdom demonstrates that voluntary approaches have limited impacts when they are relied upon to drive behaviour change alone, but that they are more

effective when applied to address environmental issues than in other areas.⁹⁴

Opportunity Statement **The use of cooperative instruments with willing resource users may complement or reduce the need for regulation in certain situations, and improve environmental outcomes.**

Persuasive instruments

Persuasive instruments are commonly implemented around the country and, like cooperative arrangements, often have lower levels of control by agencies. These instruments aim to coerce or inspire the public, business, private landowners and other stakeholders to care about biodiversity. Persuasive instruments can be effective in building social capital around a particular environmental problem (or a problem related to any common pool resource). They can be targeted at any players or society at large. They can be designed to galvanise communities to address a particular issue or demand that regulatory agencies or political parties do so. They are low-cost, but have uncertain capacity to induce behaviour change. There are some pertinent examples where they have made a significant difference. For example, Fish and Game’s Dirty Dairying campaign (Case Study 6) helped prompt Fonterra to launch the Dairying and Clean Streams Accord in 2003.⁹⁵

Information-based instruments are commonly employed to increase the target audience’s understanding of an issue of importance. A large proportion of interviewees noted the lack of knowledge of many practitioners, developers and decision-makers about biodiversity. The provision of guidance and information designed to set out the reasons

Soil

Soil is a natural capital stock that can take many millions of years to form. It is the foundation of many ecosystem services and indeed entire ecosystems and the species that live within them. Soil in New Zealand also has a rich flora and fauna of its own. Despite its critical importance, soil is rarely well considered in development planning, and even less so in environmental impact assessment. Rapid and widespread conversion of land to agricultural uses, particularly in recent years, has often entailed high inputs of fertiliser and irrigation water. Adverse effects such as pugging of soil and the build-up of cadmium and other pollutants have become evident. It is increasingly apparent that increasing the profit from land through conversion has often resulted in severe soil degradation. Landcare Research is presently developing molecular techniques to rapidly assess species diversity from samples of soil as part of the Next Generation Biodiversity Assessment project. This should help inform better management of development impacts on soil biodiversity.⁹⁶



Grazed soils in parts of the Mackenzie Basin have become degraded. Raewyn Peart

why biodiversity protection is important, the relevant statutory mandates for taking action, and the range of tools that can be deployed could considerably improve the consideration of biodiversity in development. In particular, a digest of tools currently available to protect biodiversity, accompanied by information on where and in what cases they are best applied, would likely increase their usage and help ensure that the right tool is used for the right task.

Opportunity Statement Enhancing the knowledge of practitioners, decision-makers and resource users about the range of tools available to safeguard biodiversity would help increase their usage and ensure the best tool is used for each task.

Case Study 5 Gecko conservation on the Otago Peninsula

A community-owned threatened species plan is being implemented on the Otago Peninsula – a project shared by the Department of Conservation, private landowners, businesses and conservation groups. The project focuses on the moko-kākāriki/jewelled gecko (*Naultinus gemmeus*), a species that is classified as being 'At Risk-Declining'. The jewelled gecko occupies two main habitats on the Otago Peninsula - coprosma scrub and kanuka/manuka scrub. They are rarely found in broadleaf forest. All the best jewelled gecko sites are in patchy *Coprosma propinqua*/*crassifolia* scrub on farmland. Areas fenced off from livestock (including coprosma scrub) have either no geckos or much fewer geckos.

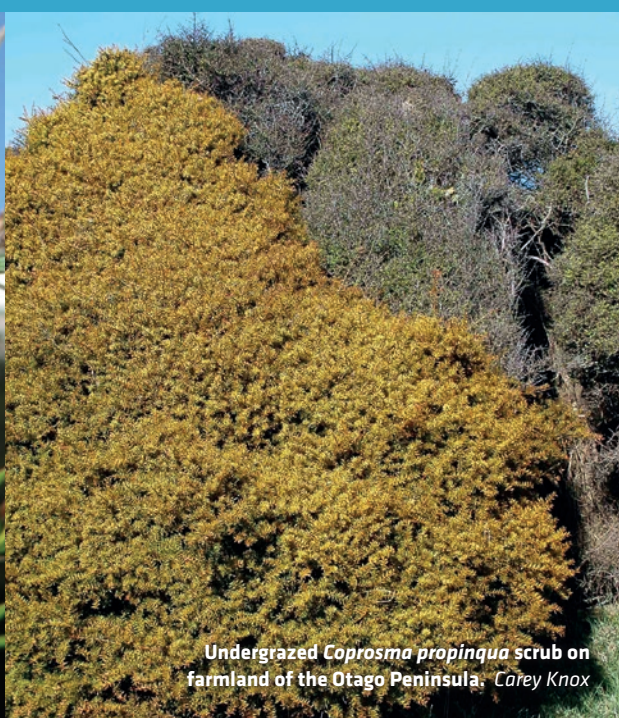
Research by Knox (2011) demonstrated that livestock grazing reduced mice and rat populations and this appeared to enable geckos to reach much higher densities on farmland (typically four to five times greater) in coprosma scrub. *Coprosma* can provide the geckos with everything they require (food, protection from the weather, some protection from larger mammal predators due to the density/complexity of the shrubs, suitable basking spots, etc.). In contrast, the gecko populations seem to struggle where rank grasses surround the shrubs, presumably because the mice and rats are abundant, and can climb up through the dense coprosmas and eat the geckos.⁹⁷

Many of the key jewelled gecko sites on the Otago Peninsula have no legal protection and although this habitat is critically important for jewelled geckos, it is not seen by most of society (or even most ecologists) as being of much value to native wildlife. The importance of coprosma scrub was often overlooked in the past; areas of good jewelled gecko habitat were cleared and the geckos lost out as a consequence. Now landowners, community groups and the community as a whole are more aware of the jewelled geckos and their habitats, so they should be better preserved.

A management plan for the gecko was developed by the Department of Conservation, but the community has taken ownership of implementation since its launch in 2011 as the Department did not consider it a priority. The plan took two to three years to formulate and included 60 voluntary actions that the public could take to improve the fate of the gecko. These include being the eyes and ears of the Department to reduce poaching of the attractive reptiles (a common threat to their survival), controlling pests and carrying out monitoring tasks.



Jewelled gecko (*Naultinus gemmeus*). Carey Knox



Undergrazed *Coprosma propinqua* scrub on farmland of the Otago Peninsula. Carey Knox

Case Study 6 Dirty Dairying campaign

In 2001, Fish and Game initiated a coordinated campaign that hit out at the environmental impacts of the New Zealand dairying industry. The primary tools used in the campaign were media messages and advertisements containing the tagline 'Dirty Dairying' targeted primarily at farmers and secondarily at the wider public. Quantitative measures of the campaign's success are difficult to determine but its influence on public policy is unquestionable. While the use of strong 'shaming' tactics made critics uncomfortable, evaluation demonstrates that the Dirty Dairying campaign was instrumental in provoking a rapid industry response.⁹⁸ The University of Waikato's Professor of AgriBusiness Jacqueline Rowarth later noted that the campaign likely harmed the relationship between urban and rural populations, but she did also concede that it had lifted public awareness of the issues.⁹⁹



The impacts of dairy farming on the environment are significant, particularly in freshwater ecosystems. Raewyn Peart

Pīngao/golden sand sedge (*Ficinia spiralis*)

Pīngao was once a very common species in sand dune environments around New Zealand's extensive coast. This sedge is rugged and hardy, generally preferring seaward sides of foredunes where many plants cannot survive, and is thus critical for the stabilisation of coastal dune systems. Pīngao has experienced dramatic declines since the arrival of Europeans and threats are numerous (including burning, disturbance from introduced animals, competition from the invasive marram grass, damage from vehicles and sand mining). Coastal development is a key modern threat which has acute impact and also generates ongoing disturbance from the domestication of the coastline. This has put severe pressure on pīngao. The Dunes Restoration Trust has developed a range of advocacy and information materials on the importance of dune conservation including protecting and expanding pīngao distributions.¹⁰⁰ This has increased public knowledge and has encouraged coastal communities to undertake significant dune restoration projects. Despite these great efforts, there is still widespread pressure on pīngao nationally. The presence of species like pīngao along the coastline actively supports the accumulation of coastal dunes. In the absence of such species, dunes are highly susceptible to erosion.



Pīngao/golden sand sedge (*Ficinia spiralis*). Rewi Elliot





There have only been two recorded sightings of *Thambotrica vates*, a threatened moth, in the last 25 years . Bryce McQuillan

Opportunities for innovation and improvement

Summary of opportunities

Throughout the analysis above, opportunities for policy innovation have been identified. Table 5 summarises these, and the solutions proposed to address them, under the following key headings:

1. Considering biodiversity in strategy, policy and planning
2. Strengthening national leadership on biodiversity in development
3. Enhancing the use of existing tools
4. Developing fresh tools to address gaps

There are many potential tools that could address the above problems and those proposed below are only some of them. They have been selected as the most promising candidates after an assessment of a longlist identified during the research process and set out in Appendix 2. In the next section the tools are evaluated against key criteria to determine their suitability in the New Zealand context.

Implementing opportunities

Our discussion now moves on to how such innovations and improvements could be implemented and in what form (from the range of possibilities). The analysis of each tool begins with setting out how it could improve outcomes and considering whether it is an effective means of addressing the problem identified. Further analysis is set out against the criteria below.

1. The innovation is able to be implemented within New Zealand's existing legal settings
2. An agency or responsible party exists that could implement the innovation

3. The innovation is unlikely to result in perverse outcomes in practice
4. The innovation is additional and does not duplicate existing mechanisms
5. The cost of the proposed tool is not prohibitive on the suggested lead organisation/s

The overall findings are then summarised with a recommendation for each tool, including the suggested organisation/s that could lead it. Where an example of the tool's use exists in New Zealand or overseas, a case study is included to demonstrate the practical application of the concept. One size does not fit all, of course, and where an alternative but more appropriate or useful application can be used, then flexible implementation is encouraged.

The basis for the assessment criteria chosen relate to the practical and tactical nature of the solutions proposed. To review: tactical solutions act to better balance the public interest in nature protection with the private interest in resource extraction, while practical tools improve the outcomes from resources already available. Neither creates substantial changes in the system, but they can assist in improving the fate of biodiversity¹⁰¹ and in increasing efficiencies for resource users. Therefore the assessment criteria do not anticipate solutions that would require substantial systemic change.

To make the cost and effort involved in implementing the solutions set out worthwhile, they need to generate outcomes additional to those which are already possible with existing tools; be an effective means of addressing the given problem; and not entail a high likelihood of generating significantly worse outcomes.

Opportunities to improve management of biodiversity in development	Proposed solutions
Planning for good outcomes	
Uncertainties in biodiversity management would be reduced if decision-makers and stakeholders could more easily access accurate and relevant biodiversity information to inform the management of development.	Improving biodiversity information
Tools to support strategic planning for biodiversity would contribute to improved outcomes from policy and better recognition of cumulative effects.	Tools for strategic planning
The use of cooperative instruments with willing resource users may complement or reduce the need for regulation in certain situations, and improve environmental outcomes.	Regional biodiversity plans
Leadership on biodiversity	
The provision of national leadership and direction would help to focus biodiversity policy, improve consistency and reduce costs of policy formulation and implementation.	National policy on indigenous biodiversity
Provision of national policy on impact management and adaptive management would help improve consistency, better manage risk to nature, increase the number of projects that fully address their impacts, and improve environmental outcomes over time.	Impact management policy
	Adaptive management policy
Refining current practice	
Enhancing the knowledge of practitioners, decision-makers and resource users about the range of tools available to safeguard biodiversity would help increase their usage and ensure the best tool is used for each task.	Improving knowledge of existing legal mechanisms to protect biodiversity
Providing comprehensive guidance on environmental impact assessment for activities that may harm vulnerable taxa and ecosystems would help to protect biodiversity by improving practice and lowering consenting risk for developers.	Guidelines for biodiversity impact assessment on vulnerable taxa and ecosystems
Ensuring that consent conditions focus on measurable outcomes would improve their clarity and effectiveness, and therefore likelihood of implementation.	Outcomes-based conditions
Strengthening the implementation of RMA bonds, by ensuring they are used when appropriate, and that they are properly costed and administered, would secure better resource management outcomes for biodiversity.	Enhanced use of RMA bonds
Investigating new approaches	
Providing a flexible process for resource users to contribute to third party delivery of offsets and compensation in lieu of delivering their own requirements would enable the gains to be delivered by actors with relevant skills, and would increase the likelihood of compliance while not necessarily alienating developers from the outcomes of their contributions.	Third party delivery of mitigation, offset and compensation requirements
Provision of guidance and support for the use of economic instruments and ecosystem services accounting in resource management and business planning could improve the management of effects on biodiversity, particularly cumulative effects.	Ecosystem services assessments in decision-making
Improving the ecological outcomes achieved by mitigation trusts by carefully directing funding and monitoring the difference made would maximise the positive impacts of their funding.	Conservation tenders for mitigation trust funding
The use of cooperative instruments with willing resource users may complement or reduce the need for regulation in certain situations, and improve environmental outcomes.	Safe Harbour agreements

Table 5: Summary of opportunities and tools

To the extent possible what each policy innovation might cost, or what savings it might enable, is considered. The analysis of costs is not to the depth of a regulatory impact statement or RMA section 32 analysis: this is not feasible

for an organisation that sits outside government. The cost estimation is therefore restricted and based on broad estimation only. Costs are generally reduced where existing legislation already provides for, or at least forms no serious

barrier to, the introduction of the policy tool and where an existing actor can take responsibility for the innovation within their current role. Proposals that entail significant costs are only proposed where their impact would be commensurate with those costs: many tools were discarded because the impact of the tool relative to its prospective high cost rendered it likely unviable in the near term. All tools that proffer the ability to better control impacts on biodiversity are recognised to generate opportunity costs from limiting extraction and use opportunities permanently or temporarily. But in most cases, the increased private costs of lost opportunities are outweighed by the public benefit of increased biodiversity protection.

A key consideration in the advent of new, or the improvement of existing, policy approaches in New Zealand is whether they are cognisant of Māori perspectives and compatible with the Treaty of Waitangi. While this analysis does not presuppose the perspective of Māori in respect of any one tool, the importance of acceptability to Māori is acknowledged. Both throughout New Zealand and in other jurisdictions, a failure to consider the impact of policies on indigenous people has led to low uptake and ultimately failure. An example of where poor design has resulted in low engagement by indigenous people is the New South Wales (NSW) BioBanking Scheme, which commenced in July 2008.

The NSW BioBanking Scheme provided an opportunity for landowners to 'sell' on a market restoration credits that resource users would purchase to offset the impacts of their activities. The large traditional land holdings appeared to be excellent candidates for the funding of ecological restoration and provision of other financial benefits. Unfortunately, Aboriginal landowners have been reluctant to participate. The NSW Aboriginal Land Council (a state oversight body for 120 Aboriginal Land Councils) advised that Aboriginal communities were sceptical of BioBanking agreements because of the need for perpetual protection of the subject land and the very strict conditions of such protection that are incompatible with traditional resource management views. An underlying mistrust of agencies was also highlighted as a deterrent, in addition to background risks of non-compliance resulting in government takeover (see Case Study 14).¹⁰²

Lessons from this and other jurisdictions demonstrate that new tools can run counter to traditional notions of resource management and this should be heeded in the design of any fresh New Zealand tools. Māori demonstrate a desire to innovate and apply new approaches to the protection of the environment, with two of many examples being their uptake of custom geographic information systems that combine modern technology with traditional ecological knowledge¹⁰³ and the development of the 'Mauri Model' decision-support tool (profiled later).

Dryland ecosystems

Dryland ecosystems cover 19 per cent of New Zealand's land area and host more than half of our threatened plants.¹⁰⁴ Drylands are characterised by distinctive climate patterns that give rise to unique biodiversity values. For example, the Mackenzie Basin is a national stronghold for threatened plants (60 species, such as *Lepidium solandri*), nationally threatened and at-risk ecosystems, and naturally rare ecosystems. Since the arrival of Māori, the Mackenzie Basin has endured significant habitat loss and constant anthropogenic disturbance.¹⁰⁵ Recent pressure to convert dryland areas to intensive dairying has seen the tussocks and sheep give way to ryegrass and cows at a rapid rate. Unfortunately, dryland ecosystems are the least protected of all ecosystems in New Zealand (three per cent of extent). Long-held public perceptions that drylands have low biodiversity value, coupled with current pressure for conversions for intensive dairy farming, mean their biodiversity is in a parlous state.



Lepidium solandri. John Barkla



Planning for good outcomes

Improving biodiversity information

Problem

Biodiversity information is incomplete, poorly managed and difficult to access for decision-makers, applicants and submitters.

Implications

In order to effectively manage biodiversity, and make decisions that address adverse impacts on it, decision-makers must know where biodiversity is located. There are significant gaps in the information available on biodiversity in New Zealand, as discussed at length in *Vanishing Nature*. Of the biodiversity information that is available, much is in multiple formats and held by separate agencies, often behind technical gateways that many practitioners do not understand. Lack of information affects the quality of decision-making and can lead to decisions (in hindsight) being overly risky or indeed overly cautious.

Concerns regarding availability, accessibility and management of national data on biodiversity have been highlighted innumerable times, particularly since the advent of the Biodiversity Strategy. For example, in the final report of the Ministerial Advisory Committee on Biodiversity and Private Land (2000), the following recommendations were made in respect of biodiversity data:

Support and facilitate better national and local tools for information collection and management in partnership with local government through:

- *Guidance to local government for the collection and use of biodiversity information and describing and*

monitoring the extent and character of terrestrial and aquatic biodiversity within an area

- *National and regional coordination of existing biodiversity information*
- *National and regional biodiversity forums.*¹⁰⁶

There are three key and closely related concerns:

- The data are not available
- The data are not managed in a coordinated way
- The data are not easily accessed by decision-makers

Many planning agencies have a limited understanding of the biodiversity resources within their jurisdiction, including where they occur and how this could or should influence development patterns and conservation goals. Funding for collection of this data is difficult to access for many agencies, so the gaps at regional and district levels tend to persist, despite the best efforts of staff within those organisations. National datasets are unlikely to adequately 'fill the gap' due to the need for biodiversity data to be collected at an appropriate level to inform decision-making.

It is clear that a lack of coordinated data management is a long-running issue, although improvements have certainly been made. For example, the Environmental Monitoring and Reporting project (often referred to by its acronym, EMaR) promises to achieve greater consistency in biodiversity state and trend monitoring, which will have valuable relevance to decision-making. It is managed by regional councils as a collective, with some support from the Ministry for the Environment and Envirolink. However, the resources required to collect and manage the data are significant and likely

beyond many of the participating agencies at present. For this reason a dual approach of improving coordination and resourcing collection and monitoring is proposed.

There are opportunities to add value to national datasets by incorporating other sources of data. For example, many data are collected by consultant ecologists under instruction from resource users for the purpose of consent applications or monitoring of outcomes. At present there is no collective repository where that information can be stored, with confidentiality protected where appropriate, and made available for wider management purposes. Further, data may not reflect the full range of values, and other sources of information – such as traditional ecological knowledge – should be included and drawn upon to improve decision-making.

Many data that are available are difficult for decision-makers to access for a variety of reasons: cumbersome access requirements, a lack of user-friendly interfaces, absence of technical support to guide usage, and an underlying lack of expertise as to what information is available and needed to impact a decision. The data may also exist only in the grey literature and be difficult to trace. Providing a solution that would both bolster the amount of information available and ensure it is accessible where needed is essential.

Suggested solution

Ongoing and multi-stream investment in biodiversity data and information systems is crucial to safeguarding biodiversity. There is a need for continuous improvement in the collection and dissemination of biodiversity-relevant data and information. *Vanishing Nature* set out the need for this in some depth and recommended significantly more investment in the collection of robust biodiversity data.¹⁰⁷ The development of a structured repository or ‘clearing house’ for biodiversity information, including that collected in a development context, would draw together existing data and provide a valuable means of collating data in the future. The development of this solution – and with some urgency – is strongly recommended, in line with multiple earlier calls for the same. In addition to the development of the clearing house, support is needed for end-users and data collectors struggling with funding shortfalls.

How would it improve outcomes and is it an effective means of doing so?

Efficient and accessible collation and storage of biodiversity information, coupled with coherent, fast and accurate delivery of information when needed, would improve the protection of biodiversity in development in many ways, including:

1. Ensuring decision-makers have access to information about the biodiversity values on a site or in an area that may be affected by development activities

2. Providing resource users with a repository for the ecological information collected at a number of scales (including monitoring data) to enable it to be accessible to the public and used for other purposes
3. Providing resource users with information to assist in the formulation of applications for consents, and potentially reducing the scale of baseline information necessary
4. Enabling agencies to have access to the data collected under other mandates, and for other purposes, to inform their management activities and reduce duplication of collection effort

Data collation and management is costly and requires significant expertise, the development of quality assurance standards, and sustained financial support due to low capacity for cost recovery. The Terrestrial Freshwater Biodiversity Information System Fund was introduced in July 2000 and was designed to provide assistance with this function, but it was frozen in 2014 and would appear to have been disestablished, with no surrogate available. This means that information necessary for decision-making is often simply not available. The cost of improving the situation will likely fall on public agencies primarily, but it will also entail a cost incurred by applicants and submitters due to the need to provide their information in an appropriate format. Training would be required to assist users in accessing the information and maximising its utility.

Evaluation

1. The innovation is able to be implemented within New Zealand’s existing legal settings

Improving the collection of biodiversity data, its storage and accessibility sits outside regulatory biodiversity mandates but is crucial to fulfilling them. There is no legislative impediment to this tool being implemented.

2. An agency or responsible party exists that could implement the innovation

The identification or establishment of a lead entity to establish a data repository or clearing house would ensure that the combined knowledge of different agencies, resource users, landowners, research organisations and others was available to professionals managing biodiversity in a development context. There are several existing agencies and programmes which could perform this task including the Department of Conservation, Ministry for the Environment, a collective of regional councils, and a civil society initiative. Of highest importance is that the lead entity would be adequately resourced for the task, including for collaboration with existing data managers and provision of support to end-users.

3. The innovation is unlikely to result in perverse outcomes in practice

Perverse outcomes would eventuate if the system was not compatible with existing datasets; could not manage confidentiality matters sufficiently; could not incorporate the data in a manner that is easily accessible to all users; and/or had cumbersome administrative protocols.

4. The innovation is additional and does not duplicate existing mechanisms

There is a wide range of information sources on biodiversity, and many systems and databases have been developed for the purpose of storing and distributing biodiversity data. An example of an existing and significant data storage innovation is NatureWatch NZ, which incorporates citizen science data for public consumption.¹⁰⁸ The clearing house should work to incorporate or interface with such datasets.

5. The cost of the proposed tool is not prohibitive on the suggested lead organisation/s

The development of an independent clearing house which manages biodiversity data at a national scale from a range of sources, and supports end-users to utilise it effectively, will incur significant setup costs and maintenance costs over time. In addition, funding would need to be made available for information-collection and monitoring purposes, particularly to smaller agencies with limited funding. Cost savings are possible for resource users and biodiversity managers by reducing duplication of data collection and enhancing information sharing.

Summary

The development of a central repository of biodiversity data could draw together the scattered efforts over many decades, streamline access to information, better inform decision-making and potentially reduce the need for repeat surveys for different purposes.

Recommendation

*That a **multi-agency effort** is undertaken to establish a central national database or databases that collate and make available biodiversity data to help inform management and better safeguard biodiversity in development (including technical support for usage by decision-makers).*

That the Terrestrial Freshwater Biodiversity Information System Fund be relaunched with increased funding, recognising that its purpose remains critical and relevant in enabling agencies to contribute to the information database.

Tools for strategic planning

Problem

Strategic planning for biodiversity commonly fails to address cumulative and long-term impacts on biodiversity as well as the knock-on effects throughout ecosystems.

Implications

Biodiversity protection inevitably occurs in a complex sociopolitical context with many competing values and priorities. Extra effort is required, in a policy context, to front-end current and long-term biodiversity considerations. Without such effort, serious conflicts can arise at the time of consenting individual projects. A lack of planning tools to help address such complexities constrains the ability of planning processes to both provide coherent leadership on key issues and manage cumulative and long-term effects on ecosystems and species. There is widespread evidence of our failure to manage such effects, even where an explicit statutory mandate exists to do so.

Suggested solution

Planning processes must better incorporate the public interest in nature conservation to address this issue and this can be achieved through the adoption of strategic planning that provides for biodiversity considerations at the appropriate scale. Tools are available to help integrate environmental considerations into decision-making by providing a structured analysis within which many different goals and points of view can be considered. The growing focus upon collaborative planning approaches has been criticised at times for not considering the viewpoints of 'weaker' stakeholders. Structured analysis can protect those views and ensure that all values are assessed and considered in a transparent way. Such processes also help to anticipate environmental effects at a project level, resulting in more efficient decision-making and better capacity to address matters such as cumulative effects.¹⁰⁹

In respect of this solution, two potential tools to guide decision-making in complex scenarios are outlined: Sustainability Appraisal (see Box 5) and the Mauri Model (see Box 6). Tools that guide strategic decision-making use high-level goals (usually sustainability-related goals) and assess how likely they are to be achieved under a range of future scenarios. Such an analysis can be undertaken either at the outset of the planning process or retrospectively to monitor effectiveness of implementation and progress towards stated objectives.

How would it improve outcomes and is it an effective means of doing so?

Tools which assist strategic decision-making and help monitor implementation enable the structured analysis of the long-term outcomes likely to arise from decisions. They are inherently outcome-oriented and make explicit the

trade-offs that are likely under a range of future scenarios. They enable the consideration of values that are often deprioritised in policy processes (e.g. nature protection and traditional indigenous knowledge). Tools such as these are highly scalable, guide stakeholders through often complex territory, and provide a framework for linking action with outcome and evaluating subsequent results.

Evaluation

1. The innovation is able to be implemented within New Zealand's existing legal settings

There are no regulatory barriers to vetting decisions against a strategic framework. More comprehensive introduction or formalisation of mandatory strategic environmental assessment is a longer-term undertaking that will necessitate legislative reform.

2. An agency or responsible party exists that could implement the innovation

A range of agencies and community organisations can utilise this approach, as can proponents of development. Strategic decision-making tools and frameworks have broad application to a range of situations that relate to biodiversity in development. For example, they can help to evaluate development proposals or groups of proposals, suggested policy approaches, and the expected outcomes of strategies and plans.

3. The innovation is unlikely to result in perverse outcomes in practice

The risks of implementing strategic decision-making are limited from the perspective of the public interest. Any additional rigour and attention paid to assessing the efficacy of policies and plans is likely only positive, as it will focus attention on the most sustainable option in the long term. However, strategic assessment has the potential to draw greater attention to the costs that will be carried by the public, which may mean that it is sometimes not supported by development interests. If recommendations and caveats resulting from the

assessment are ignored in practice, the process may not realise its goals.

4. The innovation is additional and does not duplicate existing mechanisms

Conflicts between the need to safeguard biodiversity and the need for economic development are best addressed at a strategic level, something which is not commonly achieved. These tools provide an additional mechanism that complements existing policy and planning processes by providing a strategic lens through which such conflicts can be assessed.

5. The cost of the proposed tool is not prohibitive on the suggested lead organisation/s

Undertaking strategic assessment will incur additional costs which are inherent in any planning process. These include the cost of facilitation, time of the participants, and the provision of technical support. However, the tools enable the consideration of issues and potential conflicts at a strategic level, likely reducing the amount of effort required at the consenting stage of individual projects.

Summary

The use of strategic decision-making support tools such as Sustainability Appraisal and the Mauri Model would assist in a more long-term and sustainable approach to planning and development. They would enable a more holistic account to be taken of biodiversity and ecosystem services considerations as well as their cultural and social importance.

Recommendation

*That **councils and other decision-makers** consider the use of strategic decision-making tools such as Sustainability Appraisal and the Mauri Model to support plan-making, policy development and other collaborative decision-making processes by front-ending considerations of biodiversity.*

Box 5 Sustainability Appraisal

Sustainability Appraisal is highly scalable and the level of detail can be adjusted to suit its application. It can be applied at a project level and all the way up to a national-scale policy process. Sustainability Appraisal has been demonstrated to work well in a strategic planning context (see Case Study 7). Once participants have determined the desired goal, a process of vetting alternatives and determining the favoured approach occurs, usually under expert guidance.

The first stage of screening is an analysis of whether the proposal on the table (it may be a policy, a proposed plan or even an individual project) provides opportunities to contribute to sustainability or entails risks to environmental bottom lines or other parameters. This is followed by the scoping stage, which compares the proposal and alternatives with these goals and minimums. The impact analysis stage enables alternate courses of action to be analysed to determine what they will mean in practice and what trade-offs they will drive. This stage informs a decision that takes account of the need to achieve the desired sustainability goal and ensures that critical thresholds are not breached. Finally, monitoring and evaluation processes 'truth' the process over time and enable adaptive management.¹¹⁰

Case Study 7 Canterbury Water Management Strategy appraisal

Structured analysis tools were first applied in New Zealand during the assessment of the Canterbury Water Management Strategy in 2010.¹¹¹ The appraisal was carried out over a two-day workshop attended by a wide range of stakeholders, and was based on the Sustainability Appraisal framework developed by Sadler et al. (2008) for Landcare Research. The attendees collaboratively evaluated a range of alternatives against a framework while being guided through the following steps by experts:

1. Participants selected the level of sustainability to evaluate strategic options between weak, moderate and strong (participants preferred moderate sustainability as a goal in this case).
2. Participants were required to define the six most important capital assets under each of the pillars (these were highly diverse and varied from 'ecosystem resilience' through to 'trust in institutions and processes').
3. Participants considered intergenerational and subregional issues to ensure the evaluation criteria could take account of spatial and temporal issues and constraints.
4. Draft evaluation criteria were reviewed and revised. Process criteria related to equity, feasibility and resilience were added before a final suite of criteria were agreed upon.
5. Top (preferred objective position) and bottom (lowest acceptable value) were collaboratively defined.
6. The options (including 'business as usual') were scored against the evaluation criteria. This was based on likelihood of a strong negative impact (-2) to strong positive impact (+2) for each criterion in order to draw a sustainability profile for each option.
7. Application of the options to subregional contexts was also considered.

The Sustainability Appraisal demonstrated the value of a holistic framework and showed that viewing a strategy through a single lens (development or environmental protection only) rarely yielded the most efficient and effective management of resources:

The process indicated a need to consider existing infrastructure, uses and their effects in an integrated way with new infrastructure and parallel environmental restoration activities to achieve a sustainable development pathway for the management of water resources.¹¹²

The process also demonstrated that the 'business as usual' scenario was not sustainable. Recent reporting on progress towards fulfilling the multiple goals of the Canterbury Water Management Strategy, however, suggests that the likelihood of even weak sustainability being achieved is quite low.¹¹³

Box 6 Mauri Model

The Mauri Model is a decision-making framework that is designed to assess the sustainability of policy and planning proposals. It is available online (www.maurimeter.com). The Mauri Model uses a participatory action research approach that involves all stakeholders as active contributors and owners of the outcomes. It has been verified as a *useful process for developing an indicator set to measure a community's progress towards sustainability* in a report by the Cawthron Institute.¹¹⁴

The Mauri Model provides a culturally based template within which indigenous values are explicitly recognised alongside scientific considerations.^{115, 116} It fosters the incorporation of culturally relevant knowledge which is commonly ignored in decision-making based solely on a Western scientific perspective.^{117, 118} The Māori concept of mauri is a useful means of measuring the impacts of biodiversity loss and of establishing the need for its protection. Mauri is variously explained as a binding force, the power of the gods,¹¹⁸ existing in all things,¹²⁰ and the intrinsic value of ecosystems (something which is explicitly recognised in the RMA). Mauri is the fusion that makes it possible for everything to exist by holding the physical and metaphysical elements of a being or thing together in unison.

Although it does not specifically use the term 'mauri', the language of the RMA accommodates the underpinnings of the Mauri Model. Section 5(2)(b) refers to the *life-supporting capacity of air, water, soil, and ecosystems* and this approximates the concept of mauri as a life force.¹²¹ The reference to *safeguarding* that life-supporting capacity in the same section and identification of 'kaitiakitanga' in section 7 as a matter to be given particular regard to alongside the intrinsic values of ecosystems also provides some alignment. Kaitiakitanga is the active protection and enhancement of the mauri of ecosystems.¹²² When the indigenous concepts of mauri and kaitiakitanga are coupled with science-based systems thinking, a powerful means of assessing the sustainability of decisions is provided.¹²³ The Mauri Model decision-making framework is therefore well suited to assist in the efforts to preserve biodiversity.

Regional biodiversity plans

Problem

While goals such as achieving 'no net loss' of biodiversity at a regional scale, halting the decline of biodiversity, and 'maintaining' biodiversity are common, clear pathways for their achievement are not.

Implications

Considering biodiversity at a strategic level means that the location and condition of biodiversity values is expressed early in a process, and this helps to avoid decision-making that makes their degradation inevitable. In spite of often very sparse information, communities and governments have frequently agreed on certain goals for our natural heritage via consultation on policies and plans and through collaborative processes. For example, regional councils have a statutory requirement to maintain biodiversity and increasingly they are setting a goal of no net loss of biodiversity at a regional scale. While national and regional goals for biodiversity do exist, how development will be managed in order to enable achievement of those goals is rarely set out in any detail, and there is usually no accountability for failures to achieve such goals and targets.

Suggested solution

It is recommended that non-statutory regional biodiversity plans are devised that combine consideration of all statutory requirements that relate to biodiversity (across all relevant legislation); identify regional goals for biodiversity; and provide a clear pathway to achieving those goals, taking into account human and other pressures on ecosystems. Developers could benefit from the clarity these plans would provide across multiple statutory processes, gain insight into conservation priorities, and use them as a basis to ascertain opportunities for partnerships. Multi-stakeholder initiatives to address biodiversity, including that which is affected by development, are becoming common – some led by councils and some originating from the grass roots. An example of a recently initiated multi-faceted approach is the Nelson Nature programme, a broad programme incorporating both the regulatory and non-regulatory functions of Nelson City Council in collaboration with the wider community (Case Study 8).

How would it improve outcomes and is it an effective means of doing so?

Comprehensive biodiversity planning at a regional scale could be used as a decision-support tool for several processes including allocating areas for development purposes; coordinating existing and new conservation efforts; prioritising conservation spending; directing mitigation, offset and compensation requirements where they are obligatory; and providing a basis for monitoring and reporting of the extent and condition of biodiversity.

The development of regional biodiversity plans, informed by appropriate data and engagement, would also enhance the likelihood that other plans and decision-making

processes will consider biodiversity sufficiently. A plan that is predicated on the goal of 'maintaining' biodiversity is likely to provide for greater recognition of cumulative impacts such as disturbance, the protection of corridors and connections, the location and state of highly vulnerable biodiversity, and the protection of migratory species' habitats. A plan with clear and measurable objectives would create a framework for monitoring and evaluation of policy outcomes.

Biodiversity planning would need to be evidence-based and also need to accommodate the views and interests of a range of actors including iwi, the conservation community and developers. If it is only based on scientific data then it may lack social acceptance and remain unimplemented. If it is too skewed to community input or development concerns, then its priorities may not reflect those actually necessary to maintain or improve biodiversity. Much biodiversity does not gain the attention of communities and conservation organisations because it is poorly understood (e.g. bryophytes) or difficult to conserve via individual efforts (e.g. marine areas).

The development community may well resist the actual or perceived consequences of these planning processes, or directly challenge them, as they may impinge upon development opportunities. This resistance will likely be most acute where values targeted for protection are material to the pursuit of economic development opportunities. In practice, this may skew spatial identification to areas residual to present or future economic uses. Further, biodiversity is a cross-cutting policy topic that involves a wide range of interests and agencies with differing objectives and statutory roles, which makes formulating a plan that incorporates their various interests and objectives complex.

Evaluation

1. The innovation is able to be implemented within New Zealand's existing legal settings

It is likely that, given the diversity of players and issues, the development of regional biodiversity plans would be best kept non-statutory so that flexibility is retained in *how* biodiversity goals are reached and for the mix of actors to vary over time. This recognises that all communities are different. Once detailed planning for biodiversity is in place, it will likely improve the outcomes achieved from environmental legislation. It will also inform legal matters such as zoning, the location of significant infrastructure and patterns of residential development.

2. An agency or responsible party exists that could implement the innovation

The Department of Conservation, councils, iwi, non-governmental organisations, community groups and individuals all collect data on biodiversity throughout New Zealand, but there are significant differences between the amount and quality of information held by different entities. Agencies that have a significant amount of biological data in a usable format, such as the Department of Conservation and councils, are in

a much stronger position to lead regional biodiversity planning than other stakeholders due to the breadth of their statutory responsibilities, the resources they have available, and the scale at which they operate.

3. The innovation is unlikely to result in perverse outcomes in practice

Perverse consequences resulting from a regional biodiversity planning process are unlikely.

4. The innovation is additional and does not duplicate existing mechanisms

Planning for biodiversity occurs within a range of agencies including the Department of Conservation, regional councils, some district councils, and within community initiatives such as Reconnecting Northland.¹²⁴ The development of these plans would seek to draw together what are often disconnected efforts, engage a greater variety of players, and manage duplication and gaps on a landscape scale. Regional biodiversity plans would complement regional policy statements and support region-wide goals such as achieving no net loss of biodiversity.

5. The cost of the proposed tool is not prohibitive on the suggested lead organisation/s

The development of regional biodiversity plans will necessarily be highly participatory, as their purpose is to consider the full range of existing conservation issues and interests in an area, establish priorities, and focus attention on knowledge and action gaps. Much of the cost of regional plan preparations would be incurred by the lead agency in initiating and supporting the process. Significant costs can also fall on those who participate, including developers, non-profit organisations and individuals from the community (who typically are not recompensed for their time and effort). However, the benefits which can be achieved through aligning biodiversity activity across multiple stakeholders are likely to significantly outweigh these costs.

Summary

The development of regional biodiversity plans would enable the achievement of significance assessment, prioritisation of conservation tasks, and incorporation of strategic biodiversity considerations into development planning in a far more coordinated and cost-effective way.

Recommendation

That regional councils in association with the Department of Conservation and communities (including iwi) develop comprehensive regional biodiversity plans that demonstrate how the maintenance of biodiversity will be achieved and how development will be managed so as to achieve that goal.

Case Study 8

Nelson Nature programme

Nelson City Council is a unitary authority at the top of the South Island which has a small and affluent coastal jurisdiction. In 2014, the Council undertook a review of its biodiversity responsibilities under the RMA. The Nelson Nature programme was subsequently developed in association with the Nelson Biodiversity Forum. It has identified goals and key activities spread out over 11 work streams which cover a range of non-regulatory biodiversity programmes. Prior to the programme getting underway, the typical outlay by the Council for biodiversity functions was approximately \$50,000 per year (plus parks and reserves expenditure). The Nelson Nature programme increased this to \$6 million over a 10-year period.

The Nelson Nature programme has identified focal landscapes (e.g. limestone outcrops) that will be protected or enhanced by the eradication of wilding pines and goat control to quantifiable levels. Significant Natural Areas (SNAs) will be protected from further loss and 50 sites will be brought into active management over the course of the programme. The programme is inclusive of all stakeholders, treating them as partners rather than subordinates. A strong enveloping strategy of communication, engagement and monitoring is also provided for.



Kokorua Inlet, Nelson. Boffa Miskell 2014



Keruru/Wood pigeon (*Hemiphaga novaeseelandiae*). Manu Bird

Leadership on biodiversity

National policy on indigenous biodiversity

Problem

There is no binding national policy on biodiversity and this has reduced the prominence of biodiversity in resource management decision-making.

Implications

New Zealand has a range of national-level policy that relates to biodiversity management, but no dedicated instrument that provides direction to agencies on their roles with respect to protection of biodiversity and the environmentally sympathetic exercise of their wider functions. A lack of national direction on biodiversity is likely constraining the protection of our natural heritage. This gap has also resulted in the costly proliferation of different approaches to the protection of biodiversity in development and significant repeat litigation for many stakeholders. Councils have the function of maintaining indigenous biodiversity and regional and district plans are designed to assist them in undertaking this function, but the documents have a patchy record in doing so. Agencies have little direction from government regarding biodiversity in development and there is little accountability when the legislative requirements for biodiversity are not met.

Suggested solution

Our suggested solution is the promulgation of national policy on indigenous biodiversity to provide some consistency and leadership, particularly on the management of biodiversity on private land where national policy direction is sparse. The most influential national policy construct is a national

policy statement under section 45 of the RMA (Box 7). A National Policy Statement on Indigenous Biodiversity could be promulgated along similar lines to the previous draft document (being the proposed National Policy Statement on Indigenous Biodiversity 2011) and in accordance with the recommendations of the Biodiversity Strategy.

How would it improve outcomes and is it an effective means of doing so?

A national policy statement on biodiversity would improve consistency in decision-making through improving the basis for the development of objectives, policies and rules in RMA planning documents and providing more direction on how Part II matters are to be applied. Such policy would serve to highlight the importance of indigenous biodiversity and potentially help increase public understanding of its importance. Regional policy statements and regional and district plans would need to give effect to it. Consideration of it would be mandatory in the assessment of consent proposals, irrespective of the depth and scope of relevant rules in subnational instruments. National policy statements can also make processes and definitions more consistent, which may lower costs of public interest litigation and provide greater certainty to resource users.

One example where a national policy statement could reduce costs, improve consistency and increase clarity is in defining 'significance'. Significance is a keystone concept within the RMA which is particularly relevant to statutory duties such as the need to ensure *the protection of areas of significant indigenous vegetation and significant habitats of indigenous fauna* under section 6c and to some extent the *maintenance of biodiversity* under section 30 (Case Study 9).

Box 7 What is a national policy statement?

The purpose of national policy statements is to *state objectives and policies for matters of national significance that are relevant to achieving the purpose of this Act (RMA)*.¹²⁵ The criteria in section 45(2), which are set out below, are used to determine whether a national policy statement is a desirable policy instrument. They clearly apply to biodiversity.

- 2) *In determining whether it is desirable to prepare a national policy statement, the Minister may have regard to—*
- (a) the actual or potential effects of the use, development, or protection of natural and physical resources:*
 - (b) New Zealand's interests and obligations in maintaining or enhancing aspects of the national or global environment:*
 - (c) anything which affects or potentially affects any structure, feature, place, or area of national significance:*
 - (d) anything which affects or potentially affects more than 1 region:*
 - (e) anything concerning the actual or potential effects of the introduction or use of new technology or a process which may affect the environment:*
 - (f) anything which, because of its scale or the nature or degree of change to a community or to natural and physical resources, may have an impact on, or is of significance to, New Zealand:*
 - (g) anything which, because of its uniqueness, or the irreversibility or potential magnitude or risk of its actual or potential effects, is of significance to the environment of New Zealand:*
 - (h) anything which is significant in terms of section 8 (Treaty of Waitangi):*
 - (i) the need to identify practices (including the measures referred to in section 24(h), relating to economic instruments) to implement the purpose of this Act:*
 - (j) any other matter related to the purpose of a national policy statement.*

Case Study 9 'Significance' under the RMA and the need for national policy

Assessments of significance underpin resource management in New Zealand, and in particular planning and decision-making under the RMA. Assessments of significance feed into regulatory processes in two ways: first to identify whether anticipated environmental impacts are consistent with the sustainable development purpose of the Act and second to identify significant habitats and habitats of indigenous fauna in accordance with section 6. Both these determinations usually trigger more restrictive regulatory approaches (e.g. the requirement for a biodiversity offset to address significant residual adverse effects) and so are highly influential. The principal Act provides no definition of 'significance'. Assessments are currently carried out in the absence of definitive guidance as to what constitutes significance, and there is a substantial amount of variation in how they are carried out and the criteria used.

Logically, development interests will often seek to underestimate significance values, while conservation interests may seek to overestimate them. But such ambiguity has been demonstrated to generally favour development interests by resulting in an underestimation of conservation values.¹²⁶ The devolution of the decision on what constitutes 'significant' has three key impacts: debate is confined to a local context and not at the more visible national scale; local needs tend to take precedence over national goals; and the multiple definitions arising locally and regionally make it more costly and difficult for all parties to engage in the process.¹²⁷



Rātāpihihi temperate rainforest, Taranaki. Catherine Kirby

Evaluation

1. The innovation is able to be implemented within New Zealand's existing legal settings

The RMA provides a framework for national policy statements and environmental standards to be produced by central government on matters of national importance. National policy statements can apply to all land, freshwater and marine areas out to the 12 nautical mile limit. This jurisdiction can be limited however: the Proposed National Policy Statement on Indigenous Biodiversity 2011 did not bind the Department of Conservation.

2. An agency or responsible party exists that could implement the innovation

Depending on the form of the policy – and the recommendation is that it is a national policy statement – it would be prepared by the Minister for the Environment and implemented as required by the Act. Unfortunately, national policy statements prepared under the RMA cannot apply to the Exclusive Economic Zone and the Extended Continental Shelf, although analogous policy could be adopted by regulation under the EEZ Act, which applies to that area.¹²⁸ Provision for EEZ policy statements is proposed under the Resource Legislation Amendment Bill.

3. The innovation is unlikely to result in perverse outcomes in practice

Perverse outcomes from national-level policy are possible if it is poorly designed, rendered ambiguous for the purpose of achieving consensus, or not implemented effectively through the planning and resource consenting processes. The potential shortcomings of this policy approach are little different to those of any other legislation. Standard definitions may have an effect on the functionality of lower planning instruments as definitions are often integral to them and closely intertwined with how an individual plan is to be interpreted and implemented.

4. The innovation is additional and does not duplicate existing mechanisms

There is currently no dedicated statutory guidance on indigenous biodiversity. However, the new policy would need to interface sensibly with existing national instruments including the National Policy Statement on Freshwater Management and the New Zealand Coastal Policy Statement.

5. The cost of the proposed tool is not prohibitive on the suggested lead organisation/s

The cost of developing national-level policy for any substantive subject is usually significant for both the administering agency and stakeholders contributing via submissions. Due to the earlier work in developing a

national policy statement on biodiversity, a proportion of the development costs have already been expended and the 2011 version could be used as a basis for further development. Stronger national direction on biodiversity could serve to restrict economic development opportunities for resource users and increase compliance costs. Consistent policy is also likely to provide areas of cost-saving because it reduces the likelihood of, or need for, repeat litigation over the content and interpretation of lower-level instruments.

Summary

A national policy statement on indigenous biodiversity would provide valuable clarity and direction for resource managers and decision-makers and certainty for resource users. It would be most effective if it addressed key points of ambiguity and areas of significant inconsistency in lower-level instruments.

Recommendation

*That the **Ministry for the Environment** develops a national policy statement that provides clarity on key issues that affect the protection of indigenous biodiversity.*

Impact management policy

Problem

The weak national policy settings for impact management generate ambiguity and uncertainty for all stakeholders.

Implications

Impact management refers to the implementation of the mitigation hierarchy (described above) at the jurisdictional and project levels. It spans the entire hierarchy, from identifying where impacts need to be avoided through to determining any offsetting or compensation arrangements and what conditions they must meet to be considered adequate. Economic development will generally entail some form of ecological harm and tools to ameliorate these effects are essential to effective environmental management. Compensatory actions, such as biodiversity offsets, also enable additional conservation gains to be leveraged from development activity. Cautious impact management, whereby unavoidable impacts are carefully and thoroughly addressed, is essential to protecting biodiversity in a development context.

However, used inappropriately and without safeguarding against their numerous risk factors, impact management tools such as mitigation, offsets and compensation can lock in environmental decline and generate significant transaction costs for developers. A lack of clarity on terms, expectations and bottom lines in policy has long been recognised as

constraining the potential of impact management to help stem the tide of biodiversity loss. While positive outcomes for the environment are certainly currently possible (see Case Study 10), more reliable outcomes would arise from consistent, principle-based implementation with defined minimum standards and a requirement to clearly and explicitly set out relative losses and gains. Variable application of the mitigation hierarchy and the ad hoc use of mitigation, offsets and compensation mechanisms, as well as an overarching lack of follow-up and enforcement, could be partially arrested by a robust regulatory underpinning for impact management.

Suggested solution

National policy on impact management could be implemented in New Zealand by central government in a similar way to the approach in other jurisdictions, defining key factors to trigger and frame decision-making. Policy for biodiversity offsets was first proposed in 2010, and Baker-Galloway and Christensen (2013) reflected on this three years later:

We ... suggested the introduction of both a National Policy Statement which provides national policy direction and a National Environment Statement which

Case Study 10 Waikato-Tainui Mitigation Plan

The Huntly section of the Waikato Expressway is a 15 kilometre stretch of four-lane highway to be constructed over a four-year period from 2015 to 2020 at a cost of more than \$450 million.¹²⁹ The road includes 180,000 square metres of paving, 4 million cubic metres of earthworks and nine bridges.¹³⁰ The whole Waikato Expressway project is one of seven Roads of National Significance. The proposed road crosses four waterways: the Mangawhara, Whangamaire, Komakorau and Mangatoketoke Streams (known as the 'Red Rivers').¹³¹

The mitigation package addresses the impact of the road itself, the project's impacts on two areas of conservation land, and a range of other concerns such as the route of the road going directly through a culturally significant maunga (mountain) just outside the township of Taupiri. Tangata whenua were closely involved in the impact assessment and in the determination and negotiation of the mitigation measures.

The project initially aimed for no net loss of biodiversity with an emphasis on avoidance where possible.¹³² But as the assessment progressed, the experts involved (including those engaged by the New Zealand Transport Agency) acknowledged that demonstrating no net loss was not possible at a whole project level with the available data. The final outcome was negotiated between the parties and addresses archaeological, cultural and ecological concerns in a holistic mitigation package that combined onsite and offsite approaches.¹³³

Most of the agreed mitigation requirements are included within the Raahui Pookeka Ki Taupiri (Waikato-Tainui Mitigation Plan) and resource consent conditions. Other documents, such as the Red Rivers Cultural and Ecological Restoration Strategy, were produced to guide part of the 'ecological offset mitigation'. The Red Rivers Cultural and Ecological Restoration Strategy includes a range of requirements that the New Zealand Transport Agency had agreed upon with the Tangata Whenua Working Group and other stakeholders, and which are enshrined in the conditions for the development.

Consultation, particularly with iwi, was extensive and highly innovative. A review of the process to date concluded (in respect of the planning process) that this [consultation] avoided the costs and time associated with major hearings and avoided the risks of appeals to the Environment Court. It demonstrates that RMA processes are highly adaptable and can be successfully used in the context of a constructive, collaborative approach that is also legally and technically robust using traditional consenting processes in an innovative way.¹³⁴ Work has now started on the project, with continuing involvement from stakeholders.



Looking north from the Gordonton Interchange towards the Taupiri range.
New Zealand Transport Agency

*sets out an appropriate methodology or methodologies which can be used. In the four years since that was written, the need for national direction has become even clearer... While the decisions overall are leading to a clearer understanding of the concepts, there remain some different interpretations and opinions on various issues. The overall result remains, with respect, somewhat inconsistent and confusing ... In our view, it remains highly desirable that a national statutory offsetting framework be established which sets the 'rules of the game' with respect to issues such as the mitigation hierarchy, like-for-like, metrics, acceptable gain, no net loss etc. The system would then mostly operate by administration rather than litigation.*¹³⁵

Many jurisdictions around the world have formalised these approaches after initial ad hoc implementation. Formalisation has typically consisted of the introduction of national- or state-level policies which include goals, minimum requirements and monitoring specifications. The purpose of such formalisation has been to standardise and improve the quality of decision-making. Where formal policy is in place, further clarification and greater specificity is often recommended by reviews. For example, an Australian Senate review of the Environmental Biodiversity and Conservation (EPBC) Act 1999 (Case Study 11) in 2014 recommended express recognition of offsets and the principles that should underpin them.¹³⁶

Many common challenges are being faced by policy makers around the world and there is much to learn from the experience of other jurisdictions in drafting policy to implement the mitigation hierarchy. New Zealand should engage deeply with those learnings. For example, McKenney and Kiesecker (2010) compared biodiversity offset policies

throughout the world and provided an analysis of the key issues policy must address in order to be effective (e.g. equivalency, spatial proximity and additionality of offset proposals).¹³⁷ Since that time, further models and experience have developed such as the Standard on Biodiversity Offsets in 2012 by the well-known US multi-stakeholder initiative the Business and Biodiversity Offset Programme.¹³⁸

New Zealand national policy on impact management should reflect such international experience and could also draw on existing provisions in regional policy statements and the New Zealand government guidance. This policy should include the following:

- Clarification of what avoidance means, the different kinds of avoidance (strategic, geographic, temporal and technical) and direction for when it is appropriate and compulsory (e.g. conservation status thresholds)
- Burdens of proof to demonstrate that opportunities for avoidance in all relevant forms have been thoroughly canvassed at both a policy and consent level
- Principles that proposals for compensation must comply with, such as equivalency and the achievement of no net loss or preferably net gain (particularly in recognition of inherent uncertainty)
- Minimum requirements for the timing of offsets relative to impact, security, delivery and follow-up monitoring and enforcement
- Clarification of the role of traditional knowledge
- Principles for the appropriate participation of stakeholders

Case Study 11 Australian federal environmental offsets policy

The formalisation of biodiversity offsetting in Australia occurred in response to concerns over variable outcomes arising from ad hoc implementation. The policy only applies to species and ecosystems that are recognised at a national level: all other adverse effects are managed at a state or local level. The policy clarifies a number of matters including:

1. The mitigation hierarchy is formalised
2. The use of a calculator is enshrined
3. Offsetting in advance is formalised to reduce uncertainty, and reduce the need for discount rates to be applied
4. Indirect offsets (offsets for entirely unrelated actions such as research) are limited to 10 per cent of the total quantum

The policy has a built-in calculator that assesses relative loss and gain of biodiversity. Offsets literature generally casts aspersions upon the use of calculators, for very legitimate reasons – biodiversity is complex and very hard to measure accurately.¹³⁹ Notwithstanding the obvious technical risks, most interviewees regardless of their organisation, reported that the role of the calculator had been positive. This was because it had improved transparency and had helped to 'depersonalise' the assessment to some extent. Impacts on very vulnerable biodiversity can generate a high likely cost when run through the calculator or result in a 'red flag'. This can assist in sending a price signal to the developer that may promote redesign. The ability for potential applicants to remodel their proposals around the biodiversity potentially affected means the calculator can help at the conceptual stage of the process, rather than only at the time of applying for permission.

How would it improve outcomes and is it an effective means of doing so?

A clear and coherent mandatory policy context for impact management would help to streamline decision-making processes and reduce litigation by indicating societal expectations. Policy that effectively addresses key issues can help to better manage the risks such tools pose while maximising the benefits they can offer. The policy could provide a very useful evaluative framework against which decision-makers, resource users and the wider community can vet proposals for impact management.

Evaluation

1. The innovation is able to be implemented within New Zealand's existing legal settings

Given that much impact management takes place through RMA processes, the most appropriate vehicle for policy on this topic is likely a national policy statement or national environmental standard, neither of which already exist for this or a near purpose. National direction on impact management could be incorporated into a national policy statement on indigenous biodiversity, in a similar way to the National Objectives Framework within the National Policy Statement on Freshwater Management 2014.

2. An agency or responsible party exists that could implement the innovation

Impact management policy would likely be developed by the Ministry for the Environment and implemented under the RMA by councils and other decision-makers in the course of their existing consenting functions.

3. The innovation is unlikely to result in perverse outcomes in practice

There is a risk that national policy on impact management could be insufficiently robust so that inappropriate development is facilitated and further loss of biodiversity results rather than its protection. Even where a coherent policy framework is present, good outcomes are not necessarily guaranteed on the ground, as application can be patchy. To reduce these risks, the policy should draw on international experience, be comprehensive, and cover the entire impact management process rather than focusing on one element, such as offsetting.

The political vulnerability of offset policy has been demonstrated in several recent examples from Australia. In Queensland, an incoming right-wing government made substantial modifications to existing programmes as part of facilitating a 'clear and simple offsets framework', with the explanatory note for the amendment legislation denoting that the changes reflect the government's 'commitment to cutting green tape'.¹⁴⁰ Amendments elsewhere have generally sought to reduce the requirements of the scheme to render them more 'pragmatic' and 'workable' at a cost to biodiversity and the public interest in vibrant nature.

4. The innovation is additional and does not duplicate existing mechanisms

No national-level policy exists to guide agencies, resource users and communities on the management of ecological impacts for biodiversity. The proposed National Policy Statement on Indigenous Biodiversity 2011 contained some guidance but was never made operative. The 2014 whole of government document *Guidance on Good Practice Biodiversity Offsetting in New Zealand* is also not a policy document, although at the time of release some businesses including Genesis Energy were concerned that it would have de facto policy status.¹⁴¹ While regional policy statements are increasingly addressing impact management, their approaches are varied and inconsistent. In addition, they tend to focus on the latter end of the impact hierarchy, placing little focus on the avoidance aspect. This tool will be most successful in increasing good outcomes for nature if rigorous environmental bottom lines are observed and gateway tests are implemented in practice.

5. The cost of the proposed tool is not prohibitive on the suggested lead organisation/s

National policy processes, including statutory consultation requirements and any additional collaborative processes, are costly. It may be most effective to combine the consideration of impact management policy with the wider process of developing a national policy statement on indigenous biodiversity, possibly as a national environmental standard. The policy once enacted will also have important cost implications for both agencies (increased costs of policy implementation through additional complexity and increased expectations to give effect to existing mandates) and resource users (pressure to reduce externalities on biodiversity and increased quality – and therefore cost – of assessment). A stronger emphasis on avoidance may invite greater restrictions on resource users, leading to opportunity costs of non-implementation or delayed extraction or development.

Summary

A national policy on impact management would provide important context, parameters and guidance for decision-makers, all of which is currently missing at a national level, and improve the management of impacts on biodiversity.

Recommendation

That the Ministry for the Environment develops policy to guide impact management in a regulatory context, potentially as part of a national policy statement on biodiversity.

Adaptive management guidance

Problem

The application of adaptive management is inconsistent, lacks clear limits, and thus may not adequately manage uncertainty in resource consenting.

Implications

Biodiversity is at risk from inappropriate and ineffective management of uncertainties in development projects. Adaptive management is an approach that is being increasingly applied to address such uncertainties. Adaptive management allows for the staged implementation of a project, with each stage increasing in intensity.

Progression between stages, or continuation at a particular stage, is dependent upon the environmental limits set in the conditions being maintained. Trigger points can be established to determine when a management response is required, as in the case of the Wilson Bay marine farming zone consents (Case Study 12).

Despite its increasing application in a resource management context, there remains a lack of clarity about what is expected of applicants implementing an adaptive management approach. Adaptive management requirements have been in dispute in many consent processes, including those outside the RMA, such as in the Trans-Tasman Resources and Chatham Rock Phosphate cases under the EEZ Act.

Case Study 12 Wilson Bay marine farming zone

The Wilson Bay marine farming zone covers an area of 1,106 hectares. It is zoned for mussel farming, and is an example of an adaptive management regime that has been in operation for over a decade. Adaptive management relies on measurable trigger points within a consent management framework that require a management response to address environmental effects.

Over a number of years a consultative process established Limits of Acceptable Change (after Zeldis et al., 2005) for key environmental parameters and a range of other triggers.¹⁴² Trigger conditions were defined upfront, although they were subject to review by committee on a three-yearly cycle.¹⁴³ A key trigger was the threshold of 20 per cent of phytoplankton depleted over more than 10 per cent of the Firth of Thames. Monitoring data published in 2012 indicated that no significant depletion of phytoplankton (enough to trigger a management response) had occurred in 12 years.¹⁴⁴



Wilson Bay marine farming zone. James Bentley

Previous cases have seen applicants raise concerns about adaptive management and the resulting consent conditions of approval (e.g. staged triggers, thresholds, and requirements for temporary or permanent discontinuation in the event of a breach) amounting to frustration of the consent. But legally, the breach of a limit and the associated pulling of a trigger do not equate to a situation where the consent is unable to be implemented. Where a limit is breached, the consent has been implemented and has reached a point where the consent conditions are breached. In this instance, as with any other consent, the activity must either reduce its impact to stay within the designated limit or cease to operate and apply for a new consent. In *Sustain Our Sounds v King Salmon* the Supreme Court confirmed conditions requiring exactly this course of action.¹⁴⁵

Suggested solution

A framework for adaptive management, based on existing experience and jurisprudence, could be developed within a national policy instrument. This could set out requirements that would need to be met for an adaptive management approach to be permissible and for it to act as a reasonable risk management tool. The purpose of the policy would be to clearly set out what an adaptive management approach entails and what parameters it must meet before it can be considered to be capable of managing risk.

How would it improve outcomes and is it an effective means of doing so?

Robust implementation of adaptive management is critical to more fairly balancing the risk of environmental effects falling outside expected parameters. Clear and consistent policy or guidance would refine practice and increase certainty for resource users wishing to apply for consents where impacts and the success of mitigation strategies are uncertain.

Evaluation

1. The innovation is able to be implemented within New Zealand's existing legal settings

Given the range of situations in which adaptive management is relevant, a single definition is unlikely to be found and may introduce additional complications (such as conflicting with existing statutory definitions or not being applicable in certain situations). A national policy or guidance document could transcend those differences and provide direct assistance to decision-makers and other stakeholders on the underlying principles of adaptive management and strategies for optimal implementation.

2. An agency or responsible party exists that could implement the innovation

The Ministry for the Environment, the Department of Conservation and councils have the ability to develop and release guidelines to help achieve the purpose of the RMA and other legislation. Adaptive management is an approach taken under a range of consenting scenarios and by individual stakeholders within specific projects. The development of common guidance should ideally be taken up by an entity with national reach and a statutory leadership role in consenting such as the Environmental Protection Authority or the Ministry for the Environment.

3. The innovation is unlikely to result in perverse outcomes in practice

The primary perverse consequence of overarching guidance on the principles of adaptive management, and assistance to decision-makers, is that it too strongly conflicts with existing definitions that have been developed for particular purposes or is not sufficiently detailed to be of practical assistance.

4. The innovation is additional and does not duplicate existing mechanisms

There is no existing framework to assist decision-makers or applicants in implementing adaptive management, other than case law and wording within the EEZ Act.

5. The cost of the proposed tool is not prohibitive on the suggested lead organisation/s

Development of policy for adaptive management will incur similar costs to those inherent in any policy development. It may also be more costly to administer than some current approaches and may increase costs for the development community by requiring greater expenditure to improve the quality of any adaptive management frameworks, including additional baseline surveys and monitoring. However, it would also make requirements clearer, potentially reduce litigation, and guide agencies to improve the level of long-term monitoring they undertake.

Summary

Useful and authoritative guidance to provide a clear framework for the robust implementation of adaptive management would better manage risks to nature in cases of development proceeding in the face of uncertainty.

Recommendation

That the **Ministry for the Environment in association with consenting agencies** develops common policy direction for the implementation of adaptive management approaches, particularly in the context of significant uncertainty.



Improving knowledge of existing legal mechanisms to protect biodiversity

Problem

A wide range of legal protection tools for biodiversity are available, many of which are not well known by practitioners.

Implications

Throughout the interviews conducted for this report, it became clear that many practitioners are unaware of the existence of the range of legal mechanisms that can be used to protect biodiversity, and the range of situations in which they are best implemented. Examples of tools often not implemented or well understood include limited-life covenants (e.g. 'life of trees' covenants under the Queen Elizabeth II National Trust Act 1977); heritage protection orders under the RMA; voluntary management agreements under the Reserves Act 1977 and protection strategies which can be negotiated directly with utility operators and other land users. The implication of this lack of knowledge is that suboptimal tools are used for particular purposes, or new tools are proposed or developed where existing tools should already suffice. More full and robust implementation of the tools already available will help mitigate the impacts of human activities by addressing acute impacts, disturbance and cumulative effects more effectively.

Suggested solution

Education for practitioners, including developers and submitters, is necessary to ensure that the swathe of existing tools is used whenever possible. Using the full range of potential protection mechanisms will maximise the

value of the current suite of tools and better demonstrate where gaps exist.

How would it improve outcomes and is it an effective means of doing so?

A range of protection mechanisms have been devised throughout our legal system. Improving the information basis about the range of tools available and the best situations in which they can be used would assist agencies, developers and the wider community to protect biodiversity in a cost-effective and fit-for-purpose way.

Evaluation

1. The innovation is able to be implemented within New Zealand's existing legal settings

No changes in law are necessary to provide information and advice to practitioners.

2. An agency or responsible party exists that could implement the innovation

Statutory responsibility to protect threatened taxa and ecosystems lies primarily with the Department of Conservation, although crucial aspects of that role (e.g. some forms of habitat protection and management of extraction) involve or are confined to other agencies. The protection of biodiversity in development is the responsibility of regional and district councils under the RMA, which is overseen by the Ministry for the Environment. Cooperation of these agencies will be essential to improving knowledge about existing legal mechanisms and in refining them over time. The development of the guidance could be a multi-agency

effort or be led by one agency while being supported by the others, and inclusive of the many professional organisations involved in this space.

3. The innovation is unlikely to result in perverse outcomes in practice

Provision of education material and advice is low-risk and unlikely to result in perverse outcomes.

4. The innovation is additional and does not duplicate existing mechanisms

Improving the knowledge basis on the range of tools already available would lead to reduced duplication in policy and practice and help inform decision-making about the best tools available for each job. A range of information sources that help guide decision-makers (e.g. the Quality Planning website¹⁴⁶, the Environment Guide website¹⁴⁷ and various information pamphlets, booklets and guidance from the Ministry for the Environment and the Department of Conservation) could be added to in order to better communicate this information.

5. The cost of the proposed tool is not prohibitive on the suggested lead organisation/s

Provision of information is not costly and may lead to significant efficiencies in policy and practice. Costs are likely to be borne by public agencies and the priority of this work may be limited within constrained budgets.

Summary

New Zealand has a wide range of tools to safeguard biodiversity in development, but knowledge of what they are, where they are best applied, and their relative negative and positive aspects appears limited in practice.

Recommendation

*That the **Department of Conservation in association with regional and district councils** develops educational material to outline the potential tools available to safeguard biodiversity and investigates potential improvements to them.*

Peka peka/long-tailed bats (*Chalinolobus tuberculata*)

Bats are our only native land mammals and there are two taxonomically indeterminate subspecies of the long-tailed bat, one in the North Island and one in the South. Once common and widespread, bat populations have declined due to loss of habitat, disturbance, competition for prey and predation.¹⁴⁸ A key threat to bats in a development context is the removal of mature trees. Bats roost and nest in the cavities of aged trees, both exotic and native.¹⁴⁹ If mature trees are removed, bats experience a total loss of habitat, even if the trees are replanted with young saplings. In some cases, both in New Zealand and around the world, artificial roost boxes have been installed to replicate the habitat that bats require. However, success of these fixtures has been highly variable.¹⁵¹



Peka peka/long-tailed bat (*Chalinolobus tuberculata*). Ben Paris

Guidelines for biodiversity impact assessment on vulnerable taxa and ecosystems

Problem

There are no comprehensive guidelines for biodiversity impact assessment on vulnerable taxa and ecosystems.

Implications

Indigenous biodiversity is put at risk by poor assessments of environmental effects that do not reflect best practice, exclude key taxa and key impacts, and/or undertake surveys at the wrong time of day or year. They therefore result in the underestimation of ecological impacts. Nature is likely put at risk by poor assessments that do not accurately describe the type and magnitude of effects, particularly on important biodiversity.¹⁵¹ Development proposals can be put at risk through insufficient AEEs that are rejected or questioned by agency staff or submitters, leading to increased and unnecessary costs and delays for resource users.

Without clear guidelines for matters such as survey effort during the application phase, detailed surveys are often left to the post-granting stage of the consent. Unfortunately, if these surveys detect important fauna, it is too late. Approval has already been granted for the habitat to be modified or disturbed in some way. In such cases, translocation becomes the only possible method for potentially avoiding effects. But there is little robust monitoring data to support the success of this approach for many species. Success is often assumed rather than being based on actual data and translocations have an often low and variable success rate.

Suggested solution

An expert-led process should be established to develop guidelines for ecological impact assessment for vulnerable taxa and ecosystems, and to highlight gaps where no suitable survey methods are available. Similar documents are found throughout the world such as under the EPBC Act in Australia for a range of taxa including reptiles (e.g. *Survey Guidelines for Australia's Threatened Reptiles*).¹⁵² The Research and Science Information Standard for New Zealand Fisheries is an example of a document prescribing generic requirements for information quality and the like which may prove a useful model to follow.¹⁵³

Guidelines could cover the following matters as examples:

- The appropriate times of year to conduct a survey for specific taxa and ecosystems
- Approved or acceptable methods for conducting surveys (i.e. minimum survey effort)
- When survey information should be required to be submitted to council

- Minimum information standards and a requirement to demonstrate that the best available information has been used (as in the EEZ Act)

How would it improve outcomes and is it an effective means of doing so?

The public interest in biodiversity protection would be better maintained by more robust assessment methods and agencies, and resource users would benefit from greater certainty that methods used are acceptable to agencies. Clear expectations on the nature and extent of biodiversity impact assessment, particularly for threatened fauna, is essential to better safeguard biodiversity in development.

Guidelines for biodiversity impact assessment will be most successful if they introduce minimum standards, are sufficiently flexible to enable innovation in practice, and are adopted in practice by agencies. Resource users would have a clearer understanding of what their consultants need to deliver and more certainty that the AEE they have purchased, via a relevant professional, covers the necessary subjects and uses approved methodologies to ascertain impacts.

The development of clear and consistent guidelines would result in the consolidation of existing best practice and provide an opportunity for multi-stakeholder engagement. It should enable considerable cost-savings through reduced need for repeat surveys, fewer requests for additional information where methods are considered insufficient, and fewer rejections of applications on the basis of insufficient information.

Evaluation

1. The innovation is able to be implemented within New Zealand's existing legal settings

A standard for AEEs as they pertain to biodiversity could be implemented in a number of ways. It could first be an amendment or addendum to existing and less specific guidelines from the Ministry for the Environment or other sources. A second option is to develop new stand-alone national guidance and release it on a non-statutory basis like the 2007 *Protecting Our Places* document which guides national priorities for conservation on private land. A final option is to develop a statutory tool that outlines key requirements, such as via a national environmental standard under the RMA or a New Zealand standard. In any case, such a standard should be developed in consultation with key experts and set out minimum standards of survey.

2. An agency or responsible party exists that could implement the innovation

The Ministry for the Environment, the Department of Conservation and councils have the ability to develop and release guidelines to help achieve the purpose of

the RMA and other legislation. Consenting agencies, resource users and participants in policy and planning processes would likely readily incorporate directive guidance on the scope and depth of AEEs required. It is essential, however, that neither consenting agencies nor the private sector should lead on this project, because of the potential for capture.

Biodiversity impact assessment, as it applies to taxa, is a largely technical process and guidelines should therefore be developed by experts in relevant fields. A similar process is undertaken to establish the New Zealand Threat Classification Lists and appears to work well. Other benefits include professional development of consultant ecologists, agency staff and resource users through requiring best practice environmental impact assessment.

Given the history and experience of the Department of Conservation in managing threatened species and ecosystems, and its existing statutory advocacy role, it is the logical lead organisation for the development of this material. However the strong relationship of this proposal to existing guidance from the Ministry for the Environment, and the Ministry's oversight role for councils, indicate that a joint initiative is likely to be most effective, and this will ensure that the science and policy dimensions link together well.

3. The innovation is unlikely to result in perverse outcomes in practice

The scale of biodiversity impact assessment must be appropriate for the scale of effects and the scale of the project itself. The risk of capture by private interests seeking to limit suggested survey effort to unreasonably low levels is serious. Guidance that demands excessive levels of investment will deter uptake and lead to insufficient impact assessment. If the guidance is too weak it will put nature at risk by not asking enough of resource users. Guidance that is too prescriptive may fail to incorporate the sometimes rapid innovation in survey methods and improvements in ecological knowledge. A careful balance, taking into account costs and benefits, should enable robust and practically workable guidelines to be developed.

4. The innovation is additional and does not duplicate existing mechanisms

While a range of guidance material is available on ecological impact assessment, much is out of date (e.g. the Ministry for the Environment's general guidelines on AEEs) or does not deal explicitly with wildlife (e.g. the Environmental Institute of Australia and New Zealand's *Ecological Impact Assessment Guidelines*).¹⁵⁴ These documents may provide a basis or starting point for more relevant and directive guidelines for vulnerable taxa and ecosystems.

5. The cost of the proposed tool is not prohibitive on the suggested lead organisation/s

Depending on the scale and depth of the exercise, an expert-led process would require recognition of the time and expertise of those involved, and administrative support and oversight would be required. Such costs would only be moderate.

Stricter standards for impact assessment will have cost implications on resource users and may yield information that will constrain economic development opportunities. Cost savings are also possible for agencies and applicants in a consenting process by making requirements clear upfront and reducing the time delays associated with differences of expert opinion.

Summary

Minimum standards for biodiversity impact assessment in New Zealand would provide certainty and consistency to the development community and their advisors (e.g. consultant ecologists) on what expectations are regarding the scope and depth. Standards would ensure that where surveys are conducted and methods applied they are considered appropriate and effective for that application.

Recommendation

That the **Ministry for the Environment** together with the **Department of Conservation**, other **consenting agencies** and **relevant experts** coordinate the formulation of mandatory biodiversity impact assessment guidance for vulnerable species.



Outcomes-based conditions

Problem

Resource consent conditions often do not mandate desirable ecological goals but instead focus on process-oriented issues that can be unrelated to the achievement of tangible outcomes.

Implications

Resource consent conditions often fail to include clear and measurable outcomes. Matters such as cost, timing, methodology and monitoring requirements usually receive more attention than the outcome that needs to be achieved. It is not uncommon for a development to be fully compliant with all conditions of consent and still achieve a very suboptimal ecological outcome. The fact that 'compliance does not equal success' is a finding echoed in a study of US wetland mitigation banking for the same reasons – unclear goals.¹⁵⁵

Suggested solution

Professional practice in New Zealand, and if necessary policy and guidance, should focus on identifying the environmental outcomes to be achieved in preference to the means by which they should be achieved or other matters of less relevance. A focus on process-oriented conditions can mean that conditions that make a meaningful ecological difference have less oversight. This reflects experience in New Zealand where compliance rates for administrative conditions were much higher (82 per cent) than active conditions (49 per cent).¹⁵⁶ Trends towards using regulatory instruments to define endpoints, and innovating to refine methods by which those outcomes will be achieved, are apparent here and overseas. In Australia, specific policy has been developed to encourage the use of outcomes-based conditions (Case Study 13).

How would it improve outcomes and is it an effective means of doing so?

The broader use of outcomes-based conditions would enable the ecological goals of avoidance, remediation, mitigation and offset/compensation measures to be clearly set out in the consent. It would focus the efforts of decision-makers and other participants in the planning processes on clearly articulating outcomes. Clarity on goals would improve enforceability, as well as provide sufficient flexibility within the consent process for proponents or their consultants to determine the most effective and efficient method for achieving a certain outcome, having regard to innovations. The use of outcomes-based conditions is highly scalable and provides for flexibility in how an outcome is achieved.

This approach may not be possible in circumstances where outcomes cannot be easily measured, or the link between the activity and outcome is not clear. Basing compliance

and enforcement on outcomes in those circumstances, as opposed to practices that are believed to lead to those outcomes, for example, could be an issue (i.e. it places all the risk/liability on the resource user). A wide range of conditions will likely be necessary within a given consent, including those focused on procedure and outputs.

Evaluation

1. The innovation is able to be implemented within New Zealand's existing legal settings

The nature of conditions is generally within the discretion of decision-makers, providing they comply with general principles and the parameters set down within the RMA, which form no barrier to focusing a condition on outcome instead of, or in addition to, a process or output.

2. An agency or responsible party exists that could implement the innovation

The Ministry for the Environment, the Department of Conservation and councils have the ability to develop and release guidelines to help achieve the purpose of the RMA and other legislation. Outcomes-based conditions can be used by individual staff members in agencies, and proponents of development can include them in suggested conditions of consent that are commonly submitted.

Case Study 13

Outcomes-based conditions

In Australia, the Productivity Commission recommended that an outcomes focus should be present in development consents to promote innovation and reduce regulation of minutiae.¹⁵⁷ A focus on outcomes-based conditions in Australia intended to set out the environmental outcome the approval holder must achieve for a matter of national environmental significance without prescribing how that outcome is to be achieved.

In 2015, an outcomes-based conditions policy was released.¹⁵⁸ Three types of conditions are used by the Australian federal government: administrative conditions, prescriptive conditions and achievement or outcomes-based conditions. By its own admission, the follow-up and compliance was strongest for the first category of conditions. The purpose of the change was to recognise that there are multiple ways to reach a given endpoint and to reduce the administrative burden of highly prescriptive conditions. Effort in recent years has been focused on improving condition drafting and the use of outcomes-based conditions that ensure ecological goals are met.

3. The innovation is unlikely to result in perverse outcomes in practice

There is a risk that a focus on outcomes-based conditions may mean that attention is diverted too much from appropriate planning and risk management processes up front. A healthy balance is required to ensure the environment is protected and does not bear all the risk. Detailed and effective management planning for projects, particularly where there are significant risks to the environment, and the inclusion of outcomes-based conditions are not mutually exclusive. In addition, outcomes-based conditions should only be used where the outcome is time-bound, measurable and where progress towards the outcome can be systematically assessed and attributed to what the developer is doing (attribution).

4. The innovation is additional and does not duplicate existing mechanisms

This innovation merely refines the use of consent conditions. Greater security and clarity about what a project will deliver on will improve environmental outcomes and enhance transparency. Securing gains via consent conditions, provided they are sensible, feasible and enforceable, would also help to orient consent monitoring towards addressing ecological outcomes instead of administrative compliance.

5. The cost of the proposed tool is not prohibitive on the suggested lead organisation/s

There should be no additional cost implications arising from agencies including outcomes-based conditions. They may in some instances necessitate further preparatory work by applicants but the ability to innovate in order to meet outcomes-based conditions may result in lower compliance costs for industry on a per-condition or per-consent basis.

Summary

Using consent conditions to articulate clear endpoints, and leaving room for innovation, may improve efficiencies in consenting from the application stage through to final stages of implementation. The success of outcomes-based conditions relies largely on their appropriate usage and the availability of suitable monitoring metrics to track compliance over reasonable timeframes.

Recommendation

That **consenting agencies** adopt wide use where appropriate of outcomes-based conditions that are clear, measurable and enforceable.

Enhanced use of RMA bonds

Problem

RMA bonds are often poorly administered, constraining their ability to increase the security of resource consent outcomes. In other cases, they are not used where they could help improve the security of outcomes.

Implications

A bond under the RMA which is not accurately costed is unable to secure an outcome as intended under the Act. In the event of non-compliance it cannot be uplifted to complete the full scope of works. If agreed works are not undertaken then the uncertainty and loss primarily falls on the environment and the wider public interest. On some occasions, councils and other organisations will undertake the works but constrained budgets generally mean they remain undone.

Suggested solution

Our suggested solution is to enhance the use of the bonding mechanism under the RMA to enable it to deliver additional security and better manage the risk to the public interest. Enhanced practice may include directive guidance on the following matters:

- A requirement to cost bonded works and identification of what parameters to use
- A standard calculation method for bond quantum based on the above costings
- A requirement to future-proof the sum of the bond to recognise inflation
- The format of a bond (cash, bank guaranteed, etc.) and when each form may or may not be appropriate
- Requirements for the administration and release of bonds (including maximum timeframes)
- Standard procedures for when bonds must be uplifted to guide agencies, resource users and other stakeholders
- Standard procedures for when bonds are not claimed after a certain period of time

How would it improve outcomes and is it an effective means of doing so?

RMA bonds against ecological conditions, if sufficiently costed and appropriately administered, provide both security that outcomes will be met and the ability for the agency to uplift the bond and undertake the works when they are not. Ensuring that they deliver on these outcomes when required will maximise the utility of this mechanism.

Evaluation

3. The innovation is able to be implemented within New Zealand's existing legal settings

Section 108 of the RMA enables consenting agencies to require a bond in the form of cash or bank guarantee to hold against the completion of a particular consent requirement (e.g. a planting programme).

2. An agency or responsible party exists that could implement the innovation

Provision of guidance on the use of RMA bonds is in line with the responsibilities of the Ministry for the Environment. Consenting agencies routinely gather and administer bonds under the RMA and other legislation.

3. The innovation is unlikely to result in perverse outcomes in practice

Enhancing the use of RMA bonds is unlikely to have perverse consequences unless they are used where they are inappropriate and cost estimates balloon unreasonably.

4. The innovation is additional and does not duplicate existing mechanisms

Refining the use of an existing mechanism provides for better outcomes that may not have eventuated under the current ad hoc arrangements for RMA bonds. Many agencies have existing policies to guide the collection of bonds under the RMA (e.g. Thames Coromandel District Council)¹⁵⁹ and for other purposes.

5. The cost of the proposed tool is not prohibitive on the suggested lead organisation/s

Provision of accepted, but not necessarily statutory, guidance on the valuation and administration of bonds under the RMA is a confined issue and would likely have very limited cost implications on the agency/agencies developing the guidance. For agencies that increase their use of this mechanism, there may be additional costs in administering bonds and determining how to manage them in case of the consent-holder forfeiting them. For resource users there may be additional costs in providing more and larger bonds if the use of the mechanism and quantum required generally increase.

Summary

RMA bonds which ensure that consent outcomes are achieved are a simple and effective form of insurance for some common consent requirements. Their utility and validity is entirely dependent on accurate costing and astute administration. Guidance to enhance their implementation would be valuable. This initiative would work most effectively if it was combined with administrative improvements to ensure that councils managing bonds have appropriate systems in place to do so.

Recommendation

*That **consenting agencies** collaborate together, with the assistance of the **Ministry for the Environment**, to produce common guidance on appropriate costing of bonds.*



Investigating new approaches

Third party delivery of mitigation, offset and compensation requirements

Problem

Resource users are generally responsible for the delivery of compensatory requirements leading to sometimes poor compliance and poor ecological outcomes.

Implications

Mitigation, offsets and compensation can be demanding to deliver, both in resources and expertise. Many proponents of development may be unwilling or unable to fulfil these requirements. The uncertainty of delivery, where there is a lack of enthusiasm or expertise, is usually borne by the environment and the wider community through non-compliance and resulting ecological effects. Extractive industries, for example, often want their contribution to be meaningful but generally do not wish to incur a perpetual commitment. Commodity price shifts can also have substantial impact on the resource and enthusiasm devoted to environmental measures, even where they form part of a legal agreement. Research into compliance with compensation requirements demonstrates that the longer the time period after an approval is given the less likely a requirement will be met.¹⁶⁰ A system that assumes the proponent of development will also be the proponent of conservation has obvious inadequacies.

Agencies may not accept ad hoc transfer arrangements as they are often deemed unreliable. For example, in reference to resource users donating to community groups in lieu of doing compensation projects, a Taranaki Regional Council practice note states the following:

*... donations to community groups undertaking restoration in lieu of direct work is generally not ideal as there will be less surety that the restoration is undertaken to the degree that Council would seek. In addition, community groups may change their objectives or priorities, or may dissolve and fail to undertake the work.*¹⁶¹

It would seem therefore that for some situations developer-led conservation projects are not ideal and that a robust alternative is required to improve compliance and reduce the risk to the biodiversity affected. Schema to address this need have been developed around the world, including in New South Wales (Case Study 14).

Suggested solution

New Zealand should provide a consistent platform for third party delivery of mitigation, offsets and compensation requirements. It should be administered by a dedicated independent entity. Third party delivery is common internationally, but is only carried out on an ad hoc basis in New Zealand. There is a wide diversity of potential methods available to implement such an approach, from the current context through to a highly formalised, market-based biobanking system. Other methods include a less formal brokering market where willing conservationists are paired with resource users with externalities to address.

An independent and dedicated civil society initiative to formalise third party delivery and address matters of compliance and inefficiencies of delivery would provide crucial objectivity in this often vexatious space. The suggested structure is a centralised not-for-profit entity that delivers conservation gains on behalf of the development community, separate from the policy context. The entity

Case Study 14 NSW BioBanking Scheme – the story so far

The NSW BioBanking Scheme is administered under the Threatened Species Amendment Act 2006. A BioBanking agreement requires management in perpetuity and recognises the opportunity cost of not pursuing alternative and likely more environmentally damaging uses for the land. A BioBanking agreement, once formulated, provides in perpetuity for a similar level of protection as a National Park and is usually better resourced with respect to having dedicated funds available for management. Most BioBank agreements prioritise an improvement in ecological condition (e.g. pest control or underplanting) over an increase in extent (buffer planting or habitat creation). Parties who wish to ‘sell’ credits generally have to bring a degraded system up to a certain level before it can be registered. Ongoing management actions must be carried out or annual payments can be withheld. Many would-be sellers are sceptical of the scheme which requires site survey and assessment at their cost before they know if their candidate site will be accepted.

Payments are divided into two amounts. Part A is an upfront payment that recognises the value of the land and the opportunity cost of protecting it. This figure is wholly determined by the market. Part B payments are made from an Environment Trust administered by the Office of the Environment and Heritage and cover ongoing management payments that are based on the true cost of necessary management actions. Monies are paid up front by the developer and are invested in a specific fund managed by government, protecting them against changing priorities and cost-shifting. Parties that provide BioBanking credits can do the work themselves and invoice for the costs, enabling them to raise valuable additional and untapped funding to support core functions.

The Western Sydney Parklands Trust has been an active participant in the NSW BioBanking Scheme. The Trust formed in the 1960s and is a publicly owned entity operating under its own legislation. It administers 5,280 hectares of land near Parramatta and has protected approximately 1,000 hectares of that, with a goal of achieving 40 per cent over the long term. A range of revenue streams are used by the Trust from renting out commercial property to recreation management, and BioBanking is becoming an increasingly important among these.

Interest in the scheme has been slow to build, however, and only certain sectors of society have been interested to date. Approximately 30 agreements exist with a further 15 proposed. Long-term landowners such as trusts and agencies such as councils have shown the greatest interest. The private sector, particularly mining companies and residential subdivision proponents, have also shown some interest. Additional incentives exist, due to the fact that by addressing impacts via a BioBanking purchase appeals to the Land Environment Court are avoided. There are concerns about councils being active participants in the programme, as management of existing protected areas may be non-additional when statutory mandates are considered. Many interviewees in Australia voiced concern at agencies using BioBanking and offsets more generally to fund core business. Only one agreement has been formulated with traditional owners due to tenure complexities and cultural concerns about agreements in perpetuity. Successive landowners with BioBanking agreements on their site will only have management funded and cannot enjoy the upfront bonus of the Part A payment.

could support selection of appropriate exchanges through good science-based metrics. The focus of this proposal is on providing a consistent, reliable alternative pathway for conservation gains to be delivered by a capable entity.

A third party delivery structure could work as follows (see also Figure 3). Resource users with existing regulatory requirements would approach the entity for assistance with delivering the conservation gains required. Liability would be transferred, and the developer would relinquish

all responsibility, save for funding the conservation gains to adequate levels. The entity would determine whether or not it wishes to assume liability and the feasibility of the requirement will come into that (i.e. difficult or unproven techniques may create an operating risk and may be inappropriate).

To ensure compliance, the funding would be provided up front and secured via an endowment fund or similar secure financial mechanism. The conservation gains would be

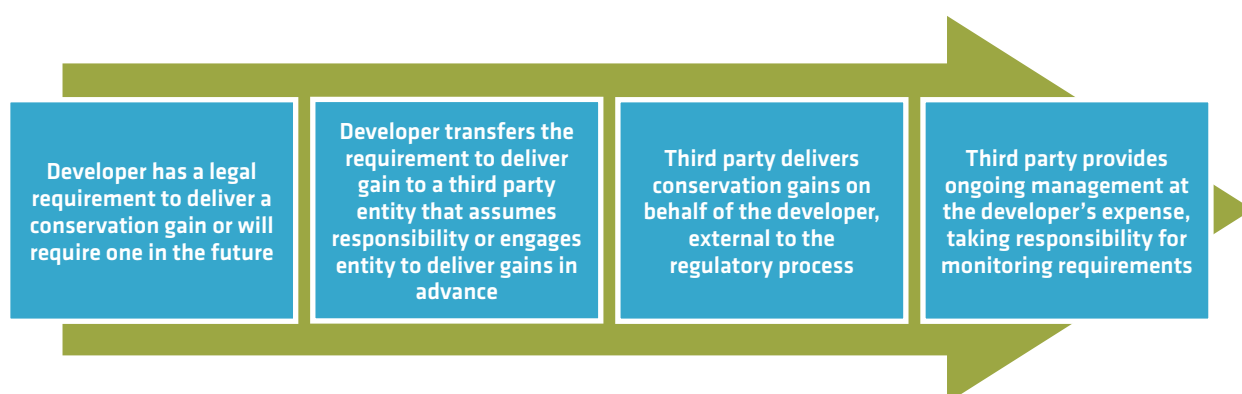


Figure 3: Suggested flow diagram of the operation of a third party delivery system for offsets and mitigation

delivered over the agreed timeline and the third party entity would report directly to the relevant agency on outcomes. Where projects are similar, the entity can be responsible for aggregating them to improve the landscape-scale contribution of developer-funded conservation. Furthermore, where mitigation, offsets or compensation is known to be required in the future (such as offsetting for an existing but as yet unimplemented approved designation), the entity can secure and implement these requirements in advance.

How would it improve outcomes and is it an effective means of doing so?

Third party delivery would potentially enable four important improvements to occur:

1. Compliance would be improved because dedicated entities are undertaking conservation as core business
2. The quality of ecological outcomes is likely to improve with more expertise and oversight available, which will be particularly crucial for situations where a high level of expertise is required to deliver gains (e.g. net increases in the population of a given species or conservation in a difficult context such as a remote area)
3. Conservation on private land will have access to an additional source of funding where appropriate
4. Advance gains can be registered to reduce uncertainty

One of the most significant improvements that a third party delivery programme would enable is the ability to secure ecological gains in advance of a project taking place. Best practice for biodiversity offsets is that gains are secured in advance of impact to reduce both ecological time lags and uncertainty – a scenario that usually decreases the overall cost of meeting requirements for resource users. Recent research demonstrated that 95 per cent of requirements in New Zealand needed to be completed after the damage has taken place. This is likely at least partly due to the absence of a reliable mechanism to secure mitigation, offset or compensation projects in advance of development and to have them appropriately recognised by agencies.¹⁶²

Offsets and compensation, in some instances, may rely on independent third party delivery of requirements due to remoteness or technical constraints. As one example, Bos et al. (2014) recommend that marine offsets should be carried out by specialised third parties, consolidated to maximise gains, and monitored separately but alongside regional monitoring programmes, with the data being made publicly available.¹⁶³ The structure of an entity in New Zealand should be designed to maximise transparency, cost-efficiency and good outcomes while at the same time providing minimal incentive to weaken requirements designed to safeguard nature. The entity could also be a vehicle for philanthropic and crowdfunded conservation work.

Evaluation

1. The innovation is able to be implemented within New Zealand's existing legal settings

Third party delivery is legally possible in New Zealand and practiced already on an ad hoc basis. The range of potential methods to implement it is significant, from the current context through to a highly formalised, market-based biobanking system. No legal amendment would be necessary to establish a third party entity.

2. An agency or responsible party exists that could implement the innovation

Existing agencies, landowners and conservation organisations are already regularly delivering mitigation, offset and compensation requirements. Examples are numerous but include the following situations: payments by mining companies to the Department of Conservation to carry out pest control; funding for a kiwi conservation project implementing predator control in the bush adjacent to a quarry; and landowners hosting threatened species within existing covenanted areas while habitat restoration elsewhere is in progress. While ad hoc implementation of this transfer of liability is probably sufficient in some cases, some standardisation could reduce overall costs.

Public agencies may not be the optimal administrators of third party delivery. If offsets are viewed as a funding stream for government agencies, they become an opportunity for formalised cost-shifting, and gains are often non-additional. The public interest therefore pays twice – first in suffering the loss of natural capital and secondly by having public agency mandates funded by mitigation. An example of where this has occurred is the decision of a local council in the Australian Capital Territory to register land a community group had restored over decades as an offset, and then sell it.¹⁶⁴ As a civil society initiative, third party delivery would be subject to less political pressures and be more likely to highlight and address issues of non-additionality.

Private sector third-party delivery may result in a reduction of controls on exchanges in order to maximise the number of initiatives undertaken and therefore revenue. Experience internationally is generally critical of private sector schemes, due to the pressure to enhance the bottom line, sometimes at the cost of maximising biodiversity outcomes. Private sector administrators may also struggle to gain access to relevant areas and to manage complex tenure arrangements. A civil society initiative, operating on a not-for-profit basis, would be subject to less economic pressure and be more flexible in the incorporation of alternative funding models.

Novel governance systems will likely be necessary to robustly administer mitigation and offsets in some areas. Complex tenure systems such as multiple

ownership land, leasehold land and coastal and marine environments all introduce additional complications, particularly for perpetual arrangements for biodiversity protection. An effective scheme would need to retain full discretion on the activities it engaged in and flexibility in approach for each specific situation.

3. The innovation is unlikely to result in perverse outcomes in practice

For New Zealand to implement an optimal version of third party delivery we must acknowledge the lessons of other jurisdictions. Some weaknesses are identified, along with potential solutions, in Table 6.

4. The innovation is additional and does not duplicate existing mechanisms

Third party delivery is already carried out in New Zealand in various ways (as described above). A civil society initiative could be undertaken on a national basis with regional representation. Such an entity would provide an alternative to mitigation, offset and compensation requirements being delivered on an ad hoc basis by the consent holder. For other stakeholders who wish to engage in conservation, funding can be difficult to obtain and sustain over time, and funding which would become available through a third party mechanism may represent a new or enhanced source of revenue. This mechanism enables new opportunities for providing conservation outcomes and for incentivising them.

4. The cost of the proposed tool is not prohibitive on the suggested lead organisation.

A plethora of different mechanisms for delivering mitigation, offset and compensation requirements increases their transaction costs. A single entity with established processes and systems may reduce the cost of administration on a project basis and in aggregate. Enabling advance mitigation may also reduce the costs of offsets and compensation for applicants as it will diminish the need for multipliers or discount rates to be applied. As a civil society initiative, initial seed funding will be necessary, but it is likely that such a system would be self-funding within a few years.

Summary

Third party mitigation, offset and compensation delivery would help address non-compliance with consent requirements, reduce uncertainty and introduce an additional innovative mechanism to fund long-term conservation activities.

Recommendation

That the third party delivery of mitigation, offsets and compensation is investigated by an appropriate entity established for this purpose.

Concern or risk	Potential to address via design	Suggested method to address risk
To determine equivalency, simplified metrics to describe biodiversity are generally required (although not always). Simplified accounting metrics may lead to losses of biodiversity that are not accounted for.	Medium	Biodiversity is extraordinarily complex and managing the risks to it to a technically accepted level is unlikely or at least very difficult to demonstrate. This technical constraint is relevant to all environmental impact management and is likely impossible to fully address. A calculator that could be adapted for this purpose is available, produced by the Department of Conservation. ¹⁶⁵ Alternatively, another strong and science-based metric could be developed to guide operations.
Failure to observe principles of exchange such as temporal equivalency, ecological similarity and additionality.	High	The proposed entity could retain full discretion as an extra-regulatory agency to ensure these principles are met. Long term, a failure to maintain integrity will erode credibility, which will be an important driver for robust exchanges.
Lack of transparency of compliance and outcomes obscures failures.	High	The monitoring and transparency of data and outcomes can be made available on an individual and aggregated basis. Monitoring requirements on an individual project basis are unlikely to change, and the ability to monitor at greater scales is likely to increase.

Table 6: Solutions to risk issues with third party delivery of offsets and compensations

Ecosystem services assessments in decision-making

Problem

Ecosystem services assessments are rarely implemented, even though there is potential for them to add value to decision-making in resource management.

Implications

Poor environmental management is extremely costly to society, but the cost can be difficult to articulate in terms that can be considered in decision-making. A common means of framing the role of nature in human society is through the concept of ecosystem services. There is a lack of guidance as to how consideration of ecosystem services can contribute

to resource management, despite broad recognition for several decades that the incorporation of ecosystem services into decision-making is essential to achieve sustainable development.¹⁶⁶

A lack of policy guidance, broadly accepted methods and knowledge of ecosystem services has meant that their application in resource management has been limited, inconsistent and unclear. This is a lost opportunity as ecosystem services tools can help bolster the breadth and accuracy of impact assessment and refine decision-making. For example, land degradation alone has been estimated to cost US\$10.6 trillion globally every year.¹⁶⁷ Ecosystem services assessments can also highlight where industries or activities are dependent on ecosystem services and therefore have business drivers to help safeguard them.

Case Study 15 The Ecosystem Services Review

The Sustainable Business Council is an *executive-led group of companies that catalyses the New Zealand business community to have a leading role in creating a sustainable future for business, society and the environment*. 'Ecosystem Health' is one of five key priority areas for the Council and its member organisations. In association with Landcare Research, the Council funded the application of a tool developed by the World Business Council for Sustainable Development to a New Zealand setting. The tool – the 'Ecosystem Services Review' – enables business to articulate impacts and dependencies on nature. It has been trialled by several organisations including Contact Energy and Aotearoa Fisheries.

Loss of kelp in the Marlborough Sounds is a serious threat to local pāua and other marine life which rely on the seaweed for food and habitat. Potentially as much as 50–70 per cent of kelp has died off in some areas and one of the main causes may be land-based sedimentation. This is one of several eye-opening issues that Aotearoa Fisheries, New Zealand's largest Māori-owned fishing company, discovered when it completed an ecosystem review of its pāua fishery with the Sustainable Business Council. Armed with a stronger understanding of some of the risks faced by this taonga (treasure), the company is now working closely with the Department of Conservation's chief scientist, Crown Research Institutes like Landcare Research, and other primary industry businesses to address the loss of kelp in the area. Addressing this issue means Aotearoa Fisheries will be able to continue its kaitiakitanga (stewardship) of New Zealand's pāua fishery for many years into the future.

The Ecosystem Services Review enables companies to assess ecosystem services in terms of their impacts on them as well as to determine what ecosystem services the company is dependent upon. The analysis has proven powerful for the companies that have participated to date. Participants were able to choose the nature and scale of the analysis, applying it to all or part of their business.¹⁶⁸ Many participants in the Sustainable Business Council's programme are or will be making changes to business practices in light of the review's findings.¹⁶⁹ The future work programme will see many more businesses encouraged and supported to use the tool, and to factor ecosystem services into their decision-making.



Ecosystem services are poorly understood but have big implications for economic wellbeing. *Manu Bird*

Suggested solution

Ecosystem services assessments offer additional ability to ascertain and describe ecological effects of proposals and to demonstrate the long-term consequences of different choices. However, limited knowledge of how they can be applied to best effect means their uptake is limited in practice. Resource management guidance, for where and when ecosystem services approaches are appropriate and necessary, would enable agencies, community groups and the private sector to make full use of this tool where it can make positive contributions to decision-making.

Of primary focus in this report is the role of ecosystem services assessment in business planning and impact assessment. The incorporation of ecosystem services in this way is increasingly common worldwide on both a voluntary basis (see Case Study 15) and in response to new requirements from organisations such as finance providers.

How would it improve outcomes and is it an effective means of doing so?

Ecosystem services assessments provide a structured and comprehensive means of articulating what environmental impacts might mean to vulnerable stakeholders, ecological interactions and other ecosystem services. More comprehensive and realistic assessment of effects over greater time scales will enable impacts to be more clearly set out to decision-makers.¹⁷⁰ The dependencies of business activities on the natural world are also able to be demonstrated through these tools.

Taking more explicit account of the impacts of development decisions will enable the more effective conservation of critical ecological elements which are often overlooked in conventional impact assessment and decision-making (e.g. lower plants, soil and less visible components of biodiversity).¹⁷¹ This technique can enable the costs of environmentally damaging activities to be compared to the benefits of carrying out those activities and there are many examples around the world of the difference this could make to business decision-making.¹⁷² Considering ecosystem services within impact assessment has been demonstrated to provide additional insights to those yielded from traditional business planning and impact assessment approaches.

Evaluation

1. The innovation is able to be implemented within New Zealand's existing legal settings

Ecosystem services assessments are versatile and can be used in voluntary contexts (e.g. internal business decisions and investment choices) or mandatory contexts such as resource consenting. Their usage in impact assessment can be within the regulatory process

or outside it to inform decisions made by the applicant or agency.

2. An agency or responsible party exists that could implement the innovation

Incorporating ecosystem services considerations into resource management decision-making is something that any organisation can undertake. It is unlikely, however, that it will be widely used in impact assessment unless organisations accept the information and recognise its value.

3. The innovation is unlikely to result in perverse outcomes in practice

Not all ecosystem services can be described to a sufficient level of detail to contribute to resource management decision-making. The most acute example of where monetary valuation techniques may fall down is where the techniques are used to describe intrinsic and other non-market values. However, the body of knowledge on ecosystem services is steadily improving. The wider exploration of relevant ecosystem services, regardless of valuation, is able to generate more far-reaching exploration of causation and potential solutions.

4. The innovation is additional and does not duplicate existing mechanisms

Traditional business planning and impact assessment processes would be enhanced with the additional consideration of ecosystem services. The added value of ecosystem services to existing procedural policy such as environmental impact assessment is already evident.¹⁷³ A recent research article demonstrates that ecosystem services valuation can make important contributions to impact assessment by better enabling provisioning services to be taken into account, and through more precisely predicting how proposed impacts may alter ecological processes and supporting services. The article goes on to caution that ecosystem services valuation cannot replace traditional impact assessment processes, but should instead be seen as an opportunity to augment analysis.¹⁷⁴

5. The cost of the proposed tool is not prohibitive on the suggested lead organisation

Incorporating ecosystem services into impact assessment and other business decision-making processes may cost more than present approaches but is highly scalable to conditions. However, tools like the Ecosystem Services Review can be applied with as little or as much detail (i.e. cost) as an organisation/decision may need. The potential for this kind of assessment to identify the wider array of risks and opportunities, refine business practices, lower overall environmental impacts, and maintain and improve the social licence to operate will likely render it money well spent. Agencies

and industry thought leaders could provide advice and support to members of the development community to assist, as the Sustainable Business Council has done. Businesses can opt to use tools such as the Ecosystem Services Review and otherwise incorporate consideration of ecosystem services at a scale of their choosing.

Summary

Ecosystem services valuations can demonstrate the cost of degradation of natural capital, the economic value of the ecosystem services upon which people and businesses rely, and the value of ecosystem restoration.

Recommendation

*That **resource users and consenting agencies** work together to establish a community of practice around the incorporation of ecosystem services assessments in a development context. Long term, policy guidance could be developed to guide their application in impact assessment.*

Conservation tenders for mitigation trust funding

Problem

Mitigation trust funding is often not targeted to maximise conservation gains, nor do fund managers always report on ecological outcomes.

Implications

Mitigation funding is commonly negotiated for large projects where effects are diverse and affect the wider community. Mitigation trusts are commonly established to distribute these funds via a (usually) contestable process. For example, Contact Energy's geothermal extraction project at Wairakei saw the establishment of a trust that distributed an agreed sum of money as mitigation for consent impacts. The key purpose of mitigation trusts is to address the adverse effects of a consented activity. The distribution of that funding may lack direction and be of limited conservation value. Depending on the nature and magnitude of effects, weakly targeted funding can undermine the ability of a trust to deliver on those basic compensatory requirements. The outcome of this for biodiversity is often substantial net loss that is largely obscured by assessments of compliance, which are generally confined to whether the agreed quantum has been spent or not.

Further, outcome monitoring can be very limited and targets poorly defined, meaning that the aggregated outcomes of funds are poorly understood. Funding that is not spent wisely is unlikely to sufficiently address adverse effects

and may squander an opportunity to achieve broader conservation outcomes. This can result in significant net losses of biodiversity despite an impression of negotiated gain, and a failure to equitably address the adverse effects.

Suggested solution

It is recommended that a conservation tendering model could be implemented to assist in the distribution of mitigation funding. Conservation tendering is a contestable allocation process that allows funding bodies to distribute money to small-scale projects according to their intended outcomes, over defined areas and in accordance with previously determined strategic goals. They can be used in a variety of ways and can have a range of co-benefits over and above traditional allocation strategies. For mitigation trusts, conservation tender models would require the following:

- Robust conservation goals preferably informed by spatial planning processes and clearly articulated at an appropriate management scale, e.g. regional or district council boundaries
- Clear and measurable targets, preferably able to be revised over time to address different conservation problems
- An understanding of relative priorities of different tasks
- Sufficient 'competition' in the subject area to drive efficiencies
- Sufficient administrative capacity within the trust to communicate and advertise the tendering process, administer the process and collect and collate aggregated results

Proposals can be vetted for quality, expertise involved and cost-effectiveness and can have monitoring requirements imposed. This approach may also enable greater landscape-level or organisation-level monitoring of ecological outcomes and increase ability to calibrate strategic focus over the life of a fund. Conservation tendering enables expenditure to more closely reflect overarching aims on an annual basis, as demonstrated by Case Study 16. It also enables more effective and efficient use of conservation funds. Here the application to the administration of mitigation trusts is contemplated, but its potential to improve allocation efficiency and outcomes of conservation funding more generally is obvious.

How would it improve outcomes and is it an effective means of doing so?

Conservation tenders allocate funds in a way that can maximise positive outcomes and enable landscape-scale outcome reporting. These mechanisms will be most successful where the priorities on which they focus are evidence-based and reviewed at regular intervals. The capacity to demonstrate success will rely on monitoring

Case Study 16

Corangamite coastal funding

The Australian Corangamite Catchment Management Authority administers funding to support coastal conservation in important coastal habitats such as saltmarsh. The federal government provides annual funding to private landowners, local government and conservation organisations in order to support conservation in these important areas. Applicants 'tender' for the funding and must commit to restoring a minimum of one hectare of a defined stretch of coastline.¹⁷⁵

The process begins when the manager of the land or organisation lodges a formal expression of interest. The administrator then carries out a site assessment based on the Environmental Benefit Index. This is calculated by the following equation:

$$\text{Preference (P)} \times \text{Gain (G)} / \text{Bid price (\$)}$$

P quantifies relative biophysical parameters within a landscape context, while G describes the anticipated environmental improvement compared to the status quo baseline. This equation was established through expert scientific panels and practitioner workshops.

Tenders that proceed do so on a competitive basis and successful bids are set for a period of five years. Management actions undertaken must be additional to business as usual scenarios. Tenders are evaluated according to a range of criteria including value for money and any longer-term commitments made by bidders. A total of AU\$2,459,430 has been spent to date, distributed over 58 sites and 1,148.16 hectares of primarily remnant vegetation. This has led to the planting of more than 200,000 plants.¹⁷⁶



Funding can support the protection of important habitats such as saltmarsh. Corangamite Catchment Management Authority

frameworks that gather appropriate data and make it available to decision-makers and the community.

Evaluation

1. The innovation is able to be implemented within New Zealand's existing legal settings

Implementation of conservation tenders would be relatively simple and likely lead to administrative efficiencies and enhanced reporting capacity. Their use could be incorporated into existing trust structures, with the agreement of trustees, and they can be mandated within future deeds of trust.

2. An agency or responsible party exists that could implement the innovation

A mitigation trust is generally established to distribute funding. Mitigation, offset and conservation monies may also be distributed by councils, the Department of Conservation and other entities, all of whom would be able to implement conservation tenders with minor revision of existing business practices.

3. The innovation is unlikely to result in perverse outcomes in practice

Selection solely on a cost basis, when criteria should be multi-dimensional, is a key risk. Multi-criteria analysis, which assesses the ability of projects to contribute to a diverse range of values, should be implemented to address this. Notwithstanding that consideration, the ability to drive costs and efficiencies down is a key function of these tools and should be retained as a focus of analysis.

4. The innovation is additional and does not duplicate existing mechanisms

A tendering process would serve to alter the manner in which conservation funding is dispensed to outside parties in order to increase its effectiveness in achieving priority conservation outcomes. In this sense it would improve an existing process rather than add to or duplicate it. Additionality may be achieved through spatially targeting priority areas and therefore maximising gains and increasing the scale of improvements to the environment. A tendering process may also increase participation when potential participants see that others around their land or project are taking part. Improving the strategy behind, and flexibility of, these funding mechanisms may also improve the long-term outcomes of contestable conservation funding, particularly where robust monitoring frameworks are in place and enforced

through linking payment to accurate and timely reporting of outcomes.

5. The cost of the proposed tool is not prohibitive on the suggested lead organisation/s

The administration structures and costs of mitigation trusts vary considerably. Introducing a conservation tendering framework would initially increase the oversight and potentially the expertise required to vet applications. The monitoring costs – if not already required – may increase as they focus on measurable ecological outcomes rather than only expenditure or other more easily quantifiable aspects.

Summary

Conservation tenders could help allocate mitigation, offset and compensation funding based on a principle of maximising the difference made by the expenditure and enabling aggregated reporting of whole-fund ecological outcomes.

Recommendation

*That **consenting agencies, resource users and mitigation trusts** incorporate tools that maximise the outcomes of funding for conservation arising from development and ensure that monitoring is focused on outcomes on both a project and a landscape level.*

Safe Harbour agreements

Problem

Resource users that demonstrate high levels of compliance and best practice management on a consistent basis, and administer activities potentially consistent with conservation goals, have limited incentive to maintain good performance as compliance costs remain constant.

Implications

Only providing for conservation in areas set aside for that purpose potentially disregards the potential to both limit adverse effects and promote positive ecological outcomes as an integral part of resource use and development. This is particularly the case with ongoing and long-term activities such as farming. Few mechanisms recognise positive stewardship of land in New Zealand and there is no clear link between this recognition and any relevant regulatory requirements. Equally, wholly voluntary conservation programmes often have uncertain value and unreliable outcomes.

Suggested solution

Cooperative policy instruments could be developed that enable development activities to continue while achieving

agreed conservation outcomes within and alongside the usual activities. These could be modelled on the 'Safe Harbour' agreements in the United States, which is one of a number of jurisdictions that use such agreements to engage private landowners in broader conservation tasks.¹⁷⁷ Safe Harbour agreements are conservation plans which mandate certain management activities and ecological outcomes in exchange for financial incentives, advice, social recognition and sometimes reduced regulatory requirements (see Case Study 17). While this pluralistic approach is common throughout Europe and many other jurisdictions, there is only very limited policy recognition of the value of the concept in New Zealand.¹⁷⁸ These types of programmes are recommended within the as yet unimplemented Mackenzie Accord.

Conducting development activities in a more sympathetic way would reduce the loss of biodiversity and ecosystem services usually inherent within development. Safe Harbour agreements may be most applicable and effective in New Zealand if used to recognise early adopters of high environmental standards ('beyond compliance'), particularly where regulatory requirements may increase in the future. They may also be applicable in situations where the activity being conducted on the site is compatible with conservation outcomes.

Given that our economy is heavily reliant upon agriculture, it would seem logical to investigate concepts that seek to reduce the overall impacts of farming and garner additional conservation gains in the process. A potential immediate application may be in areas subject to ongoing expansion and intensification of dairy farming. Safe Harbour agreements may provide a mechanism by which willing landowners are incentivised to reduce the overall environmental impacts of their farming activities, and increase associated biodiversity benefits, without the need for or as a prelude to regulation. Landowners or resource users could sign up to conservation agreements that entail a waiver from district or regional plan controls on certain relevant activities.

How would it improve outcomes and is it an effective means of doing so?

Safe Harbour agreements have demonstrated their utility in driving behaviour change, building goodwill and achieving sustained conservation outcomes. In some circumstances, a Safe Harbour agreement could secure wins for nature by providing landowners with an incentive to be environmentally responsible. Where they have been deployed, Safe Harbour agreements have yielded impressive conservation outcomes, particularly where administrative burdens on landowners are not too cumbersome.¹⁷⁹

A mechanism that recognises a 'trusted entity' capable of delivering conservation gains within their usual operations

Case Study 17 Safe Harbour agreements

The US Endangered Species Act explicitly provides for 'Safe Harbour' agreements.¹⁸⁰ This mechanism came about because so many endangered species predominantly existed on private land (just as in New Zealand). A Safe Harbour agreement under the Act is required to provide a net conservation benefit that contributes to the recovery of the species in question relative to a baseline scenario described and agreed upon initially.

There are two forms of Safe Harbour agreements. In the first, landowners are engaged by the US Fish and Wildlife Service and the Environmental Defense Fund to help protect threatened or endangered species. Such an agreement binds the agency to not increase certain regulatory burdens on the landowner for the period of the agreement in exchange for the landowner agreeing to certain conservation outcomes. Enhancement of Survival Permits are issued to landowners. These guarantee that no further action will be required by them for the life of the permit, even if populations of endangered species increase on their property.¹⁸¹ The second form sees an agency administer a single Safe Harbour agreement that may operate over a large geographical area and multiple properties.

These mechanisms mean that landowners can carry out conservation activities on their property that may result in increased populations of important species without fear of added regulatory requirements. The activities can include proactive conservation measures and also the avoidance of activities with deleterious impacts on the subject species (e.g. habitat fragmentation). There is a backstop in the legal requirements whereby if the landowner is likely to undertake or be undertaking works that may have a deleterious impact on the species, the permit may be revoked. Landowners can renew agreements for as long as they wish, and subsequent landowners can choose to uplift or continue the agreement. While the agreements do not require protection in perpetuity, net benefits to the species must be demonstrable.¹⁸²

would provide a valuable pathway to partnership and collaboration. There is significant potential to restore ecosystems on private land, were the right incentives in place. In many cases this can be achieved with minimal implications for productive use in the rural sector. For example, Dymond et al. (2013) demonstrated that there are 2 million hectares of grassland and 0.7 million hectares of regenerating shrubland with a high ratio of biodiversity benefit over loss of agricultural production, meaning that it was more cost-effective to retain it than convert it for productive purposes.¹⁸³

Evaluation

1. The innovation is able to be implemented within New Zealand's existing legal settings

Most relevant legislation in New Zealand has some means of absolving certain players, activities or impacts from regulatory requirements: 'permitted activities' under the RMA and EEZ Act being an example. Such an approach can be formulated in regional and district planning documents. For example, the Operative Waipa District Plan provides for certain dispensations from consenting requirements for some activities (e.g. the activities of an 'approved' conservation organisation within 50 metres of a peat lake). This, however, is not subject to any agreed or monitored conservation gains.¹⁸⁴

2. An agency or responsible party exists that could implement the innovation

All consenting agencies, and agencies with statutory responsibilities for biodiversity, could implement this initiative. In order to link it to potentially relaxed regulatory requirements, a consenting agency would have to be involved. In Ontario, Canada, the Ministry of

Resources is the management agency that administers Safe Harbour agreements.¹⁸⁵ Success of the instruments would likely in part depend on the level of ecological expertise and understanding of conservation priorities within the agency, which indicates that the Department of Conservation should be involved. A multi-agency effort would be required to implement this approach at a national scale, although it could be put in place by an individual agency, potentially inclusive of a Schedule 1 process under the RMA (in the case of a council). For example, a plan change could render activities 'in accordance with an approved Safe Harbour agreement' permitted activities.

3. The innovation is unlikely to result in perverse outcomes in practice

There is a clear and serious risk of poor outcomes, particularly if the risk to biodiversity is poorly managed when absolving landowners of regulatory requirements. This method is likely to be most appropriate for long-term landholders or resource users, and is unlikely to be appropriate where the area is vulnerable to intensive development activity of very limited duration (e.g. residential subdivision or gravel extraction).¹⁸⁶ Landowners may experience a 'change of heart', or successive landowners may keep the agreement in place without fully understanding the implications. For this reason, a robust and transparent management and monitoring system would be essential to the success of this measure. Measures of success will need to be commensurate to the time period of the agreement to be able to track compliance and success over time at reasonable intervals.

In addition, there is also a serious risk of agency capture, whereby private interests pressure the (usually small and underfunded) regulatory agency into agreeing to

the delivery of certain conservation gains that may not align with conservation priorities or best practice. It is suggested that the Department of Conservation or an independent entity is engaged as a third party in these arrangements to ensure they are sensible and reasonable and do deliver concrete ecological benefits. The requirements for the agreements could also be linked to regional or national conservation priorities.

4. The innovation is additional and does not duplicate existing mechanisms

Recognising positive stewardship is a growing focus area for many New Zealand regulatory agencies, and this positive interface with the community can have wider social benefits in addition to ecological gains. Safe Harbour agreements may provide some financial relief from onerous consenting requirements for a range of landowners including iwi trusts and those undertaking ecologically sensitive developments and ecological restoration programmes. Over large sites or multiple important habitats they may yield very significant results and feed into national conservation priorities. Safe Harbour agreements may build goodwill and positive relationships between communities, agencies and landowners and generate additional ecological gains from which the community directly and indirectly benefits.

5. The cost of the proposed tool is not prohibitive on the suggested lead organisation/s

Provision of Safe Harbour agreements will generate additional costs in the form of survey, liaison, development of conservation agreements and ongoing

costs of monitoring. These may fall on agencies or on resource users in different ways, depending on how the programme is designed. If agencies absorb the costs and do not have a cost recovery pathway available, it may have serious budget implications. If landowners and resource users are required to cover the costs of the agreement, as well as the resultant conservation activities, it may be more expensive than a traditional consenting pathway and thus be unattractive. Costs should be fairly shared across participants and not fall unreasonably on the agency or the landowner. They should be lower and more effective than traditional consenting pathways.

Summary

Providing formal recognition of sites that combine development activities and conservation activities by reducing regulatory requirements where appropriate has the potential to generate positive ecological outcomes over and above those which can be achieved by setting areas aside for conservation. The success of these measures will depend on rigorous monitoring of conservation achievement and clear mechanisms to ensure that regulatory requirements are reinstated in the event of non-achievement.

Recommendation

*That the **regional and district councils in association with the Department of Conservation** investigate the use of Safe Harbour agreements as an alternative to regulatory requirements in appropriate situations.*

Tūturiwhatu/northern New Zealand dotterel (*Charadrius obscurus aquilonius*)

This slight wading bird is the northern subspecies of the New Zealand dotterel. Once common and widespread, the northern dotterel numbers just 1,700 individuals distributed throughout much of the North Island. The other subspecies, the southern New Zealand dotterel, is now generally confined to Stewart Island/Rakiura and numbers just 250.¹⁸⁷ The tuturiwhatu inhabits the coastline, much of which has been heavily modified to create settlements, ports and recreational facilities. Males and females both guard the nest, which is usually located out in the open in a small bowl-shaped hole in the sand. If a perceived threat is sensed by the adult birds, they will feign injury to distract attention from the nest.

Human disturbance is one cause of decline, with people, vehicles and animals on the beaches frightening birds and destroying their nests.¹⁸⁸ However, most of the decline relates to introduced predators. Where predators are controlled, fledging is generally regular. The tuturiwhatu is adaptable and in some situations it can benefit from new structures. One of the most successful breeding sites for this species is the fuel tank compound at the Marsden Point Refinery in Northland. They have also colonised a tailings dam and begun nesting on the new golf course (Tara-iti) at Te Arai.¹⁸⁹



Tūturiwhatu/northern New Zealand dotterel (*Charadrius obscurus aquilonius*). Manu Bird

Opportunities for improvements and recommendations to address them

The opportunities for improvement that this report has identified are summarised in Table 7, with the resulting recommendations set out in the second column. To provide a sense of priority, each recommendation is given a number between 1 and 3. The number 1 applies to solutions that are high priority and should be implemented as soon as possible; 2 applies to areas needing rapid

investment but likely sustained, long-term undertakings; and 3 applies to recommendations that are discretionary and may serve to help address issues in some situations but not in others. Different stakeholders will ascribe different levels of priority to these tasks and will be differently able to implement them. As a result, a linear prioritisation is neither possible nor appropriate.

Fischers Egg (*Claustula fischeri*)

Fischers egg is a puffball-like New Zealand fungus that was thought to be endemic when first described in 1927. However *Claustula fischeri* was found to also exist in Tasmania in 1997. This fungus is classified as 'Nationally Critical' on the Department of Conservation Threat Classification List and has been recently submitted to the International Union for the Conservation of Nature (IUCN) for inclusion on the Red List. Just two locations are known for the fungus in New Zealand: one in Fringed Hill overlooking Nelson City and another (but only a single record) in Croydon Bush near Gore. The Fringed Hill site is therefore the most important, and is also the location of a planned mountain bike development. Mycology experts met with staff from Nelson City Council when development plans were being processed and discussed various strategies for avoiding harm to the species. But poor knowledge of fungi means that it is difficult to assess population size and vulnerability, and therefore the risk from development activities.



Fischers Egg (*Claustula fischeri*). Landcare Research

	Problem	Recommendation
BIODIVERSITY IN STRATEGY	Biodiversity information is incomplete, poorly managed and difficult to access for decision-makers, applicants and submitters.	That a multi-agency effort is undertaken to establish a central national database or databases that collate and make available biodiversity data to help inform management and better safeguard biodiversity in development (including technical support for usage by decision-makers). That the Terrestrial Freshwater Biodiversity Information System Fund be relaunched with increased funding, recognising that its purpose remains critical and relevant, to enable agencies to contribute to the information database. (2)
	Strategic planning for biodiversity commonly fails to address cumulative and long-term impacts on biodiversity, and the knock-on effects throughout ecosystems.	That councils and other decision-makers consider the use of strategic decision-making tools such as Sustainability Appraisal and the Mauri Model to support plan-making, policy development and other collaborative decision-making processes by front-ending considerations of biodiversity. (3)
	While goals such as achieving 'no net loss' of biodiversity at a regional scale, halting the decline of biodiversity, and 'maintaining' biodiversity are common, clear pathways for their achievement are not.	That regional councils in association with the Department of Conservation and communities (including iwi) develop comprehensive regional biodiversity plans that demonstrate how the maintenance of biodiversity will be achieved and how development will be managed so as to achieve that goal. (2)
NATIONAL LEADERSHIP	There is no binding national policy on biodiversity which has reduced the prominence of biodiversity in resource management decision-making.	That the Ministry for the Environment develops a national policy statement that provides clarity on key issues that affect the protection of indigenous biodiversity. (1)
	The weak national policy settings for impact management generate ambiguity and uncertainty for all stakeholders.	That the Ministry for the Environment develops policy to guide impact management in a regulatory context, potentially as part of the national policy statement on biodiversity. (1)
	The application of adaptive management is inconsistent, lacks clear limits and thus may not adequately manage uncertainty in resource consenting.	That the Ministry for the Environment in association with consenting agencies develops common policy direction for the implementation of adaptive management approaches, particularly in the context of significant uncertainty. (1)

Table 7: Digest of recommendations



	Problem	Recommendation
REFINING CURRENT PRACTICE	A wide range of legal protection tools for biodiversity are available, many of which are not well-known by practitioners.	That the Department of Conservation in association with regional and district councils develops educational material to outline the potential tools available to safeguard biodiversity and investigates potential improvements to them. (3)
	There are no comprehensive guidelines for biodiversity impact assessment on vulnerable taxa and ecosystems.	That the Ministry for the Environment together with the Department of Conservation, other consenting agencies and relevant experts coordinate the formulation of mandatory biodiversity impact assessment guidance for vulnerable species. (2)
	Resource consent conditions often do not mandate desirable ecological goals but instead focus on process-oriented issues that can be unrelated to the achievement of tangible outcomes.	That consenting agencies adopt wide use where appropriate of outcomes-based conditions that are clear, measurable and enforceable. (3)
	RMA bonds are often poorly administered, constraining their ability to increase the security of resource consent outcomes. In other cases, they are not used where they could help improve the security of outcomes.	That consenting agencies collaborate together, with the assistance of the Ministry for the Environment , to produce common guidance on appropriate costing of bonds. (3)
INVESTIGATING NEW TOOLS	Resource users are generally responsible for the delivery of compensatory requirements leading to sometimes poor compliance and poor ecological outcomes.	That the third party delivery of mitigation, offsets and compensation is investigated by an appropriate entity established for this purpose. (2)
	Ecosystem services assessments are rarely implemented, even though there is potential for them to add value to decision-making in resource management.	That resource users and consenting agencies work together to establish a community of practice around the incorporation of ecosystem services assessments in a development context. Long term, policy guidance could be developed to guide their application in impact assessment. (3)
	Mitigation trust funding is often not targeted to maximise conservation gains, nor do fund managers always report on ecological outcomes.	That consenting agencies, resource users and mitigation trusts incorporate tools that maximise the outcomes of funding for conservation arising from development and ensure that monitoring is focused on outcomes on both a project and a landscape level. (2)
	Resource users that demonstrate high levels of compliance and best practice management on a consistent basis, and administer activities potentially consistent with conservation goals, have limited incentive to maintain good performance as compliance costs remain constant.	That the regional and district councils in association with the Department of Conservation investigate the use of Safe Harbour agreements as an alternative to regulatory requirements in appropriate situations. (3)

Firth of Thames. Catherine Kirby

Conclusions

This report has demonstrated that there is significant potential and opportunity to innovate in the policy space to better protect biodiversity in a development context. Marshalling the enhanced implementation of existing tools, together with devising new approaches, will enable better outcomes for the environment and for stakeholders. Consenting agencies, resource users and communities should work together to make best use of policy instruments and ensure that they are fit for purpose. A common theme emerging from our analysis of existing approaches is the absence of national guidance, and we consider this to be a matter of primary importance which needs to be addressed urgently. The creation of a clear and cogent national policy context for indigenous biodiversity management is high priority, including a protocol for impact management and the use of adaptive management to address uncertainty.

To support high-level policy direction, detailed implementation planning and protection is required to ensure that where important biodiversity exists these approaches provide for its persistence. There is considerable opportunity to refine the use of existing tools, where improved implementation could increase their contribution to good outcomes (e.g. accurately costing RMA bonds). Where no fit-for-purpose tool exists, we encourage the development of new tools to address the need for adaptive and flexible methods to achieve good outcomes for biodiversity. There is considerable scope for the application of new tools within the current statutory context.

Although much can be achieved through the adoption of tactical and practical approaches, as recommended in this report, we also remind readers of the key message of *Vanishing Nature* which is that truly transformative change will only be achieved through strategic solutions, such as the implementation of novel economic institutions. Powerful new strategic approaches would help to shift the default approach from 'what do we need to save?' to a more sustainable 'what do we need to harm?'. In the absence of strategic change, the implementation of lower-level tools will help to slow biodiversity loss and environmental degradation, but their achievement is only likely to be modest.

But the improvements these tools do promise are essential to stemming the tide of biodiversity loss in the near term. Our precious biodiversity is in crisis, and better management of development impacts is well within our reach. More clarity and flexibility through adaptive policy approaches and a willingness to try new ways of doing things will result in better outcomes and greater efficiencies. By closing the implementation gap through enhancing current practice and introducing new tools, New Zealand can look forward to a prosperous future with a thriving environment and robust economy.



Appendix 1

Agency	Relevant legislation	Relevant role
Ministry for the Environment	Resource Management Act 1991 (RMA)	Lead agency for implementing sustainable management, developing national instruments, and providing an important oversight role of the RMA (s 24)
	Environment Act 1986	Research and advisory role (s 31)
Councils	Resource Management Act 1991	Regional councils have a statutory goal to maintain biodiversity under the RMA ¹⁹⁰ while district councils have a more limited but complementary mandate to control the effects of the use, development, or protection of land, including for the purpose of maintaining indigenous biological diversity. ¹⁹¹ They achieve this through developing policies and plans and acting as consenting agencies.
Department of Conservation	Conservation Act 1987	Overarching responsibility for biodiversity conservation, formulation of conservation management strategies, and a statutory advocacy role nationwide
	Wildlife Act 1953 and Marine Mammal Protection Act 1978	Lead implementation agency including all permitting and enforcement of wildlife law as it applies to indigenous biodiversity
Environmental Protection Authority	Exclusive Economic Zone and Continental Shelf (Environmental Effects) Act 2012 (EEZ Act)	Deciding marine consents (which are considered by Decision-Making Committees) and monitoring and enforcing their requirements under the EEZ Act
	Resource Management Act 1991	Processing of 'proposals of national significance' under the RMA which are heard by Boards of Inquiry appointed by the Minister
Māori Land Court	Te Ture Whenua Act 1993	Managing development impacts on the approximately 6.1 per cent of New Zealand's land area under the Te Ture Whenua Act 1993. ¹⁹² Māori land is defined as land that is held by Māori in accordance with tikanga Māori. ¹⁹³
Land Information New Zealand	Various	Overarching role in managing eight per cent of New Zealand's land which is used for a variety of purposes including forestry (licences for land held for Treaty settlement), farming (South Island High Country), mining (e.g. Waihi gold mine) and navigable riverbeds
Overseas Investment Office and relevant Ministers of subject Ministries ¹⁹⁴	Overseas Investment Act 2005	Processing of approvals for foreign purchase of 'sensitive New Zealand assets' which can include conservation requirements
Fish and Game Council of New Zealand	Conservation Act 1987	Statutory duty to advocate generally and in any statutory planning process the interests of the New Zealand Fish and Game Council and, with its agreement, of any Fish and Game Council in the management of sports fish and game, and habitat ¹⁹⁵
Ministry for Primary Industries	Fisheries Act 1996, Forest Act 1949 (including Sustainable Forest Amendment Act 1993)	Principal industry regulator for primary industries including managing the fisheries Quota Management System and biosecurity
	Biosecurity Act 1993	

Table 8: Legal responsibilities related to biodiversity in development

Appendix 2

A wide range of tools were canvassed for their potential contribution to protecting biodiversity in development in New Zealand. Many were discarded. Table 9 provides a summary of those tools that were considered but not ultimately progressed and gives the reason/s each was excluded from the recommendations in this report.

Potential tool and what it could achieve	Reasons for not considering it as a recommendation from this report
Universal rates relief to improve incentives for conservation on private land and avoid damage to biodiversity	Rates relief is usually minimal and insufficient to encourage further conservation work (i.e. additional to that already occurring). In addition, economic incentives for nature protection were set aside for the purpose of this report, as they warrant specific and extensive discussion.
Specific consenting mechanism for adaptive management	Further work is required to assess the potential utility of a custom mechanism for adaptive management projects.
Enshrining rights for nature within legislation and practice	Significant legislative changes are required to fully recognise rights for nature in New Zealand, and further work is required to embed this.
Mandatory strategic environmental assessment as in many international jurisdictions	An overarching strategic assessment framework that coordinates existing law and policy is likely necessary for New Zealand, but the magnitude of legislative change puts it beyond the scope of this report.
Limited-life covenants	Limited life covenanting mechanisms are available already in New Zealand so it was unclear what another version would contribute.
Long-term monitoring systems for councils and other consenting agencies	It is clear that the capacity of agencies and their staff to monitor long-term projects needs to be improved, but further work is needed to establish the most effective way to achieve that.
Further tools for managing disturbance on vulnerable species	A wide range of tools are available or able to be implemented within existing instruments (e.g. district plan rules) to limit human-induced disturbance. Further work is required to determine other methods and additional information requirements needed to limit adverse impacts of disturbance.
Comprehensive threatened species and ecosystems legislation	<i>Vanishing Nature</i> recommended this solution, and it is supported. However, the scope of changes required which necessitate legislative reform is outside the ambit of this report.
Key threatening processes regime	<i>Vanishing Nature</i> set out the advantages of this approach compared with present efforts, but it will likely require significant changes to both policy and practice which is outside the scope of this report.
Legal protection for threatened plants	<i>Vanishing Nature</i> recommends this solution and work is underway by the Department of Conservation to address this.
Integrated planning and/or consenting process between major legislation (e.g. RMA and Conservation Act) to reduce duplication	Although commonly suggested by resource users, the merit of this is unclear and further work is needed to establish the costs and benefits of this proposal. The Resource Legislation Amendment Bill 2015 indicates that something of this nature is currently proposed.

Table 9: Summary of solutions considered but discarded for the purpose of this report

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New Zealand's world-renowned natural heritage is in crisis, and efforts to safeguard it in a development context can be fraught. New Zealand relies on thriving biodiversity as it underpins our prosperity and substantial change is needed to turn the negative trends around. In the near term, much opportunity exists to enhance outcomes through improvements in existing policy and practice and other innovations.

This report describes present approaches to managing development and proposes a range of policy and practice improvements to enhance environmental outcomes and regulatory efficiency. The publication is aimed at the development sector, ecological professionals, policy makers and planners. Its user-friendly format and plain language makes it accessible to a general audience as well.

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